

Every safety data sheet is now built from your data each time it is generated. The formula and each raw material's records produce the hazard classification. That classification picks the wording for most sections, from first aid through disposal and transport. Raw-material properties supply the physical data, your regulatory lists supply Section 15, and Admin Settings, product families and manufacturer records supply the supplier block, use text and disclaimer. The 16-section layout, the labels and the regulatory phrases are fixed text, translated into English, Spanish, French and German. Where an operator has typed per-product text for a language, that text prints instead of the automatic text.

Before this update, only about a third of the sheet came from data. Most paragraphs (first aid, firefighting, spills, handling, stability, toxicology, ecology, disposal, regulatory status) were the same canned sentences on every product. Several of them were English on every language sheet. Those paragraphs now change with each product's hazards, physical state and ingredients. Concentrations print as prescribed ranges instead of exact percentages. Transport is classified for the mixture as a whole, not copied from one ingredient. The editor saves only text an operator actually types, so automatic text is no longer frozen when a product is saved.

Key points

- A blank emergency phone number blocks publishing. Standard sheets read it from Admin Settings only. Private-label sheets read only the manufacturer's own number.
- A Section 14 transport result of 'Not determined' (no flash point and no hazard data) blocks publishing until the data is filled in or a UN number or hazard class is entered as an override.
- Raw-material data now drives much of the sheet: flash point, boiling point, physical state, odor, appearance, solubility, specific gravity and VOC feed Sections 5, 6, 8, 9, 13 and 14, so gaps show up as 'Not determined'.
- Keep the regulatory lists current (TSCA, SARA 313, HAP, SNUR, Prop 65, carcinogens, RCRA waste codes). They decide what Sections 11 to 15 say, and a TSCA gap prints 'not verified' and raises an operator warning.
- Maintain product-family use text and the disclaimer in each language, and manufacturer records for private label. Otherwise ES/FR/DE sheets fall back to generic or English text.
- Regulatory wording stays fixed: GHS hazard and precautionary statements, the 0.1% disclosure cut-off, the prescribed concentration ranges, 49 CFR shipping names, Prop 65 safe-harbor wording and the CFR/ANSI/NFPA citations.
- Per-product edits now hold only text someone typed and can be reset to automatic. Edits saved under the old editor may still carry outdated wording and should be reviewed.

How to read this document

The pages that follow are laid out like a generated Safety Data Sheet: the same sixteen sections in the same order, under the same banners, with the same field labels. Where the sheet would print a value, this document shows where that value comes from.

Each entry has a coloured tag naming the kind of source, a plain description (when several sources compete, they are listed as A > B > C, and the first one that has a value wins), an example of what prints, **You maintain** (the information a person keeps current) and **If missing** (exactly what prints, or what happens, when that information is blank). The small grey line is the technical reference for the system team.

FORMULA / RAW MATERIALS

Calculated from the product's formula and the data entered on its raw materials.

HAZARD ENGINE

GHS classification computed from the formula and the competent person determinations.

REGULATORY LIST

Looked up in a maintained regulatory list (Prop 65, HAP, SARA 313, TSCA, RCRA, carcinogens).

PRODUCT FAMILY

Comes from the product's family, assigned by rules or inherited from its raw materials.

TRANSPORT RULES

Derived by the DOT rules from flash point, boiling point and hazard classes.

COMPANY / MANUFACTURER

Company settings, or the private label manufacturer on private label sheets.

ADMIN SETTING

An administrator setting that applies to every sheet.

PRODUCT OVERRIDE

Text typed for one product and language in the SDS editor; replaces the automatic value.

PUBLISH RECORD

Stamped when a version is published (version number, effective date).

TEXT CHOSEN BY HAZARD

Standard wording, selected and combined according to the product's hazards.

FIXED TEXT (TRANSLATED)

Standard wording, the same on every sheet, in the sheet's language.

FIXED TEXT

The same on every sheet (labels, regulatory citations).

At a glance

Fields mapped in Sections 1-16	226
From your product data	102 (45%)
Standard wording chosen by the product's hazards	68 (30%)
Admin settings and per-product edits	6 (3%)
Fixed text (layout, labels, regulatory phrases)	50 (22%)
Conditions that block publishing	6
Languages	English, Spanish, French, German (same sources; labels and standard wording translated)

SECTION 1: IDENTIFICATION

Section 1 combines the product's code and description from the CMS import, use text from its product family (unless the product has its own text), and the supplier block and logo from Admin > Settings (or from the private-label manufacturer record), all set in fixed, translated labels and page furniture.

Product Identifier:	FORMULA / RAW MATERIALS The product code and description from the CMS import. Alias sheets use the alias code and description. Private label: custom code/description > shared alias > product code/description. <i>On a sheet: Product Identifier: BK1080 — Process Black Offset Ink</i> You maintain: Product code and description (CMS import / Finished goods screen); supplier product name on resale raw materials; alias customer code and description; private-label custom code/description when used If missing: A blank description prints a dangling "BK1080 — ". This includes a resale RM whose supplier name is stored as " and an alias with a blank description. "CODE — CODE" prints only for legacy NULL RM names. The line is never omitted. <pre>FG product_code.' - '.description (SDSGenerator:821); resale stripPack(rm.internal_code).' - '.(supplier_product_name ?? code) :374; createAliasVariant :592-604; PrivateLabelPublisher::resolveIdentity</pre>
Recommended Use:	PRODUCT FAMILY The product family's text for the sheet language. Order: per-product override > product's own text > family text (sheet language, else English) > standard sentence (a different one for resale raw materials). <i>On a sheet: Recommended Use: UV-curable offset ink for sheetfed presses.</i> You maintain: Recommended Use text in every language on each Product Family, plus family rules (recompute after an import or rule change). Leave the product's own Recommended Use blank unless needed. Keep the legacy Admin 'Default Product Use' setting blank. If missing: "Printing ink for commercial and industrial applications." (resale: "Raw material for industrial formulation."), translated. Never blank. <pre>resolveUseText(:844-864): text_overrides(1,'recommended_use',fg,lang) > finished_goods.recommended_use > product_families.recommended_use_json[lang]>['en'] > section1.recommended_use[_resale]</pre>
Restrictions on Use:	PRODUCT FAMILY The same order as Recommended Use: per-product override > product's own text > family text (sheet language, else English) > standard sentence. <i>On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.</i> You maintain: Restrictions text in every language on each Product Family. Leave the product's own Restrictions blank unless needed. Keep the legacy setting sds.default_restrictions_on_use blank. If missing: "For professional/industrial use only. Not for household consumer use." (the same text for resale), translated. Never blank. <pre>resolveUseText(..,'restrictions'): text_overrides(1,'restrictions') > finished_goods.restrictions_on_use > product_families.restrictions_json[lang]>['en'] > section1.restrictions[_resale]</pre>
Manufacturer / Supplier Information:	FIXED TEXT (TRANSLATED) A fixed translated sub-heading above the supplier block. You maintain: Nothing If missing: Always printed, even when every supplier line below it is blank <pre>labels.manufacturer_info via getLabels() -> meta.labels (SDSGenerator:3463); PDFService:373-376; preview.php:66</pre>
Company:	COMPANY / MANUFACTURER Company name from Admin > Settings. On a private-label sheet, the manufacturer's name. <i>On a sheet: Company: Your Ink Company, Inc.</i> You maintain: Company Name in Admin > Settings > Manufacturer Information; Name on each private-label manufacturer If missing: A saved blank omits the line. A key that was never saved prints the config.php placeholder. <pre>settings company.name (config.php only if key never stored) -> buildManufacturerBlock manufacturer_name; PL manufacturers.name via Manufacturer::toCompanyInfo</pre>
Address:	COMPANY / MANUFACTURER Street, city, state and ZIP, and country from Admin > Settings (or the private-label manufacturer), joined with commas and skipping blanks. <i>On a sheet: Address: 123 Industrial Blvd, Suite 100, Anytown, OH 44000, US</i> You maintain: Address, City, State, ZIP, Country in Admin > Settings; the same fields on each manufacturer If missing: Blank parts are skipped. If every part is blank, the line is omitted. <pre>formatManufacturerAddress (SDSGenerator:616-628): company.address/city/state/zip/country; PL manufacturers.address/city/state/zip/country</pre>

Phone:	COMPANY / MANUFACTURER Company phone from Admin > Settings, or the private-label manufacturer's phone. <i>On a sheet: Phone: (555) 123-4567</i> You maintain: Phone in Admin > Settings; phone on each manufacturer If missing: The line is omitted silently. Nothing warns and publishing is not blocked. settings company.phone (config fallback if never stored) / manufacturers.phone -> buildManufacturerBlock (trimmed) :645; PDFService:381
Email:	COMPANY / MANUFACTURER Company email from Admin > Settings, or the private-label manufacturer's email. <i>On a sheet: Email: sds@yourinkcompany.com</i> You maintain: Email in Admin > Settings; email on each manufacturer If missing: Line omitted settings company.email / manufacturers.email -> section1 :830; PDFService:383; preview.php:58
Website:	COMPANY / MANUFACTURER Company website from Admin > Settings, or the private-label manufacturer's website. <i>On a sheet: Website: https://www.yourinkcompany.com</i> You maintain: Website in Admin > Settings; website on each manufacturer If missing: Line omitted settings company.website / manufacturers.website -> section1 :831; PDFService:384; preview.php:59
Emergency:	COMPANY / MANUFACTURER The Emergency Phone saved in Admin > Settings, with no fallback. On a private-label sheet, only the manufacturer's own emergency number. Alias sheets auto-published from an existing sheet copy that sheet's number. <i>On a sheet: Emergency: CHEMTREC: (800) 424-9300</i> You maintain: Emergency Phone in Admin > Settings (required); emergency phone on every private-label manufacturer If missing: The line is omitted. Gated publish paths refuse to publish, and the FG and PL previews show "Publishing blocked: ...". The new-alias auto-publish copies whatever the base sheet holds. settings company.emergency_phone only (getCompanySettings skips config, :3288-3301) -> section1 :829; PL trim(manufacturers.emergency_phone) :678; gate SDSReadinessService:313-352

SECTION 2: HAZARD(S) IDENTIFICATION

Section 2 is built almost entirely from the hazard classification of the formula: ingredient hazard records (PubChem), competent-person determinations, trade-secret entries, mixture calculations, the optional per-product hazard override and the carcinogen list. That classification is worded from fixed GHS phrase and label tables in the sheet language, and two per-product text overrides (Other Hazards and the Section 8 PPE fields) can change it.

Not a hazardous substance or mixture.:	TEXT CHOSEN BY HAZARD A fixed sentence. It prints only when, after every hazard step, the product has no signal word, pictogram, hazard class or H-statement. P-statements alone do not count. <i>On a sheet: Not a hazardous substance or mixture.</i> You maintain: Nothing for the wording. Whether it shows depends on the hazard data listed under Signal word. If missing: Snapshots stored before 2026-05-22 have no is_classified key, so a re-render prints the sentence even on classified products. section2.not_classified (en.php:16); gate SDSGenerator::isClassified() :903-909; PDFService.php:400-404; preview.php:72-74
Signal word (DANGER / WARNING):	HAZARD ENGINE The most severe signal word found in the triggered ingredient hazard records, determinations, trade-secret entries and mixture rules. Order of application: computed result > product hazard override (additive mode can only raise it; replace mode sets it) > carcinogen-list upgrade. Red for Danger, orange for anything else. <i>On a sheet: DANGER (red) or WARNING (orange)</i> You maintain: Current hazard data for each ingredient CAS (Raw materials, hazard refresh); a determination for each CAS without federal data (CAS determinations); trade-secret hazards (Raw materials); optional hazard override (Finished goods); carcinogen list (Regulatory lists); physical state on every raw material. If missing: The signal word block is omitted. Known defect: a liquid whose only hazard is carbon black or TiO2 still prints 'WARNING' after its carcinogen line is removed. HazardEngine::classify :876-884 (hazard_classifications.signal_word, CPD, manual_hazard_json) > applyFinishedGoodOverride :1658-1765 > SDSGenerator applyCarcinogenFindings; PDF :406-414

Pictograms:	FIXED TEXT (TRANSLATED) A fixed label in the sheet language, shown only when pictograms are present. <i>On a sheet: Pictograms:</i> You maintain: Nothing If missing: Falls back to the sheet-language translation, then English, then the key. labels.pictograms (getLabels SDSGenerator.php:3464; en.php:544); PDFService.php:615; preview.php:83
GHS pictogram images (GHS01-GHS09):	HAZARD ENGINE All pictograms from the triggered ingredient hazard records (PubChem stores the whole substance's list on every class row), the determinations (limited to classes that trigger), trade-secret entries, mixture rules and the product hazard override. GHS07 is dropped when GHS06 or GHS05 is present; GHS08 is added by the carcinogen and carbon-black rules and prints last. <i>On a sheet: GHS02 flame + GHS07 exclamation mark diamonds</i> You maintain: The same hazard data as the signal word. Optional replacement artwork can be uploaded on Admin > Pictograms. If missing: No pictograms: the block is omitted. In the PDF, a code with no artwork is silently skipped; if no code has artwork, the text 'GHS02 (Flame), ...' prints instead. classify :910-916, :969 applyPictogramPrecedence :2275-2295; SDSGenerator :3700-3747, :4155-4157; PictogramHelper::getPngPath :63-80; PDF :597-660
Pictogram caption under each image:	FIXED TEXT (TRANSLATED) The fixed GHS pictogram name for each code, in the sheet language. <i>On a sheet: Flame / Exclamation Mark</i> You maintain: Nothing If missing: Prints the code itself, e.g. 'GHS02'. GHSStatements::pictogramName: ghs_{lang}.pictogram_names > PICTOGRAM_NAMES (EN) > code; GHSStatements.php:271-281, :346-360; PDFService.php:648-656
GHS Classification:	FIXED TEXT (TRANSLATED) A fixed heading, shown only when at least one hazard class exists. <i>On a sheet: GHS Classification:</i> You maintain: Nothing If missing: Falls back to the sheet-language translation, then the key. labels.ghs_classification (en.php:545); PDFService.php:467; preview.php:103
Physical Hazards: / Health Hazards: / Environmental Hazards:	FIXED TEXT (TRANSLATED) Three fixed headings. All three print whenever any class exists. Each class is placed under a group by an exact match of its name, and a name that does not match goes under Health Hazards. <i>On a sheet: Physical Hazards:</i> You maintain: Nothing for the headings. Class names must match the standard GHS names to land in the right group. If missing: Falls back to the sheet-language translation. An unrecognised class name (e.g. PubChem 'Flammable liquids' or 'Unclassified') is listed under Health Hazards. labels.physical_hazards/health_hazards/environmental_hazards (en.php:546-548); HazardEngine::groupByHazardType :2724-2737; PDF :469-486
None (under an empty hazard group):	FIXED TEXT (TRANSLATED) A fixed word in the sheet language, printed under any group that has no classes. <i>On a sheet: None</i> You maintain: Nothing If missing: Falls back to the sheet-language translation, then the key. labels.none (getLabels SDSGenerator.php:3516; en.php:690); PDFService.php:492-493; preview.php:149-150
Hazard class lines (H-code — Class (Category)):	HAZARD ENGINE One line per hazard class from triggered ingredient records, determinations, trade-secret entries, mixture calculations (summation/ATE/aquatic), the product hazard override, and the carcinogen-list and carbon-black rules. The engine keeps one category per exact class name (meant to be the most severe), and lines are sorted by H-code. <i>On a sheet: H226 — Flammable Liquids (Category 3): Flammable liquid and vapour</i> You maintain: Ingredient hazard data and determinations (Raw materials, CAS determinations); constituent CAS and %; physical state on each raw material; carcinogen list; hazard override classes, each with its H-code entered (Finished goods). If missing: No classes: the whole classification block is omitted. Unrecognised class or category: no H-code prefix. H-code not in the phrase table: no ': phrase'. Blank category: class name alone. classify hazard_classes :865-874, :973-989; consolidateHazardClasses :2609-2660; getDefaultsForClassCategory; SDSGenerator :3680-3700, :4130-4145; PDF :428-523

Unclassified (Health Hazards line):	HAZARD ENGINE PubChem stores a generic 'Unclassified' row when a substance has H-statements but no class it could read. At 1% or more that row triggers and prints under Health Hazards with no H-code or phrase. <i>On a sheet: Unclassified</i> You maintain: Fix the ingredient's hazard data with a determination (CAS determinations), or refresh PubChem. If missing: Not printed when the CAS has readable classes or is below 1%. PubChemConnector.php:686-695 (class_name 'Unclassified', category ''); HazardEngine::getCutoff 1% default :1190-1196; groupByHazardType default health
H-statement phrase (text after the colon):	REGULATORY LIST The standard GHS wording for the line's H-code, in the sheet language. The wording on the source record is always replaced. <i>On a sheet: Causes serious eye irritation</i> You maintain: Nothing (fixed phrase table). If missing: Code not in the table: the line ends at '(Category n)'. GHSStatements::H_STATEMENTS (EN) / ghs_{lang}.h_statements via resolveHStatements; HazardEngine.php:886-896, :995-1000; GHSStatements.php:310-323, :411-422
Precautionary Statements:	FIXED TEXT (TRANSLATED) A fixed heading, shown only when P-statements exist. <i>On a sheet: Precautionary Statements:</i> You maintain: Nothing If missing: Falls back to the sheet-language translation. labels.precautionary_statements (en.php:549); PDFService.php:536; preview.php:181
Precautionary statement lines:	HAZARD ENGINE P-codes from the triggered ingredient records (the substance's whole PubChem list), determinations, trade-secret entries, mixture-rule class defaults and the product hazard override, sorted by code. The carcinogen and carbon-black rules then append their class-default codes at the end. Wording comes from the fixed GHS table. <i>On a sheet: P210: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources.</i> You maintain: The same hazard data as the class lines; P-codes in the hazard override if used. If missing: No P-codes: block omitted. Code not in the table: the code prints alone. Withdrawn P281 still prints, fully worded, whenever source data carries it. classify :898-908; appendClassDefaultPStatements SDSGenerator :4013-4040 (P201,P202,P280,P308+P313,P405,P501); GHSStatements P_STATEMENTS/ghs_{lang}; PDF :530-550
Recommended Personal Protective Equipment (PPE):	FIXED TEXT (TRANSLATED) A fixed heading, shown only when at least one PPE field is driven by the product's H-codes. <i>On a sheet: Recommended Personal Protective Equipment (PPE):</i> You maintain: Nothing If missing: Falls back to the sheet-language translation. labels.ppe_recommendations (en.php:550); gate resolvePPE hazard_driven (tier not general/none) SDSGenerator.php:918-922; PDFService.php:556-564
PPE pictogram images (eye, hand, respiratory, skin):	TEXT CHOSEN BY HAZARD One icon for each PPE field whose level is set by the H-codes, e.g. H319 gives goggles, H315 gives gloves, H335 gives a respirator. P-codes play no part. The H-codes counted include ones that print nowhere in Section 2. <i>On a sheet: Blue eye-protection and glove icons</i> You maintain: Ingredient hazard data that yields H-codes. Optional artwork on Admin > Pictograms. If missing: No hazard-driven field: no icons. Missing artwork: the PDF drops that icon and its caption. HazardEngine::derivePPE :1098-1177 via SDSGenerator::resolvePPE :1923-1941; PictogramHelper PPE-*.png; PDFService.php:566-567, :665-725
PPE icon captions:	FIXED TEXT (TRANSLATED) A fixed caption for each active PPE icon, in the sheet language. <i>On a sheet: Wear Eye Protection / Wear Gloves</i> You maintain: Nothing If missing: Falls back to the sheet-language translation. labels.ppe_wear_eye/_gloves/_respiratory/_skin (en.php:552-555); PDFService.php:667-671, :720
Respiratory: / Hand Protection: / Eye Protection: / Skin/Body Protection:	FIXED TEXT (TRANSLATED) Fixed sub-labels, printed before each hazard-driven PPE sentence. <i>On a sheet: Hand Protection:</i> You maintain: Nothing If missing: The line is skipped when its sentence is empty. labels.respiratory/hand_protection/eye_protection/skin_body (en.php:594-598); PDFService.php:570-576; preview.php:215-228

PPE recommendation sentences:	TEXT CHOSEN BY HAZARD The same sentence Section 8 prints for that field: Section 8 per-product override (if not blank) > translated sentence for the H-code level. Printed in Section 2 only when the level is hazard-driven. <i>On a sheet: Chemical-resistant gloves (nitrile or neoprene recommended) meeting 29 CFR 1910.138.</i> You maintain: Ingredient hazard data (H-codes). Optional Section 8 PPE text per product per language (SDS editor). If missing: Baseline level (general/none): not printed in Section 2; Section 8 still prints it. UV acrylate products: Section 8 adds extra sentences that Section 2 lacks. <pre>text_overrides(sec 8, respiratory hand_protection eye_protection skin_protection) > section8.ppe.<field>.<tier>; resolvePPE SDSGenerator.php:1923-1941</pre>
Other Hazards:	FIXED TEXT (TRANSLATED) A fixed label, printed with the Other Hazards text whenever that text is not blank. <i>On a sheet: Other Hazards:</i> You maintain: Nothing If missing: Omitted together with the text when the text is blank. <pre>labels.other_hazards (en.php:551); PDFService.php:589; preview.php:236</pre>
Other Hazards text:	PRODUCT OVERRIDE Per-product, per-language override typed in the SDS editor > translated default 'None known.' <i>On a sheet: None known.</i> You maintain: Optional. Enter hazards not otherwise classified (e.g. combustible dust) in the SDS editor, per product and language. If missing: 'None known.' in the sheet language. A stored blank override row omits the whole Other Hazards line. Resale raw-material SDSs always print the default. <pre>text_overrides(fg, lang, section 2, 'other_hazards', sds_version_id NULL) ?? section2.other_hazards; SDSGenerator.php:924, :939; PDFService.php:578-590</pre>
SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS	
Section 3 is built from the formula: each raw material's listed ingredients (CAS number, name and percentage) are combined across the formula, filtered and labelled by the hazard engine, the hazard-data records and the competent-person determinations, and printed in a fixed table whose headings and notes come from the translation files. A published sheet keeps a frozen copy of this section until a new version is published.	
Type:	FIXED TEXT (TRANSLATED) Fixed label from the language file; the colon is added by the layout. <i>On a sheet: Type:</i> You maintain: Nothing If missing: Sheet language > English > the literal key "labels.type". <pre>meta.labels.type = t('labels.type') (en.php:558) via SDSGenerator::getLabels; ':' appended in PDFService::labelValue:1263 and preview.php:240</pre>
Type (value: Mixture / Substance):	FORMULA / RAW MATERIALS Substance/Mixture choice on the finished good > if set to Auto, Substance only when the formula is a single raw material marked Substance > Mixture. Resale sheets use the raw material's own choice. <i>On a sheet: Type: Mixture</i> You maintain: Substance / Mixture on the finished-good form; Substance / Mixture on the raw-material form for single-substance raws. If missing: "Mixture" prints. <pre>finished_goods.substance_mixture > SubstanceMixtureResolver::resolveForFinishedGood (1 line, raw_materials.substance_mixture) > 'mixture'; labels.substance/mixture</pre>
Hazardous-only disclosure note:	FIXED TEXT (TRANSLATED) Fixed sentence from the language file, printed above the table whenever at least one ingredient row is listed. <i>On a sheet: Hazardous ingredients and ingredients with an occupational exposure limit are listed. ...</i> You maintain: Nothing If missing: Not printed when no row is listed (the no-hazardous note prints instead). <pre>meta.labels.hazardous_only_note (en.php:562, es/fr:535, de:532); PDFService.php:732-736; preview.php:241-242</pre>
CAS Number Chemical Name Concentration H-Codes (table header):	FIXED TEXT (TRANSLATED) Fixed column headings from the language file, bold on a grey band. <i>On a sheet: CAS Number Chemical Name Concentration H-Codes</i> You maintain: Nothing If missing: Sheet language > English > literal key; old snapshots fall back to the translation. Table only prints when rows exist. <pre>meta.labels.cas_number/chemical_name/concentration/h_codes (en.php:559-561,566); PDFService.php:738-746; preview.php:243-249</pre>

CAS Number (ingredient row):	FORMULA / RAW MATERIALS The CAS number entered on each raw-material ingredient, combined across every formula line and every nested finished-good component. <i>On a sheet: 108-88-3</i> You maintain: CAS number on every ingredient of every raw material (Raw materials > ingredients table); a current formula on every finished good used as a component. If missing: An ingredient with a blank CAS is never listed. A finished-good component with no current formula contributes nothing, silently. A CAS flagged trade secret on any contributing raw material prints as "TRADE SECRET". <code>raw_material_constituents.cas_number</code> via <code>Formula::getExpandedComposition (resale: FormulaCalcService::buildResaleComposition); SDSGenerator:1084</code>
Chemical Name (ingredient row):	FORMULA / RAW MATERIALS Prop 65 list name > CAS master preferred name > name typed on the raw-material ingredient. Resale sheets use the name stored on the ingredient. <i>On a sheet: Toluene</i> You maintain: Ingredient name on the raw material; Prop 65 list and CAS master names (Regulatory lists). If missing: The name as typed on the ingredient; blank if every source is blank. Never translated. <code>COALESCE(prop65_list.chemical_name, cas_master.preferred_name, raw_material_constituents.chemical_name)</code> <code>Formula.php:291; RawMaterial::canonicalCasName</code> at save
Concentration (ingredient row):	FORMULA / RAW MATERIALS Formula line % x ingredient % in the raw material, mapped to the widest fixed band that contains it. Supplier min-max range > exact % (exact, else range maximum) on finished goods; resale sheets use exact %, else the range midpoint. <i>On a sheet: 10 - 30%</i> You maintain: Formula line percentages (Finished goods > formula); exact % or min + max % on each raw-material ingredient. If missing: Below 0.1 % prints "<0.1%"; an ingredient with no % counts as 0 and is dropped. If no band fits: "<0.1%" or "80 - 100%". <code>SDSGenerator::formatConcentration:3408-3453</code> over <code>PRESCRIBED_RANGES</code>; <code>concentration_min/max (Formula.php:362-371) > concentration_pct (resolveConstituentPct:772-788)</code>
H-Codes (ingredient row):	HAZARD ENGINE Hazard codes the hazard engine credits to that CAS: federal hazard data > competent-person determination, plus mixture-rule contributors, registry carcinogens and the powder H351 rule. <i>On a sheet: H225, H304, H315, H336</i> You maintain: Federal hazard data for each CAS, or an active competent-person determination (CAS determinations); carcinogen list; physical state on each raw material. If missing: Blank cell. <code>\$casToHCodes</code> from <code>hazardResult.hazard_classes (SDSGenerator:979-1008); getDefaultForClassCategory; MIXTURE contributors[]; applyCarcinogenFindings; applyCarbonBlackLogic</code>
Which ingredients are listed (rows):	HAZARD ENGINE An ingredient is listed when it has a CAS, is at least 0.1 %, is classified hazardous or has an exposure limit, is not flagged non-hazardous on every source, and is not an inhalation-only CAS with nothing to report. <i>On a sheet: Toluene listed; water and talc in a liquid not listed</i> You maintain: Hazard data or a CPD for every CAS; exposure limits; non-hazardous flag and physical state on raw materials; inhalation-only CAS list (Settings). If missing: A CAS with no hazard data, no CPD and no exposure limit is silently left out. Publishing is blocked only if it is at or above the missing-data threshold (default 1.0 %). <code>SDSGenerator::section3:1026-1059</code> (<code>hazardous_cas</code>, <code>exposure_limits</code>, <code>is_non_hazardous</code>, <code>sds.inhalation_only_cas</code>); <code>applySolidPowderLiquidFiltering; HazardEngine 571-1031</code>
TRADE SECRET (trade-secret row, CAS cell):	FIXED TEXT (TRANSLATED) Fixed word from the language file, printed in place of the CAS number for trade-secret ingredients. <i>On a sheet: TRADE SECRET</i> You maintain: Nothing (the row appears when an ingredient is flagged trade secret or a raw material is marked hazardous with no CAS). If missing: English text, then the literal key. <code>t('labels.trade_secret_cas') (en.php:687); SDSGenerator:1066; PDFService.php:752; preview.php:253</code>
Chemical Name (trade-secret row):	FORMULA / RAW MATERIALS The trade-secret description chosen on the raw-material ingredient > the translated words "Trade Secret". One row per distinct description. <i>On a sheet: Proprietary Resin Blend</i> You maintain: Trade-secret description on each trade-secret ingredient (Raw materials). If missing: "Trade Secret" (translated). <code>SDSGenerator::tradeSecretName(raw_material_constituents.trade_secret_description):2017-2024; pick-list settings.sds.trade_secret_descriptions</code>

Concentration (trade-secret row):	FORMULA / RAW MATERIALS Sum of the trade-secret ingredients sharing one description, mapped to a band the same way as normal rows. <i>On a sheet: 1 - 5%</i> You maintain: Formula line percentages; ingredient percentages on the raw material. If missing: As for normal rows. SDSGenerator::section3:1072-1079,1098 then formatConcentration
H-Codes (trade-secret row):	HAZARD ENGINE All hazard codes the engine attributes to trade-secret declarations in the formula; the same list prints on every trade-secret row. <i>On a sheet: H317, H319</i> You maintain: Manual hazard classification on raw materials marked hazardous with no CAS; trade-secret H-codes on trade-secret ingredients (Raw materials). If missing: Blank cell. union of h_codes on hazard_classes with cas='TRADE_SECRET' (SDSGenerator:1010-1020,1099); raw_materials.manual_hazard_json; rmc.trade_secret_h_codes
Row order:	FORMULA / RAW MATERIALS Rows sorted by the exact calculated percentage, highest first; trade-secret rows are placed by their summed percentage. You maintain: Formula and ingredient percentages. If missing: Not applicable. usort on components[].concentration_pct desc (SDSGenerator:1103-1104); FG path uses pct_max for ranged ingredients
No-hazardous-ingredients note:	FIXED TEXT (TRANSLATED) Fixed sentence from the language file, printed instead of the table when no ingredient qualifies. <i>On a sheet: No hazardous ingredients above disclosure thresholds.</i> You maintain: Nothing If missing: Only shown when no row is listed. meta.labels.no_hazardous_note (en.php:563); PDFService.php:773-776; preview.php:261-262
Trade-secret / withheld-percentage statement:	FIXED TEXT (TRANSLATED) Fixed sentence from the language file, printed below the table whenever any row is listed (bands withhold the exact percentage on every row). <i>On a sheet: Specific chemical identity and/or exact percentage ... withheld as a trade secret ...</i> You maintain: Nothing If missing: Not printed when no row is listed. sections[3].trade_secret_note = t('section3.trade_secret_note') when components non-empty (SDSGenerator:1116-1121); PDFService:778-783; preview.php:264-266

SECTION 4: FIRST-AID MEASURES

Section 4 is built almost entirely from the product's final GHS health hazard codes. Each route paragraph, the symptoms line and the notes to physician are assembled from translated sentences chosen by those codes. Two other sources can change it: an optional per-product, per-language text override (overrides replace whole fields) and a UV acrylate note driven by product family.

Inhalation:	TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H330 or H331 replaces the base paragraph; otherwise H332 adds a sentence. H334, H335 and H336 each add a sentence in every case. <i>On a sheet: Move to fresh air. If breathing difficulty persists, seek medical attention. Harmful if inhaled...</i> You maintain: Formula composition and raw-material constituents (CAS, %), the hazard data per CAS (classification sources, competent-person determinations), manual or trade-secret hazards on raws, and any FG hazard override. Optional text on the SDS edit screen. If missing: No matching codes: "Move to fresh air. If breathing difficulty persists, seek medical attention." Any stored override, including an old frozen copy, prints as-is and suppresses every hazard sentence. text_overrides(4,'inhalation',lang) > section4.inhalation[_fatal _toxic] + _harmful(H332, base only) + _resp_sensitizer/_irritant/_narcotic; SDSGenerator.php:1142-1163
Skin Contact:	TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H314 or H310/H311 replaces the base paragraph; otherwise H312 and H315 add sentences. H317 adds a sentence in every case. <i>On a sheet: Remove contaminated clothing. Wash skin thoroughly with soap and water. ... May cause an allergic skin reaction...</i> You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen. If missing: "Remove contaminated clothing. Wash skin thoroughly with soap and water. If irritation persists, seek medical attention." A stored override wins outright. text_overrides(4,'skin',lang) > section4.skin[_corrosive _toxic] + _harmful(H312)/_irritant(H315) base only + _sensitizer(H317); SDSGenerator.php:1166-1184

Eye Contact:	TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H314 (30-minute rinse) or H318 (20-minute rinse) replaces the base paragraph. H314, H318 or H319 adds the contact-lens sentence. <i>On a sheet: Flush eyes with large amounts of water for at least 15 minutes... Remove contact lenses, if present...</i> You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen. If missing: The base 15-minute flush sentence, with no contact-lens sentence. A stored override wins outright. <pre>text_overrides(4, 'eyes', lang) > section4.eyes[_corrosive _serious_damage] + eyes_contact_lenses(H314/H318/H319); SDSGenerator.php:1187-1199</pre>
Ingestion:	TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H304/H305 gives the aspiration paragraph, which takes priority over everything else. Otherwise H300/H301 gives the toxic paragraph; otherwise the base paragraph prints, plus a sentence for H302. <i>On a sheet: Do not induce vomiting. Rinse mouth with water... Harmful if swallowed. Call a poison center...</i> You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen. If missing: "Do not induce vomiting. Rinse mouth with water. Seek medical attention if symptoms develop." A stored override wins outright. <pre>text_overrides(4, 'ingestion', lang) > section4.ingestion_aspiration(H304/H305) _toxic(H300/H301) ingestion + _harmful(H302); SDSGenerator.php:1202-1214</pre>
Most Important Symptoms/Effects, Acute and Delayed:	HAZARD ENGINE Per-product override > the product's health hazard statements (H3xx), restated. Sensitisation, carcinogenicity, reproductive toxicity and repeated-exposure effects are grouped as Delayed; all others as Acute. <i>On a sheet: Acute: Harmful if swallowed. Causes serious eye irritation. Delayed: May cause an allergic skin reaction.</i> You maintain: Same hazard inputs as Inhalation, plus the carcinogen lists and the 'inhalation-only CAS' setting (these add H350/H351). H-statement translations in the ghs_es/fr/de files. If missing: No health codes: "No significant symptoms or effects are known or expected under normal conditions of use." A missing translation prints the English statement. A sub-coded statement (e.g. H360D) has no text and is silently left out. <pre>text_overrides(4, 'symptoms', lang) > deriveSymptoms(hazard.h_statements) :1265-1301; DELAYED_EFFECT_H_PREFIXES :71; text from GHSStatements::H_STATEMENTS / ghs_{lang}.php</pre>
Acute: / Delayed: (prefixes inside the symptoms line):	FIXED TEXT (TRANSLATED) Fixed words from the translation file. Each prints only when that group has at least one statement. <i>On a sheet: Acute: ... Delayed: ...</i> You maintain: Nothing If missing: An empty group is left out. If both groups are empty, the 'No significant symptoms...' sentence prints instead. <pre>section4.symptoms_acute_prefix / section4.symptoms_delayed_prefix; SDSGenerator.php:1293-1300; en.php:51-52</pre>
Notes to Physician:	TEXT CHOSEN BY HAZARD Per-product override > hazard sentences, in this order: aspiration (H304/H305), corrosive (H314/H318), delayed lung effects (H330/H331). The default closing sentence always follows. <i>On a sheet: Do not induce vomiting; risk of aspiration into the lungs. Treat symptomatically. Show this SDS...</i> You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen. If missing: "Treat symptomatically. Show this SDS to medical personnel." A stored override wins outright. <pre>text_overrides(4, 'notes', lang) > notes_aspiration + notes_corrosive + notes_inhalation_delayed + section4.notes; SDSGenerator.php:1230-1244; en.php:28, :54-56</pre>
UV Acrylate Information:	PRODUCT FAMILY Prints only when the UV rule pack is enabled, the product's family is flagged UV, and an acrylate is found in the composition. If the product is not classified H317, a shorter version prints without the sensitisation sentence. <i>On a sheet: UV/EB curable product containing acrylate monomers/oligomers (1,6-Hexanediol diacrylate). In...</i> You maintain: The UV flag on product families (Product families screen) and correct family assignment. The UV rule-pack setting (Settings). Constituent CAS numbers and chemical names on raw materials. Trade-secret descriptions. If missing: Not printed. Names come from prop65 > CAS master > constituent name (resale sheets: constituent name only). The built-in acrylate label is used only when the name is null; a trade secret prints its description or "Trade Secret". <pre>UVAcrylateRulePack::isApplicable (settings.uv_acrylate_rule_pack, product_families.is_uv) + detectAcrylates; section4.uv_acrylate_note[_unclassified]; SDSGenerator.php:174-182, :226-230</pre>

SECTION 5: FIRE-FIGHTING MEASURES

Section 5 is mostly translated standard wording. The system picks each paragraph from the hazard codes it calculates from the ingredients (the same classification as Section 2). The flash point comes from raw material data. A product can carry its own override text, and a UV acrylate note is added for UV-family products.

Suitable Extinguishing Media: **TEXT CHOSEN BY HAZARD** One of four fixed sentences, chosen by the product's calculated hazard codes. Order: product override > water-reactive (H260/H261) > oxidizer (H271/H272) > flammable liquid (H224-H227) > default.

On a sheet: Alcohol-resistant foam, dry chemical, carbon dioxide (CO₂), or water spray (fog).

You maintain: Ingredient CAS numbers and percentages on raw materials. Hazard codes for each CAS on the CAS determinations screen. Hazard codes for trade-secret or no-CAS ingredients. Optional finished-good hazard override H-codes.

If missing: With no matching hazard code: "Water spray, dry chemical, carbon dioxide (CO₂), foam."

text_overrides(5,'suitable_media',lang) > section5.suitable_water_reactive > suitable_oxidizer > suitable_flammable > suitable_media; extractHCodes :791-808; SDSGenerator.php:1398-1411

Unsuitable Extinguishing Media: **TEXT CHOSEN BY HAZARD** Fixed sentence chosen by hazard codes. Order: product override > water-reactive (H260/H261) > flammable liquid (H224-H227) > default. Oxidizers get the default.

On a sheet: Do not use a solid water stream; it may scatter the burning liquid and spread the fire.

You maintain: Same hazard-code data as Suitable Extinguishing Media.

If missing: "Do not use direct water stream as it may spread fire."

text_overrides(5,'unsuitable_media',lang) > section5.unsuitable_water_reactive > unsuitable_flammable > unsuitable_media; SDSGenerator.php:1413-1423

Specific Hazards - flammable liquid sentence: **HAZARD ENGINE** Printed when the calculated hazard codes include a Flammable Liquids code. The most severe code wins: H224 = Cat 1, H225 = Cat 2, H226 = Cat 3, H227 = Cat 4. The flash point plays no part in this choice.

On a sheet: Flammable liquid and vapor (Flammable Liquids, Category 3). Vapors can form ignitable...

You maintain: A Flammable Liquids H-code on a constituent CAS present at 1% or more of the finished good, on a CPD or trade-secret entry, or in the finished-good hazard override H-codes list.

If missing: Sentence omitted, along with the flash point line, when no H224-H227 is present. Omitted entirely when a Specific Hazards override exists.

section5.specific_hazards_flammable_cat{1-4}; SDSGenerator.php:1380-1390, :1431-1432; HazardEngine cutoff 1% for physical classes :1190-1197 (no flash-point logic)

Specific Hazards - Flash point line: **FORMULA / RAW MATERIALS** Same value as Section 9: Section 9 Flash Point override > lowest raw-material flash point anywhere in the formula, including sub-formulas, with a '>' prefix when that raw material is ticked 'greater than'. Printed only after a flammable-liquid sentence.

On a sheet: Flash point: 38 °C (100.4 °F).

You maintain: Flash point (°C) and the '>' tick on each raw material that has one, on the Raw materials screen. Or a Section 9 Flash Point override, entered per language in the SDS editor.

If missing: Line omitted when no raw material has a flash point and there is no Section 9 override. Never printed on products without H224-H227.

section5.flash_point_line ':fp'; resolveFlashPointDisplay :1327-1334 (text_overrides(9,'flash_point') > flashPointDisplay :1310-1320); FormulaCalcService::deriveFormulaProperties :466-474

Specific Hazards - water-reactive sentence: **TEXT CHOSEN BY HAZARD** Fixed sentence added when the hazard codes include H261. The H260 version adds 'which may ignite spontaneously'.

On a sheet: Reacts with water, moisture, and water-based extinguishing agents to release flammable gases.

You maintain: H260/H261 in the constituent CAS hazard data, a trade-secret entry or the finished-good hazard override.

If missing: Omitted when neither H260 nor H261 is present.

section5.specific_hazards_water_reactive / _water_reactive_h260; SDSGenerator.php:1392-1393, :1437-1441

Specific Hazards - closing sentence: **TEXT CHOSEN BY HAZARD** Always the last sentence. Order: oxidizer (H271/H272) 'may cause or intensify fire' > organic peroxide (H240-H242) 'may catch fire or explode' > default combustion sentence. The combustion products named are generic, not taken from the formula.

On a sheet: Combustion may produce carbon monoxide, carbon dioxide, and other toxic fumes.

You maintain: H271/H272 or H240-H242 in the hazard data where they apply. Otherwise nothing.

If missing: The default combustion sentence prints on every sheet that has no override.

section5.specific_hazards_oxidizer > specific_hazards_organic_peroxide > specific_hazards; joined with ' ' at SDSGenerator.php:1442-1449

Advice for Firefighters: **TEXT CHOSEN BY HAZARD** Fixed sentence chosen by hazard codes. Order: product override > explosive (H200-H205) > water-reactive (H260/H261) > flammable liquid (H224-H227) > default.
On a sheet: Wear self-contained breathing apparatus (SCBA)... Eliminate all ignition sources...
You maintain: The hazard codes above in the CAS hazard data. Optional per-product override.
If missing: "Wear self-contained breathing apparatus (SCBA) and full protective gear. Cool containers exposed to fire with water spray."
`text_overrides(5, 'firefighter_advice', lang) > section5.firefighter_advice_explosive > _water_reactive > _flammable > firefighter_advice; SDSGenerator.php:1452-1464`

UV Acrylate Information: **PRODUCT FAMILY** A fixed note, printed only when the UV rule pack is on, the product's family is flagged UV, and any ingredient at any percentage is an acrylate (by CAS list or name). It cannot be edited or turned off for a single product.
On a sheet: UV-curable formulations may generate acrid smoke if involved in a fire. Fire-fighters must...
You maintain: UV rule pack setting (Settings). Family assignment on the finished good. The UV flag on the family (Product families). Constituent CAS numbers and names on raw materials.
If missing: Nothing printed (no label, no text) when the pack is off, the family is not UV, or no acrylate is found.
`UVAcrylateRulePack::getSafeHandlingLanguage()[5] = section5.uv_acrylate_note; isEnabled (settings.uv_acrylate_rule_pack) + product_families.is_uv + detectAcrylates; SDSGenerator.php:172-182, :226-230...`

SECTION 6: ACCIDENTAL RELEASE MEASURES

Section 6 is assembled from fixed sentences in the translation files, and the hazard codes, physical state and product family pick which ones print; operators can replace any of the three main fields per product and language.

Personal Precautions: **FIXED TEXT (TRANSLATED)** Field label from the language file, stored with the sheet when it is generated.
On a sheet: Personal Precautions:
You maintain: Nothing
If missing: Sheet language > English > literal 'labels.personal_precautions'. Legacy snapshots print 'Personal Precautions'. No label prints when the value is empty.
`labels.personal_precautions -> meta.labels (getLabels :3458-3525); PDFService FIELD_LABEL_MAP :43; preview $genericLabelMap :571`

Personal Precautions (standard paragraph): **TEXT CHOSEN BY HAZARD** Built-in sentence used when the product has no acute toxicity category 1-3 code (H300/H301/H310/H311/H330/H331) and no H314. Operator text for this product > this sentence.
On a sheet: Use appropriate PPE (see Section 8). Avoid contact with skin and eyes. Ensure adequate...
You maintain: Hazard data on the CAS determinations and raw material screens (current classifications, competent-person determinations, trade-secret H-codes, constituent %), plus any finished-good hazard override.
If missing: This is the default; it prints whenever no severe code is found. If hazard data is missing, the standard paragraph prints even for a toxic or corrosive product.
`text_overrides(6, 'personal_precautions') > section6.personal_precautions; SDSGenerator :1500-1507; codes from extractHCodes(HazardEngine::classify) :791-808`

Personal Precautions (acute toxicity paragraph): **TEXT CHOSEN BY HAZARD** Replaces the standard paragraph when any of H300, H301, H310, H311, H330 or H331 applies. It takes priority over the corrosive paragraph.
On a sheet: Use appropriate PPE including self-contained breathing apparatus (SCBA) (see Section 8)...
You maintain: Acute toxicity classifications and percentages of constituents (CAS determinations, raw materials); finished-good hazard override.
If missing: If no acute toxicity code results, the corrosive or standard paragraph prints.
`section6.precautions_acute_toxic; SDSGenerator :1502-1503 (if branch ahead of H314 elseif)`

Personal Precautions (corrosive paragraph): **TEXT CHOSEN BY HAZARD** Replaces the standard paragraph when H314 applies and no acute toxicity 1-3 code applies. H318 alone does not trigger it.
On a sheet: Use appropriate PPE including chemical-resistant suit, gloves, and face shield...
You maintain: Skin corrosion classifications of constituents; finished-good hazard override.
If missing: Standard paragraph prints.
`section6.precautions_corrosive; SDSGenerator :1504-1505`

Personal Precautions (flammable add-on): **TEXT CHOSEN BY HAZARD** Added after whichever paragraph was chosen when H220-H226 or H228 applies. It is not added when the field has operator text, and the flash point is not consulted.
On a sheet: ... Remove all sources of ignition. Use only non-sparking tools and explosion-proof equipment...
You maintain: Flammability classifications of constituents (CAS determinations); finished-good hazard override.
If missing: Nothing is added.
`section6.precautions_flammable appended with ' '; SDSGenerator :1509-1511 (inside override null-check :1501); H227 not in list`

Environmental Precautions:	FIXED TEXT (TRANSLATED) Field label from the language file, stored with the sheet when it is generated. <i>On a sheet: Environmental Precautions:</i> You maintain: Nothing If missing: Sheet language > English > literal key. Legacy snapshots print 'Environmental'. labels.environmental_precautions (data key 'environmental'); PDFService :44; preview.php :572
Environmental Precautions (drains sentence):	FIXED TEXT Always the first sentence on every sheet. Operator text for this product > this sentence. <i>On a sheet: Prevent entry into drains, sewers, and waterways.</i> You maintain: Nothing If missing: Always prints unless the field has operator text. text_overrides(6,'environmental') > section6.environmental; SDSGenerator :1515-1517
Environmental Precautions (aquatic acute sentence):	TEXT CHOSEN BY HAZARD H400 > H401 picks the wording. It follows the drains sentence and is not added when the field has operator text. <i>On a sheet: Very toxic to aquatic life.</i> You maintain: Aquatic acute classifications and M-factors of constituents (CAS determinations). If missing: No acute sentence. section6.environmental_aquatic_acute (H400) > section6.environmental_aquatic_acute_toxic (H401); SDSGenerator :1519-1523
Environmental Precautions (aquatic chronic sentence):	TEXT CHOSEN BY HAZARD H410 > H411 picks the wording. It prints after the acute sentence when both apply. <i>On a sheet: Toxic to aquatic life with long lasting effects.</i> You maintain: Aquatic chronic classifications and M-factors of constituents. If missing: No chronic sentence. section6.environmental_aquatic_chronic_very (H410) > section6.environmental_aquatic_chronic (H411); SDSGenerator :1524-1528
Environmental Precautions (harmful sentence):	TEXT CHOSEN BY HAZARD Prints for H402, H412 or H413, but only when no acute or chronic sentence above printed. <i>On a sheet: Harmful to aquatic life.</i> You maintain: Aquatic acute 3 and chronic 3/4 classifications. If missing: Not printed. section6.environmental_aquatic_harmful; SDSGenerator :1529-1532 (only if \aquaticParts === [])
Environmental Precautions (notify sentence):	TEXT CHOSEN BY HAZARD Added last whenever any aquatic sentence printed. <i>On a sheet: Notify authorities if the product enters drains or waterways.</i> You maintain: Aquatic classifications. If missing: Not printed for products with no aquatic code. section6.environmental_notify; SDSGenerator :1533-1536
Containment and Cleanup:	FIXED TEXT (TRANSLATED) Field label from the language file, stored with the sheet when it is generated. <i>On a sheet: Containment and Cleanup:</i> You maintain: Nothing If missing: Sheet language > English > literal key. Legacy snapshots print 'Containment'. labels.containment_cleanup (data key 'containment'); PDFService :45; preview.php :573
Containment and Cleanup (solid / powder):	FORMULA / RAW MATERIALS Used when the product's physical state is Solid or Powder. Operator text > finished good's physical state > state of the raw material with the highest % in the formula > 'Liquid'. <i>On a sheet: Avoid generating dust. Sweep or vacuum up material. Collect in suitable containers...</i> You maintain: Physical State on the finished good screen, or Physical State on the dominant raw material. If missing: With no state anywhere, the product is treated as 'Liquid' and the liquid text prints. section6.containment_solid; SDSGenerator :1540-1544; resolvePhysicalState :1954-1962 (fg.physical_state > FormulaCalcService formula_props.physical_state :498-542 > 'Liquid')
Containment and Cleanup (gel / paste):	FORMULA / RAW MATERIALS Used when the resolved physical state is Gel or Paste (same order as above). <i>On a sheet: Stop leak if safe to do so. Scrape up or shovel material and collect in suitable...</i> You maintain: Physical State = Gel/Paste on the finished good, or Paste on the dominant raw material (raw materials have no Gel option). If missing: Liquid text prints. section6.containment_paste; SDSGenerator :1545-1546

**Containment and Cleanup
(liquid / everything else):**

FORMULA / RAW MATERIALS

Used for Liquid, Gas, custom states such as Aerosol, and when no state is recorded.

On a sheet: Stop leak if safe to do so. Dam or contain spill. Absorb with inert absorbent material...

You maintain: Physical state data as above. A missing state gives this text with no warning.

If missing: This is the fallback text.

section6.containment_liquid; SDSGenerator :1547-1550

UV Acrylate Information:

PRODUCT FAMILY

Prints last, only when the UV acrylate setting is on, the product's family is flagged UV, and an acrylate is found among the ingredients by CAS or name. It cannot be edited per product.

On a sheet: UV Acrylate Information: Avoid release to drains. Uncured acrylate monomers must not...

You maintain: UV flag and rules on the Product families screen (family on the finished good / raw material); ingredient CAS numbers and names (Prop 65 name > CAS master preferred name > constituent name); admin toggle in Settings.

If missing: Not printed. Acrylates entered only as trade secrets or without a CAS number are never detected.

UVAcrylateRulePack isEnabled(uv_acrylate_rule_pack) && familyIsUv && detectAcrylates :118-149 -> section6.uv_acrylate_note (:195); appended SDSGenerator :225-230/:553-557

SECTION 7: HANDLING AND STORAGE

Section 7 is built automatically from the product's GHS hazard codes: fixed base sentences come first, hazard-specific sentences are added for each hazard class, and Storage ends by repeating the Section 10 incompatible-materials list. A per-product, per-language override in the SDS editor replaces either paragraph entirely, and a UV acrylate note is appended for UV-family products that contain an acrylate.

Handling:

FIXED TEXT (TRANSLATED)

Fixed label from the language files.

On a sheet: Handling:

You maintain: Nothing

If missing: English 'Handling', then the literal 'labels.handling'. Old snapshots without meta.labels fall back to the sheet-language label, then 'Handling'.

meta.labels.handling = labels.handling (getLabels, SDSGenerator.php:3482);
PDFService.php:47,1239-1279; preview.php:574,599-600

Handling (paragraph as a whole):

TEXT CHOSEN BY HAZARD

Published sheet snapshot > the product's Section 7 Handling override in the SDS editor (used word for word) > the base sentence plus every hazard sentence that matches the product's H-codes. Alias and private-label sheets copy the base product's text. Resale raw-material sheets can never take an override.

On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing. ...

You maintain: Correct GHS classifications on each CAS (CAS determinations / federal hazard data), trade-secret hazard codes on raw materials, the finished good's hazard override mode and JSON, an optional per-language override in the SDS editor, and a republish after any change

If missing: Never blank: with no matching H-codes the base sentence prints alone. A product whose pre-update override froze the old stock paragraph still prints that old text.

snapshot_json > text_overrides(fg.id,7,'handling',lang,sds_version_id NULL) > section7.handling_base + fragments by extractHCodes(hazard_result); SDSGenerator.php:1700-1771

**Handling: base sentence
(always first):**

FIXED TEXT

Fixed text that every sheet prints unless Handling is overridden. It ends 'Wear the personal protective equipment specified in Section 8.'

On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing.

You maintain: Nothing

If missing: English, then the literal key (all four languages define it)

section7.handling_base; SDSGenerator.php:1733; en.php:130; es/fr/de.php:122

**Handling: heat and ignition
sources:**

TEXT CHOSEN BY HAZARD

Added when the product is classified flammable (H220-H226, H228), combustible (H227), pyrophoric (H250) or self-reactive (H240-H242). The formula's flash point is not used, and neither is an oxidizer classification.

On a sheet: Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources.

You maintain: Flammable / combustible / pyrophoric / self-reactive classifications on the CAS records (1% physical cutoff), or the finished good's hazard override

If missing: Not printed

section7.handling_ignition when \$ignition (SDSGenerator.php:1713-1728, 1734-1736); en.php:131

Handling: grounding and static:

TEXT CHOSEN BY HAZARD

Added for flammable products (H220-H226, H228). Not added for H227 combustible.

On a sheet: Keep container tightly closed. Ground and bond container and receiving equipment. ...

You maintain: Flammable classifications on the CAS records or the FG hazard override

If missing: Not printed

section7.handling_flammable_static; SDSGenerator.php:1713, 1737-1739; en.php:132

Handling: aerosol:	TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229). <i>On a sheet: Pressurized container: do not pierce or burn, even after use.</i> You maintain: Aerosol classification, usually through the finished good's hazard override If missing: Not printed section7.handling_aerosol; SDSGenerator.php:1715, 1740-1742; en.php:133
Handling: oxidizer:	TEXT CHOSEN BY HAZARD Added for oxidizers (H270, H271, H272). <i>On a sheet: Keep away from clothing and other combustible materials. Do not mix with flammable ...</i> You maintain: Oxidizer classification on the CAS records or the FG override If missing: Not printed section7.handling_oxidizer_combustibles; SDSGenerator.php:1716, 1743-1745; en.php:134
Handling: self-reactive:	TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). <i>On a sheet: Keep cool. Handle and open container with care. Keep only in original container ...</i> You maintain: Self-reactive / organic peroxide classification If missing: Not printed section7.handling_self_reactive; SDSGenerator.php:1717, 1746-1748; en.php:123
Handling: self-heating:	TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252). <i>On a sheet: Keep cool. Protect from sunlight. Maintain air gap between stacks or pallets. ...</i> You maintain: Self-heating classification If missing: Not printed section7.handling_self_heating; SDSGenerator.php:1718, 1749-1751; en.php:135
Handling: pyrophoric (air):	TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250 only). <i>On a sheet: Do not allow contact with air. Handle under inert gas.</i> You maintain: Pyrophoric classification If missing: Not printed section7.handling_pyrophoric_air; SDSGenerator.php:1719, 1752-1754; en.php:136
Handling: water-reactive:	TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261). <i>On a sheet: Do not allow contact with water. Handle under inert gas.</i> You maintain: Water-reactive classification If missing: Not printed section7.handling_water_reactive_water; SDSGenerator.php:1720, 1755-1757; en.php:137
Handling: moisture:	TEXT CHOSEN BY HAZARD Printed once when the product is pyrophoric (H250) or water-reactive (H260/H261). <i>On a sheet: Protect from moisture.</i> You maintain: Pyrophoric / water-reactive classification If missing: Not printed section7.handling_moisture; SDSGenerator.php:1758-1760; en.php:138
Handling: corrosive:	TEXT CHOSEN BY HAZARD Added for skin-corrosive products (H314). <i>On a sheet: Do not breathe mist, vapors, or spray. Do not get in eyes, on skin, or on clothing.</i> You maintain: Skin Corr. 1 classification (including the engine's summation) If missing: Not printed section7.handling_corrosive; SDSGenerator.php:1721, 1761-1763; en.php:139
Handling: corrosive to metals:	TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290). <i>On a sheet: Keep only in original container.</i> You maintain: Met. Corr. 1 classification If missing: Not printed section7.handling_corrosive_metals; SDSGenerator.php:1722, 1764-1766; en.php:140
Handling: sensitizer:	TEXT CHOSEN BY HAZARD Added for skin or respiratory sensitizers (H317, H334). <i>On a sheet: Avoid breathing vapors, mist, or spray. Contaminated work clothing must not be ...</i> You maintain: Skin / respiratory sensitizer classification If missing: Not printed section7.handling_sensitizer; SDSGenerator.php:1723, 1767-1769; en.php:141
Storage:	FIXED TEXT (TRANSLATED) Fixed label from the language files. <i>On a sheet: Storage:</i> You maintain: Nothing If missing: English 'Storage', then the literal 'labels.storage'. Old snapshots fall back to the sheet-language label, then 'Storage'. meta.labels.storage = labels.storage (SDSGenerator.php:3482); PDFService.php:48,1167-1182; preview.php:575,599-600

Storage (paragraph as a whole):	TEXT CHOSEN BY HAZARD Published sheet snapshot > the product's Section 7 Storage override (used word for word, and it also replaces the incompatibles list) > base sentence + hazard sentences + 'Store locked up.' + the Section 10 incompatibles sentence. Alias and private-label sheets copy the base product; resale raw-material sheets can never take an override. <i>On a sheet: Store in a cool, dry, well-ventilated area. ... (see Section 10): strong oxidizing ...</i> You maintain: CAS GHS classifications, the FG hazard override, an optional Section 7 Storage override, an optional Section 10 Incompatible Materials override (SDS editor), and a republish after changes If missing: Never blank. The minimum is the base sentence plus 'Store away from incompatible materials (see Section 10): strong oxidizing agents, strong acids, strong bases.' A frozen pre-update override still prints the old text. snapshot_json > text_overrides(fg.id,7,'storage',lang,NULL) > storage_base + fragments + storage_locked + storage_incompatible(:list); SDSGenerator.php:1773-1814
Storage: base sentence (always first):	FIXED TEXT Fixed text that every sheet prints unless Storage is overridden. <i>On a sheet: Store in a cool, dry, well-ventilated area. Keep container tightly closed when not ...</i> You maintain: Nothing If missing: English, then the literal key section7.storage_base; SDSGenerator.php:1776; en.php:142; es/fr/de.php:134
Storage: heat and ignition sources:	TEXT CHOSEN BY HAZARD Same trigger as the Handling ignition sentence: flammable, combustible (H227), pyrophoric or self-reactive. Oxidizers are not included. <i>On a sheet: Keep away from heat and ignition sources.</i> You maintain: Physical-hazard classifications on the CAS records If missing: Not printed section7.storage_ignition when \$ignition; SDSGenerator.php:1728, 1777-1779; en.php:143
Storage: flammables storage area:	TEXT CHOSEN BY HAZARD Added for flammable products (H220-H226, H228). Not added for H227. <i>On a sheet: Store in an approved flammable-materials storage area. Ground and bond containers ...</i> You maintain: Flammable classifications If missing: Not printed section7.storage_flammable_area; SDSGenerator.php:1780-1782; en.php:144
Storage: aerosol:	TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229). <i>On a sheet: Protect from sunlight. Do not expose to temperatures exceeding 50 °C / 122 °F.</i> You maintain: Aerosol classification If missing: Not printed section7.storage_aerosol; SDSGenerator.php:1783-1785; en.php:145
Storage: oxidizer:	TEXT CHOSEN BY HAZARD Added for oxidizers (H270-H272). <i>On a sheet: Store separately from combustible materials, reducing agents, and flammable substances.</i> You maintain: Oxidizer classification If missing: Not printed section7.storage_oxidizer_separate; SDSGenerator.php:1786-1788; en.php:146
Storage: self-reactive:	TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). <i>On a sheet: Store at the recommended temperature. Keep away from heat and sunlight.</i> You maintain: Self-reactive classification If missing: Not printed section7.storage_self_reactive; SDSGenerator.php:1789-1791; en.php:147
Storage: self-heating:	TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252). <i>On a sheet: Keep cool and protect from sunlight. Maintain air gap between stacks or pallets.</i> You maintain: Self-heating classification If missing: Not printed section7.storage_self_heating_cool; SDSGenerator.php:1792-1794; en.php:148
Storage: pyrophoric (air):	TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250). <i>On a sheet: Protect from air.</i> You maintain: Pyrophoric classification If missing: Not printed section7.storage_pyrophoric_air; SDSGenerator.php:1795-1797; en.php:149
Storage: water-reactive:	TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261). <i>On a sheet: Protect from water.</i> You maintain: Water-reactive classification If missing: Not printed section7.storage_water_reactive_water; SDSGenerator.php:1798-1800; en.php:150

Storage: inert gas and moisture:	TEXT CHOSEN BY HAZARD Printed once for pyrophoric (H250) or water-reactive (H260/H261) products. <i>On a sheet: Store under inert gas. Protect from moisture.</i> You maintain: Pyrophoric / water-reactive classification If missing: Not printed section7.storage_inert_moisture; SDSGenerator.php:1801-1803; en.php:151
Storage: corrosive to metals:	TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290). <i>On a sheet: Store in a corrosion-resistant container with a resistant inner liner.</i> You maintain: Met. Corr. 1 classification If missing: Not printed section7.storage_corrosive_metals; SDSGenerator.php:1804-1806; en.php:152
Storage: Store locked up:	TEXT CHOSEN BY HAZARD Added when the product carries one of a fixed list of codes: H300, H301, H310, H311, H330, H331, H304, H314, H335, H336, H370 or H371. Carcinogen, mutagen and reproductive codes and H305 are not on the list, although Section 2 can print P405 for them. <i>On a sheet: Store locked up.</i> You maintain: Acute toxicity 1-3, aspiration, Skin Corr. 1 and STOT SE classifications If missing: Not printed section7.storage_locked when \$lockUp (hard-coded list, SDSGenerator.php:1724-1727, 1807-1809); en.php:153
Storage: incompatible materials (see Section 10):	TEXT CHOSEN BY HAZARD The product's Section 10 Incompatible Materials override (trailing full stops removed, capitalisation kept) > a list built from the hazard classes: acids and bases always, oxidizing agents unless the product is an oxidizer, and further items for each hazard class. Not printed when Section 7 Storage is overridden. <i>On a sheet: Store away from incompatible materials (see Section 10): strong oxidizing agents, ...</i> You maintain: Physical-hazard classifications; optional Section 10 Incompatible Materials override in the SDS editor (write it lower-case with no full stop so it reads well after the colon) If missing: Always printed: with no physical hazard the list is 'strong oxidizing agents, strong acids, strong bases' section7.storage_incompatible :list = incompatibleMaterialsInline(): text_overrides(fg.id,10,'incompatible') rtrim '.' > section10.incompat_<key> in INCOMPAT_ORDER; :1611-1698
UV Acrylate Information:	FIXED TEXT (TRANSLATED) Fixed label from the language files, printed only when the UV note below prints. <i>On a sheet: UV Acrylate Information:</i> You maintain: Nothing If missing: Not printed when there is no UV note. Old snapshots fall back to 'Uv Acrylate Note'. meta.labels.uv_acrylate_note = labels.uv_acrylate_note (SDSGenerator.php:3474, 3515); PDFService.php:50; preview.php:576; en.php:688
UV acrylate note:	PRODUCT FAMILY Fixed text that prints only when three conditions hold: the UV acrylate rule pack is on in admin settings, the product's family is flagged UV/LED, and an acrylate is detected anywhere in the composition at any amount. It cannot be overridden, it prints even when Handling or Storage is overridden, and alias, private-label and resale sheets carry it too. <i>On a sheet: Store away from UV light sources, direct sunlight and heat to prevent premature ...</i> You maintain: Product family UV/LED flag and membership rules (Settings → Product Families); the admin toggle 'UV acrylate rule pack'; CAS numbers or chemical names that identify acrylates (CAS master preferred name / Prop 65 name / constituent name) If missing: Nothing prints (neither label nor note) sections[7].uv_acrylate_note = section7.uv_acrylate_note via UVAcrylateRulePack::getSafeHandlingLanguage; isEnabled && family_is_uv && detectAcrylates; SDSGenerator.php:174-182,225-230

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

Section 8 combines the federal and competent-person exposure-limit data for each ingredient (with concentration bands from the formula), engineering-control and PPE sentences chosen automatically from the product's hazard codes and physical state, extra sentences for UV acrylate product families, and optional per-product text overrides.

Exposure limits table (which rows appear):

HAZARD ENGINE One row per exposure limit on file for each ingredient at 0.01% or more. Federal limits (NIOSH, ACGIH, OSHA) always print. Limits from a competent-person determination are added only when that CAS has no current federal hazard classification. Rows are then removed for inhalation-only CAS (e.g. carbon black) and for powder-only ingredients when the product contains a liquid.

On a sheet: 108-88-3 | Toluene | PEL-TWA | 754 | mg/m3 | 10 - 30% | Skin; 29 CFR 1910.1000 Table Z-1

You maintain: Keep the federal data imports current. On Admin > CAS determinations, add exposure limits for chemicals with no federal data. Set the physical state on every raw material. Keep constituent CAS numbers and percentages complete. Keep the inhalation-only CAS list in Admin Settings up to date.

If missing: When no rows remain, the table is not printed and no sentence replaces it. The section opens with Engineering Controls.

HazardEngine::classify :592-638,:703-716 (exposure_limits JOIN hazard_source_records is_current=1), :742-778 CPD (is_active=1 newest); dedupe :1050-1068; SDSGenerator applyCarbonBlackLogic/applySolid...

Exposure limits table headers: CAS | Chemical | Type | Value | Units | Conc% | Notes:

FIXED TEXT (TRANSLATED) Fixed column headings from the language file.

On a sheet: CAS | Chemical | Type | Value | Units | Conc% | Notes

You maintain: Nothing

If missing: English heading, then the key name

labels.el_cas/el_chemical/el_type/el_value/el_units/el_conc_pct/el_notes via getLabels (SDSGenerator.php:3484-3486); PDFService.php:791-798; widths [22,42,22,20,17,15,37] mm

CAS (exposure limits table):

FORMULA / RAW MATERIALS The ingredient's CAS number from the raw-material constituent. Trade-secret ingredients are not masked here.

On a sheet: 108-88-3

You maintain: CAS number on each raw-material constituent (Raw materials screen)

If missing: Constituents with no CAS never get a row

raw_material_constituents.cas_number via Formula::getExpandedComposition, copied by HazardEngine.php:704-706, :2534-2536; no masking in section8() :1833-1848

Chemical (exposure limits table):

FORMULA / RAW MATERIALS Ingredient name: Prop 65 list name > CAS master preferred name > constituent name typed on the raw material. It is never translated and not masked for trade secrets.

On a sheet: Toluene

You maintain: Preferred name in the CAS master, or the constituent name on the raw material

If missing: Blank cell

COALESCE(NULLIF(prop65_list.chemical_name,''), NULLIF(cas_master.preferred_name,''), rmc.chemical_name) Formula.php:291; HazardEngine.php:705

Type (exposure limits table):

REGULATORY LIST Limit type as stored in the federal data or entered on the determination. It is printed exactly as stored.

On a sheet: TLV-TWA

You maintain: Federal data imports; the limit type on each determination row

If missing: Blank cell (CPD rows only)

exposure_limits.limit_type: NIOSH REL-TWA/REL-STEL/REL-Ceiling/IDLH/PEL-TWA; ACGIH TLV-TWA/STEL/Ceiling; OSHA PEL-TWA/STEL/Ceiling/Action Level; CPD exp_limit_type[]

Value (exposure limits table):

REGULATORY LIST Numeric limit from the federal data or the determination.

On a sheet: 20

You maintain: Federal data imports; the value on each determination limit

If missing: A determination row with no value is dropped

exposure_limits.value (NOT NULL); CPD determination_json.exposure_limits[].value (rows with empty() value, including '0', skipped) HazardEngine.php:708, :2530-2538

Units (exposure limits table):

REGULATORY LIST Unit stored with the limit. Importers keep mg/m3 when the source gives it, otherwise ppm, so only one unit is printed per limit.

On a sheet: mg/m3

You maintain: Federal data imports; the units on each determination limit

If missing: "mg/m3"

exposure_limits.units (DB default 'mg/m3'; NIOSHConnector and load-seed-data importExposureLimitsJson prefer *_mgm3); CPD units mg/m3/ppm/f/cc/mppcf

Conc% (exposure limits table):	FORMULA / RAW MATERIALS The ingredient's concentration as a prescribed band, never the exact %. It uses the supplier min/max range when one is given, otherwise the calculated %. <i>On a sheet: 10 - 30%</i> You maintain: Exact % or min/max % on raw-material constituents, kept consistent across raw materials that supply the same CAS; formula line percentages If missing: "<0.1%" when the maximum is below 0.1% or no band fits and the minimum is below 0.1%. "80 - 100%" when no band fits and the minimum is 0.1% or more. Blank on snapshots published before #8. section8() SDSGenerator.php:1833-1848 -> formatConcentration(first composition row for CAS) :3397-3451 PRESCRIBED_RANGES; concentration_pct unset
Notes (exposure limits table):	REGULATORY LIST For ACGIH rows: the notation. For OSHA rows: the notation plus the CFR reference. For determination rows: free text. NIOSH rows always have blank notes. <i>On a sheet: Skin; A4</i> You maintain: Federal seed files; notes on determination limits If missing: Blank cell exposure_limits.notes written by scripts/load-seed-data.php:158-170 (notation [+ ';' + cfr_ref]); NULL for NIOSH; CPD exposure_limits[].notes; HazardEngine.php:711-713
Engineering Controls:	TEXT CHOSEN BY HAZARD A standard ventilation sentence, plus extra sentences for dust (solid or powder products), flammable liquids (H224-H226) and corrosives (H314/H318). Precedence: per-product override > automatic sentences. <i>On a sheet: Use local exhaust ventilation or other engineering controls to maintain airborne...</i> You maintain: Physical state on the finished good, or on the main raw material (resale sheets use the raw material's own state). The hazard data that produces H224-H226/H314/H318. Optional override in the SDS editor. If missing: Only the base sentence. A blank physical state counts as Liquid, so no dust sentence is added. text_overrides[8]['engineering'] (?? no blank check, :1873) > engineeringControls() :1893-1910: section8.engineering + engineering_dust/_flammable/_corrosive; state = resolvePhysicalState :1954-1962
Respiratory Protection:	TEXT CHOSEN BY HAZARD One of five sentences chosen from the hazard codes (SCBA, sensitizer, cartridge respirator, general, none), plus the UV acrylate sentence for UV products. Precedence: per-product override > automatic tier sentence. <i>On a sheet: If exposure limits are exceeded or irritation is experienced, use a NIOSH-approved...</i> You maintain: Hazard data that drives the H-codes (federal data, determinations, manual GHS on raw materials, finished-good hazard override). Optional override per language in the SDS editor. If missing: "No special respiratory protection required under normal conditions of use with adequate ventilation." non-blank text_overrides[8]['respiratory'] > section8.ppe.respiratory.<tier>; tier = HazardEngine::derivePPE(final h_statements) via SDSGenerator::resolvePPE :1923-1944; +section8.uv_respiratory
Hand Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the hazard codes (impervious, sensitizer, resistant, general, none), plus the UV sentence for UV products. Precedence: per-product override > automatic. <i>On a sheet: Chemical-resistant gloves (nitrile recommended) meeting 29 CFR 1910.138...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special hand protection required under normal conditions of use. Gloves may be worn as good industrial hygiene practice." non-blank text_overrides[8]['hand_protection'] > section8.ppe.hand_protection.<tier> (H310/H311/H314 impervious; H317 sensitizer; H312/H313/H315/H316/systemic resistant)
Eye Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the hazard codes (goggles and face shield, goggles, glasses, general, none), plus the UV sentence for UV products. Precedence: per-product override > automatic. <i>On a sheet: Chemical splash goggles or safety glasses with side shields meeting ANSI/ISEA Z87.1...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special eye protection required under normal conditions of use. Safety glasses may be worn as good industrial hygiene practice." non-blank text_overrides[8]['eye_protection'] > section8.ppe.eye_protection.<tier> (H314/H318 goggles_faceshield; H319 goggles; H320 glasses)
Skin Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the hazard codes (suit, clothing, general, none), plus the UV sentence for UV products. Precedence: per-product override > automatic. <i>On a sheet: Wear protective clothing (long sleeves and an impervious apron) to prevent skin contact...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special protective clothing required under normal conditions of use." non-blank text_overrides[8]['skin_protection'] > section8.ppe.skin_protection.<tier> (H310/H311/H314 suit; H312/H313/H315/H316/H317/systemic clothing)

UV acrylate sentences (added to the end of each PPE line):	PRODUCT FAMILY Added when the UV rule pack is on, the product's family is marked UV/LED, and any ingredient at any level has an acrylate CAS or an acrylate/acrylic name. It is skipped on any line that has an override, and is added whatever the hazard tier. <i>On a sheet: ... For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm...</i> You maintain: The UV/LED flag on the product family (Product families screen) and correct family assignment. Ingredient names and CAS numbers. The UV rule pack toggle in Admin Settings. If missing: Nothing is added UVAcrylateRulePack::isEnabled (settings.uv_acrylate_rule_pack) && product_families.is_uv && detectAcrylates(full composition); section8.uv_<field> appended SDSGenerator.php:1860-1867
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SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Section 9 is calculated almost entirely from the finished good's formula and the raw-material records: the dominant or lowest-value raw material, or weighted sums. Five lines can be replaced per product and per language, and the labels, title and fallback wording come from the translation files.

Physical State:	FORMULA / RAW MATERIALS Product's own physical state > physical state of the raw material with the highest total wt% in the formula > 'Liquid'. Translated for the six standard states. <i>On a sheet: Physical State: Liquid</i> You maintain: Physical state on the finished good (Finished Goods > Edit), or physical state on the largest raw material (Raw Materials > Edit). If missing: 'Liquid' (Líquido / Liquide / Flüssig). The next-largest raw is not tried. resolvePhysicalState (SDSGenerator.php:1954-1962): finished_goods.physical_state > formula_props.physical_state (FormulaCalcService.php:498-533) > 'Liquid'; localizeSection9Enum('state')
Color:	FORMULA / RAW MATERIALS The product's colour dropdown. The seven default colours are translated; custom colours print as entered. <i>On a sheet: Color: Black</i> You maintain: Colour on the finished good (Finished Goods > Edit). If missing: The whole Color line is left off. It is always left off on resale raw-material SDSs, because raw materials have no colour field. finished_goods.color (vocab setting sds.color_options) -> localizeSection9Enum('color') (SDSGenerator.php:2037-2040,2111); PDF labelValue skips ''
Appearance:	FORMULA / RAW MATERIALS Per-product override > product colour + resolved state (language word order) > largest raw material's appearance text (English, untranslated) > state alone. <i>On a sheet: Appearance: Black liquid (ES: Líquido, negro; FR: Liquide ; couleur : noir)</i> You maintain: Colour and physical state on the finished good. Otherwise the appearance text on the largest raw material. Optional override in SDS > Edit Text, per language. If missing: The state alone, e.g. 'liquid' (EN, lower case) or 'Líquido' (ES). Never blank. text_overrides(9,'appearance',lang) > section9.appearance_color_state if finished_goods.color > formula_props.appearance (raw_materials.appearance) > section9.appearance_state (SDSGenerator.php:2074-...
Odor:	FORMULA / RAW MATERIALS Per-product override > odour of the raw material with the highest wt% > 'Not determined'. Raw-material text is not translated. <i>On a sheet: Odor: Mild solvent</i> You maintain: Odor on the raw material that dominates each formula (Raw Materials > Edit). Optional override in SDS > Edit Text. If missing: 'Not determined' (No determinado / Non déterminé / Nicht bestimmt). The second-largest raw is not tried. text_overrides(9,'odor',lang) > formula_props.odor (dominant raw_materials.odor; FormulaCalcService.php:498-533) > labels.not_determined (SDSGenerator.php:2096-2102)
Initial Boiling Point and Boiling Range:	FORMULA / RAW MATERIALS Per-product override > lowest boiling point of any raw material in the full formula (sub-formulas included, wt% ignored) > 'Not determined'. <i>On a sheet: Initial Boiling Point and Boiling Range: 78.4 °C (173.1 °F)</i> You maintain: Boiling point (°C) on solvent and other volatile raw materials (Raw Materials > Edit). Pigments and resins can stay blank. If missing: 'Not determined' text_overrides(9,'boiling_point',lang) trimmed > MIN(raw_materials.boiling_point_c) (FormulaCalcService.php:476-486; mig 054) as '{C} °C ({F} °F)' > labels.not_determined

Flash Point:	FORMULA / RAW MATERIALS Per-product override (printed as typed) > lowest flash point of any raw material in the full formula, with '>' if that raw is flagged > 'Not determined'. Section 5 prints the same text. <i>On a sheet: Flash Point: > 93 °C (199.4 °F)</i> You maintain: Flash point and the '>' flag on every raw material that has one (Raw Materials > Edit). An override should give °C or °F with the degree sign. If missing: 'Not determined'. If the hazard data is also empty, Section 14 is 'Not determined' and publishing is blocked. <pre>resolveFlashPointDisplay (SDSGenerator.php:1327-1334): text_overrides(9,'flash_point') > MIN(raw_materials.flash_point_c)+flash_point_greater_than > labels.not_determined</pre>
Solubility:	FORMULA / RAW MATERIALS Per-product override > band from the formula's soluble fraction (Soluble raws count fully, Partially half, Negligible/Insoluble zero; raws without a value skipped) > 'Not determined'. <i>On a sheet: Solubility: Not soluble in water</i> You maintain: Solubility dropdown on raw materials (Raw Materials > Edit). Optional override in SDS > Edit Text. If missing: 'Not determined' when no raw material in the formula has a solubility value. <pre>text_overrides(9,'solubility') > section9.solubility_<key>, F=100·Σ(pct·w)/Σpct(valued); ≥90/≥5/≥1/else (FormulaCalcService.php:28-38,488-496,555-567)</pre>
Specific Gravity:	FORMULA / RAW MATERIALS Weight-averaged mixture specific gravity from every raw material's specific gravity. A raw with no value is silently counted as 1.0. <i>On a sheet: Specific Gravity: 1.052</i> You maintain: Specific gravity on every raw material (Raw Materials > Edit). If missing: Each missing raw counts as 1.0, and an empty formula gives '1'. 'Not determined' never prints in practice. <pre>VOCcalculator::getMixtureSG = Σw/Σ(w/SG), SG NULL or ≤0 -> 1.0 (VOCcalculator.php:236-267,582-592); round(,3) (SDSGenerator.php:2117)</pre>
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS VOC weight fraction × mixture specific gravity × 8.345. Each raw's VOC % is 0.99 if flagged '<1%', otherwise its VOC %, and 0 when blank. <i>On a sheet: VOC (lb/gal) (EPA Method 24): 2.15</i> You maintain: VOC % (or the <1% flag) and specific gravity on every raw material (Raw Materials > Edit). If missing: Missing VOC counts as 0 and missing specific gravity as 1.0, silently. A number always prints (e.g. '0'). <pre>getVOClbPerGal = (Σ pct/100·effVOC)/100 · mixtureSG · 8.345; effVOC: voc_less_than_one->0.99, voc_wt, NULL->0; mode method24_standard (VOCcalculator.php:277-294)</pre>
VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw material in the formula is flagged '<1% VOC'. Otherwise the weighted sum of raw VOC % (0.99 for flagged raws, 0 when blank). <i>On a sheet: VOC (wt%): 12.35 or VOC (wt%): <1</i> You maintain: VOC % or the <1% flag on every raw material (Raw Materials > Edit). If missing: Missing VOC counts as 0, silently. A formula with no valid lines prints '<1'. <pre>formula_props.all_voc_less_than_one ? '<1' : round(total_voc_wt_pct,2), total=Σ pct/100·effVOC (SDSGenerator.php:2052-2056; FormulaCalcService.php:461-464)</pre>
Solids (wt%):	FORMULA / RAW MATERIALS Weighted sum of each raw's solids %. When a raw has no solids %, it is calculated as 100 – VOC – exempt solvent – water. <i>On a sheet: Solids (wt%): 64.2</i> You maintain: Solids % on every raw material, or accurate VOC/exempt/water % and constituents (Raw Materials > Edit). Keep the exempt VOC CAS list current. If missing: Back-calculated silently. A number always prints. <pre>getSolidsWeightPercent=Σ pct/100·effSolids; effSolids=solids_wt else max(0,100-effVOC-effExempt-effWater); exempt_voc_list, CAS 7732-18-5 (VOCcalculator.php:654-694)</pre>
'Not determined' value:	FIXED TEXT (TRANSLATED) Fixed no-data wording from the language file. It prints for Odor, Boiling Point, Flash Point and Solubility when no data exists. <i>On a sheet: Not determined</i> You maintain: Nothing If missing: English wording, then the key text. <pre>labels.not_determined via TranslationService (SDSGenerator.php:2031); en.php:681, es/fr.php:654, de.php:651</pre>

SECTION 10: STABILITY AND REACTIVITY

Section 10 is built mostly from the product's final hazard classification (H-codes) and the nitrogen/sulfur/halogen flags on its constituent CAS numbers. The product family adds the UV item, translation files supply the wording, and a per-product, per-language override can replace any of the five lines.

Reactivity / Chemical Stability / Conditions to Avoid / Incompatible Materials / Hazardous Decomposition Products (labels):

FIXED TEXT (TRANSLATED) Fixed label text in the sheet's language. The decomposition label is wider than the PDF label column, so its value prints on the next line.

On a sheet: Hazardous Decomposition Products:

You maintain: Nothing

If missing: A label key missing everywhere prints the field key in title case, e.g. "Decomposition". A line whose value is an empty string is not printed at all.

```
meta.labels[key] (getLabels) > SDSDocumentStrings::translate(lang, 'labels.<key>') > ucwords(field key); PDFService FIELD_LABEL_MAP :51-56; preview.php:577-581
```

Reactivity:

TEXT CHOSEN BY HAZARD Comes from the product's hazard classification. Each reactive class (self-reactive H240-H242, pyrophoric H250, water-reactive H260/H261) adds its own sentence. Precedence: per-product override > hazard sentences > standard sentence.

On a sheet: Self-reactive: may undergo exothermic decomposition when heated, subjected to friction ...

You maintain: Hazard data that produces H240-H242, H250 or H260/H261. CAS-level data (hazard sources and competent-person determinations, /determinations) counts only when the constituent is at least 1% of the formula. Manual hazards on no-CAS raws and trade-secret H-codes (Raw materials screen) count at any level. The finished-good hazard override (Finished goods screen) also counts.

If missing: "No dangerous reaction known under conditions of normal use."

```
text_overrides(10, 'reactivity') > section10.reactivity_{self_reactive, pyrophoric, water_reactive} joined ' ' > section10.reactivity; SDSGenerator.php:2144-2169, extractHCodes :791-808
```

Chemical Stability:

TEXT CHOSEN BY HAZARD The same reactive hazard classes as Reactivity, listed as the conditions under which the product is unstable. Precedence: per-product override > 'Unstable under ...' sentence > standard sentence.

On a sheet: Unstable under the following conditions: contact with water or moisture (releases ...).

You maintain: Same hazard inputs as Reactivity

If missing: "Stable under recommended storage conditions."

```
text_overrides(10, 'stability') > section10.stability_unstable(:conditions = stability_cond_* joined ' ' > section10.stability; SDSGenerator.php:2171-2181
```

Conditions to Avoid:

TEXT CHOSEN BY HAZARD Built from a list of items chosen by the hazard classification: heat always; ignition sources for flammable (H220-H228), pyrophoric or self-reactive products; air, water, shock and storage temperature for the reactive classes; combustibles instead of oxidizers for oxidizers. Precedence: per-product override > generated list.

On a sheet: Excessive heat, sparks, open flames and other ignition sources, contact with strong oxidizing agents.

You maintain: Same hazard inputs as Reactivity, plus flammable (H220-H228), self-heating (H251/H252) and oxidizer (H270-H272) classifications. Flammability comes only from H-codes, never from the formula flash point.

If missing: "Excessive heat, contact with strong oxidizing agents." (never empty)

```
text_overrides(10, 'conditions_avoid') > section10.cond_<key> in COND_ORDER (:1565-1567), flags at SDSGenerator.php:2147-2152, build :2183-2218
```

Conditions to Avoid: UV light item:

PRODUCT FAMILY Added at the end of Conditions to Avoid when the product's assigned family is flagged UV/LED. It is not tied to the UV acrylate setting or to acrylate detection.

On a sheet: ..., exposure to UV light and direct sunlight (risk of premature polymerization; protect from light).

You maintain: The UV/LED flag on each UV family (Product families screen). Family rules or manual family picks so finished goods and resale raws land in the right family. Recompute families after imports or rule changes, because the stored family_id is used.

If missing: Item omitted (no family, or the family is not flagged UV). A conditions_avoid override also removes it.

```
UVAcrylateRulePack::familyIsUv = fg.family_is_uv from product_families.is_uv via attachFamily (SDSGenerator.php:874-892, ProductFamily::findById, no is_active filter); section10.cond_uv
```

Incompatible Materials:

TEXT CHOSEN BY HAZARD Built from a list chosen by the hazard classification: acids and bases always; oxidizing agents unless the product is an oxidizer; halogens for flammables; water, air, amines and metals for the reactive and H290 classes. Precedence: per-product override > generated list. The same value is repeated in Section 7 Storage.

On a sheet: Strong oxidizing agents, strong acids, strong bases, halogens.

You maintain: Same hazard inputs as Reactivity, plus oxidizer (H270-H272), flammable (H220-H228) and corrosive-to-metals (H290) classifications

If missing: "Strong oxidizing agents, strong acids, strong bases." (never empty)

```
resolveIncompatibleMaterials: text_overrides(10, 'incompatible') > section10.incompat_<key> in INCOMPAT_ORDER; SDSGenerator.php:1611-1684, :2221; Section 7 inline copy :1691-1698
```

Hazardous Decomposition Products:

FORMULA / RAW MATERIALS Comes from the nitrogen, sulfur and halogen flags on each ingredient's CAS number, at any concentration. One element names that gas; two or more list them all. Precedence: per-product override > element sentence > standard sentence.
On a sheet: Carbon monoxide, carbon dioxide, nitrogen oxides, and other toxic gases may be released ...
You maintain: Element flags for every constituent CAS (/determinations, CAS Descriptions tab, 'Flags'), and the molecular formula in the CAS registry, which the seeder uses first. Re-run the element-flag seeder (/determinations/element-flags or scripts/seed-cas-element-flags.php) after new CAS rows arrive through CAS Descriptions or TSCA saves, seed data, or direct constituent inserts, because those leave the flags at 0.
If missing: "Carbon monoxide, carbon dioxide, and other toxic gases may be released upon thermal decomposition." This prints when no constituent is flagged, the CAS is not in the registry, the flags are 0, the constituent is a no-CAS trade secret, the raw material is hazardous with no CAS, or the CAS is blank (that constituent is dropped).
`text_overrides(10,'decomposition') > decomposition_el> / decomposition_multi(:products) / decomposition; cas_master.has_nitrogen/has_sulfur/has_halogen via Formula::getExpandedComposition / buildRes...`

Decomposition base wording (carbon monoxide, carbon dioxide, other toxic gases):

FIXED TEXT Fixed wording inside every decomposition sentence. No carbon flag exists, so it prints even for wholly inorganic products unless overridden.
On a sheet: Carbon monoxide, carbon dioxide, ...
You maintain: Nothing
If missing: Always printed unless overridden
Fixed text inside section10.decomposition, decomposition_el>, decomposition_multi (en.php:236, :241-243, :263; ES omits the release clause; DE puts it mid-sentence)

SECTION 11: TOXICOLOGICAL INFORMATION

Section 11 is assembled from the hazard classification (federal data, competent-person determinations, trade-secret hazard JSON, ATE summation and finished-good overrides), the carcinogen registry, exposure-limit datasets and the formula composition, framed by translated labels and sentence templates, and any of the three main paragraphs can be replaced by a per-product override.

Acute Toxicity:

FIXED TEXT (TRANSLATED) Fixed label from the translation files.
On a sheet: Acute Toxicity:
You maintain: Nothing
If missing: Sheet language > English > literal 'labels.acute_toxicity'. The PDF skips the whole line if the value is empty; the preview prints the label anyway.
`meta.labels.acute_toxicity (getLabels SDSGenerator.php:3496) > SDSDocumentStrings::translate; PDFService.php:865 labelValue; preview.php:340`

Acute toxicity (oral):

HAZARD ENGINE Built from the product's oral acute-toxicity classification (most severe Category 1-4). Per-product override > hazard classification > 'Not classified' sentence.
On a sheet: Acute toxicity (oral): Category 4 — Harmful if swallowed (H302).
You maintain: Oral acute-toxicity data on each ingredient CAS (federal data, or a CPD on the CPD screen where no federal data exists); ingredient % on Raw materials; optional hazard override on the Finished good.
If missing: 'Acute toxicity (oral): Not classified based on available data.' Category 5 (H303) also prints this.
`text_overrides(11,acute_toxicity) > SDSGenerator::buildAcuteToxicity :2353-2484 (canonical acute_toxicity_oral; code = first H3[0-3]x in h_codes else route default; GHSStatements::hText)`

Acute toxicity (dermal):

HAZARD ENGINE Same as the oral line, for the skin route. Per-product override > hazard classification > 'Not classified' sentence.
On a sheet: Acute toxicity (dermal): Category 3 — Toxic in contact with skin (H311).
You maintain: Dermal acute-toxicity data on ingredient CAS (federal or CPD); ingredient %.
If missing: 'Acute toxicity (dermal): Not classified based on available data.'
`buildAcuteToxicity canonical acute_toxicity_dermal; route defaults H310/H310/H311/H312; section11.acute_route_dermal; en.php:288`

Acute toxicity (inhalation):

HAZARD ENGINE Same as the oral line, for inhalation. Per-product override > hazard classification > 'Not classified' sentence.
On a sheet: Acute toxicity (inhalation): Category 4 — Harmful if inhaled (H332).
You maintain: Inhalation acute-toxicity data on ingredient CAS (federal or CPD); ingredient %.
If missing: 'Acute toxicity (inhalation): Not classified based on available data.'
`buildAcuteToxicity canonical acute_toxicity_inhalation; defaults H330/H330/H331/H332; section11.acute_route_inhalation; en.php:289`

ATEmix value (after a classified route):	HAZARD ENGINE The calculated mixture acute toxicity estimate, appended only when its category matches the printed category. Explicit LD50/LC50 on the ingredient > category default estimate. <i>On a sheet: ATEmix = 1250 mg/kg bw. / ATEmix = 12.3 mg/L (4 h, vapors).</i> You maintain: LD50/LC50 (ATE) values on ingredient CAS data or in the CPD; correct line % (trade-secret RMs are counted at 100%). If missing: Nothing is appended. Inhalation always shows the 'vapors' unit; the dust unit cannot print. hazard_classes 'ate_mixture'.ate_mix > ate_results[route] (same cat); HazardEngine::applyATECalculation; ATEi = hazard_classifications.ate_* CPD ate_* CATEGORY_DEFAULT_ATES
Chronic Effects:	FIXED TEXT (TRANSLATED) Fixed label from the translation files. <i>On a sheet: Chronic Effects:</i> You maintain: Nothing If missing: Sheet language > English > literal 'labels.chronic_effects'. meta.labels.chronic_effects; PDFService.php:866; preview.php:341; en.php:629
Chronic effects: sensitisation:	TEXT CHOSEN BY HAZARD Standard sentence printed when the product carries H334 or H317. Per-product override replaces the whole Chronic Effects paragraph. <i>On a sheet: Skin sensitizer: may cause an allergic skin reaction (H317). Repeated or prolonged...</i> You maintain: Sensitisation data on ingredient CAS (federal, CPD or trade-secret hazard JSON) or a Finished good hazard override. If missing: Fragment omitted. text_overrides(11,chronic_effects) > buildChronicEffects :2263-2287 (extractHCodes; H334 chronic_resp_sens, H317 chronic_skin_sens)
Chronic effects: germ cell mutagenicity:	TEXT CHOSEN BY HAZARD Standard sentence when the product carries H340 (else H341). <i>On a sheet: Germ cell mutagenicity: suspected of causing genetic defects (H341).</i> You maintain: Mutagenicity data on ingredient CAS. If missing: Fragment omitted. buildChronicEffects :2288-2293; section11.chronic_muta_1 / chronic_muta_2
Chronic effects: carcinogenicity:	TEXT CHOSEN BY HAZARD H350 sentence > H351 sentence > 'listed' sentence when any carcinogen-registry hit survives. H350/H351 can come from hazard data, from the registry (only for a CAS with no other hazard data), or from the carbon-black rule. <i>On a sheet: One or more components are listed as carcinogens by IARC, NTP, or OSHA; see Carcinogenicity below.</i> You maintain: Carcinogen registry (seed CSV only, no admin screen); carcinogenicity data on CAS; physical state on Raw materials; Settings: sds.inhalation_only_cas. If missing: Fragment omitted. buildChronicEffects :2295-2306; chronic_carc_1/2/listed; applyCarcinogenFindings :4050-4181; applyCarbonBlackLogic :3647-3762
Chronic effects: reproductive toxicity / lactation:	TEXT CHOSEN BY HAZARD Standard sentence for H360x (else H361x), plus a lactation sentence for H362. <i>On a sheet: Reproductive toxicity: suspected of damaging fertility or the unborn child (H361).</i> You maintain: Reproductive-toxicity data on ingredient CAS. If missing: Fragment omitted. buildChronicEffects :2308-2316; chronic_repr_1/2, chronic_lactation
Chronic effects: STOT repeated exposure:	TEXT CHOSEN BY HAZARD Standard sentence for H372 (else H373); target organs are not named here, the text points to Section 2. <i>On a sheet: Specific target organ toxicity, repeated exposure (Category 2): may cause damage... (H373).</i> You maintain: STOT RE data on ingredient CAS. If missing: Fragment omitted. buildChronicEffects :2318-2323; chronic_stot_re_1/2
Chronic effects: repeated skin / eye contact:	TEXT CHOSEN BY HAZARD Skin sentence for H314/H315; eye sentence for H318/H319. <i>On a sheet: Repeated or prolonged skin contact may cause irritation, dryness, or dermatitis.</i> You maintain: Skin/eye irritation data on ingredient CAS. If missing: Fragment omitted. buildChronicEffects :2325-2331; chronic_repeated_skin / chronic_repeated_eye
Chronic effects: none known:	FIXED TEXT (TRANSLATED) Fixed sentence printed only when no chronic sentence above applies and there is no override. <i>On a sheet: None known. Based on available data for the mixture and its components, the...</i> You maintain: Nothing If missing: This is itself the fallback. buildChronicEffects :2333-2335; section11.chronic_none (en.php:337)

Carcinogenicity:	FIXED TEXT (TRANSLATED) Fixed label, always printed, text on the lines below. <i>On a sheet: Carcinogenicity:</i> You maintain: Nothing If missing: Sheet language > English > literal 'labels.carcinogenicity'. meta.labels.carcinogenicity; PDFService.php:868-871; preview.php:343
Carcinogenicity: no listed components:	FIXED TEXT (TRANSLATED) Fixed sentence printed when no registry hit at 0.1% or more survives filtering. Per-product override > this sentence. <i>On a sheet: No components present at or above 0.1% are listed as carcinogens by IARC, NTP, or OSHA.</i> You maintain: Carcinogen registry kept current (storage/data/seed/carcinogens.csv + load-seed-data.php). If missing: Default text. text_overrides(11,carcinogenicity) > section11.carcinogenicity (:threshold = CarcinogenService::LISTING_THRESHOLD_PCT); SDSGenerator.php:2684-2692
Carcinogenicity: listed-components intro:	REGULATORY LIST Printed when any ingredient at 0.1% or more matches the carcinogen registry after physical-state filters. Override > registry result. <i>On a sheet: The following component(s), present at or above 0.1%, are listed as carcinogens or...</i> You maintain: Carcinogen registry; ingredient CAS and %; physical state on Raw materials; Settings: sds.inhalation_only_cas. If missing: Negative sentence prints instead. CarcinogenService::buildSummaryText :257-272; section11.carcinogenicity_listed_intro; filters SDSGenerator.php:3843-3938
Carcinogenicity: listed component line:	REGULATORY LIST One line per listed ingredient: name, CAS, Section 3 band and each agency classification. Name: Prop 65 name > CAS master name > constituent name > registry name. <i>On a sheet: Ethylbenzene (CAS 100-41-4, 1 - 5%) — IARC: Group 2B; NTP: Reasonably anticipated...</i> You maintain: Carcinogen registry (one correct row per CAS+agency in the seed CSV); constituent name, CAS and min/max or exact % on Raw materials. If missing: Unmapped classification prints as stored; band falls back to the exact-% band; trade-secret identity is not masked. CarcinogenService::buildComponentTexts; section11.carcinogenicity_listed_line/_listing/carcinogen_class_*; formatConcentration(compByCas)
Component Toxicological Data:	FIXED TEXT (TRANSLATED) Fixed sub-heading, printed only when at least one component row qualifies. <i>On a sheet: Component Toxicological Data:</i> You maintain: Nothing If missing: Not printed when no component has an exposure limit or carcinogen listing. meta.labels.component_tox_data; PDFService.php:878-882; preview.php:346-347
Component row (name, CAS, band):	FORMULA / RAW MATERIALS Each ingredient at 0.1% or more that has an exposure limit or a carcinogen listing, with the same band as Section 3. Name: Prop 65 name > CAS master name > constituent name. <i>On a sheet: Toluene (CAS 108-88-3) — 10 - 30%</i> You maintain: Constituent CAS, name and pct_exact / pct_min / pct_max on Raw materials; formula lines on Finished goods. If missing: Row omitted. A trade-secret no-CAS bucket with limits prints 'Trade Secret (CAS TRADE_SECRET)' in English. section11() :2625-2663 over calcResult.composition; formatConcentration; literals 'CAS ', ' - ' in PDFService.php:884-894
Component carcinogen listing line:	REGULATORY LIST Agency and classification from the carcinogen registry; English sheets add the registry description, other languages a translated gloss. <i>On a sheet: IARC: Group 1 — Carcinogenic to humans.</i> You maintain: Carcinogen registry rows (agency, classification, description). If missing: No description part when empty; unmapped classification prints as stored. CarcinogenService::localiseListings :100-120; carcinogen_list.agency/classification/description; PDFService.php:896-906
Component exposure limit line:	HAZARD ENGINE All exposure limits found for the CAS, combined: federal datasets plus CPD limits (only for CAS without federal hazard data) plus trade-secret hazard JSON. Printed as stored, untranslated. <i>On a sheet: PEL-TWA: 435 mg/m3 (29 CFR 1910.1000 Table Z-1)</i> You maintain: Federal OEL dataset refresh (OSHA PEL / ACGIH TLV seed, NIOSH connector); CPD exposure limits for CAS without federal data; physical state on Raw materials. If missing: No line; '(notes)' omitted when notes are blank. hazardResult.exposure_limits by CAS: exposure_limits+hazard_source_records.is_current CPD determination_json.exposure_limits manual JSON; dedupeExposureLimits

UV Acrylate Information:	FIXED TEXT (TRANSLATED) Fixed label, printed only with the UV note. <i>On a sheet: UV Acrylate Information:</i> You maintain: Nothing If missing: Not printed without a note. meta.labels.uv_acrylate_note (SDSGenerator.php:3474,3515); PDFService.php:922-927
UV acrylate note text:	PRODUCT FAMILY Printed for products in a UV/LED family that contain an acrylate, when the admin toggle is on. Sensitizer wording when the product has H317, otherwise the unclassified wording; no override. <i>On a sheet: Contains acrylate monomers that are known skin sensitizers. Repeated exposure may...</i> You maintain: Product family assignment and its UV flag on Product families; ingredient CAS/names; Settings: UV acrylate rule pack left enabled. If missing: Not printed. UVAcrylateRulePack::isApplicable (settings.uv_acrylate_rule_pack missing or 'enabled'; product_families.is_uv; detectAcrylates); section11.uv_acrylate_note(unclassified)

SECTION 12: ECOLOGICAL INFORMATION

Section 12 is built mostly from the aquatic classifications and M-factors of each constituent, as summed by the hazard engine, and from the EPA PBT flag on the SARA 313 list; fixed translation sentences fill whatever has no data, and a per-product text override can replace any of the four paragraphs.

Ecotoxicity:	TEXT CHOSEN BY HAZARD Per-product text override > aquatic hazard statements the product is classified with (lead-in sentence + H statements + release warning) > "does not meet the criteria" sentence when components carry aquatic data > standard no-data sentence. The aquatic H codes come from summing each component's aquatic category, weighted by its M-factor, or from a finished-good hazard override. <i>On a sheet: Classified as hazardous to the aquatic environment by the GHS summation method ... H410: Very toxic</i> You maintain: Aquatic classifications and M-factors on each constituent CAS (federal hazard data, or a CAS determination when no federal data exists); constituent percentages on raw materials and formula lines; finished-good hazard override if used; review stored Section 12 text overrides. If missing: "No data available on the mixture. Avoid release to the environment." text_overrides(fg,lang,12,'ecotoxicity') > H4(0[0-2] 1[0-3]) in h_statements (section12.ecotoxicity_classified[_no_table]+env_warning) > ecotoxicity_not_classified > section12.ecotoxicity; SDSGenerat...
Ecotoxicity: lead-in when classified:	TEXT CHOSEN BY HAZARD Fixed sentence chosen by whether a component table follows. With a table it cites the summation method and the components "listed below"; with no table it states only the classification. <i>On a sheet: Classified as hazardous to the aquatic environment (GHS Rev. 7, Chapter 4.1):</i> You maintain: Nothing beyond the aquatic data. The no-table version is reached in practice only through a finished-good hazard override (mode additive or replace). If missing: Not printed; another branch of the Ecotoxicity paragraph is used. section12.ecotoxicity_classified (component_aquatic non-empty) else section12.ecotoxicity_classified_no_table; SDSGenerator.php:2768-2770; en.php:360-362
Ecotoxicity: aquatic H statements:	HAZARD ENGINE The aquatic hazard statements the product actually resolved to, in code order, using the standard GHS wording for the sheet's language. Summation alone can produce only H400, H410, H411 and H412; H401, H402 and H413 come only from a finished-good override. <i>On a sheet: H400: Very toxic to aquatic life. H410: Very toxic to aquatic life with long lasting effects.</i> You maintain: Constituent aquatic classes and M-factors; finished-good hazard override (mode + JSON) where used. If missing: A non-English sheet with no translation for the code prints the English statement. A code unknown to both tables prints only the code, e.g. "H413." h_statements codes ^H4(0[0-2] 1[0-3])\$ ksort; EN text GHSStatements::H_STATEMENTS (resolveHStatements); non-EN ghs_{lang}.php > EN; SDSGenerator.php:2719-2763
Ecotoxicity: release warning:	TEXT CHOSEN BY HAZARD Fixed warning sentence added after the H statements, only when the product is aquatic-classified. <i>On a sheet: Avoid release to the environment. Prevent entry into waterways, sewers, and soil.</i> You maintain: Nothing If missing: Not printed on unclassified products. section12.environmental_warning; SDSGenerator.php:2771-2773; en.php:358

Ecotoxicity: not-classified sentence:	TEXT CHOSEN BY HAZARD Fixed sentence printed when components carry aquatic data but the product has no aquatic H code. This happens below the thresholds, when components carry only Acute 2/3 or Chronic 4 (which are never summed), or when a replace-mode override removes the codes. <i>On a sheet: Based on the component data below, the mixture does not meet the GHS criteria ...</i> You maintain: Constituent aquatic classifications. If missing: Falls back to the no-data sentence when no component has aquatic data. section12.ecotoxicity_not_classified when no aquatic H-code and aquatic_components non-empty; SDSGenerator.php:2774-2778; en.php:363
Ecotoxicity: no-data sentence:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, used when no component has aquatic data and there is no override. <i>On a sheet: No data available on the mixture. Avoid release to the environment.</i> You maintain: Nothing If missing: This is the fallback itself. section12.ecotoxicity; en.php:345; es/fr/de.php:329
Component Aquatic Hazard Data:	FIXED TEXT (TRANSLATED) Fixed sub-heading, printed only when at least one component has an aquatic classification. <i>On a sheet: Component Aquatic Hazard Data:</i> You maintain: Nothing If missing: The heading and table are omitted when no component has aquatic data. labels.component_ecotox_data; PDFService.php:943-947 (adds ':'), preview.php:399-400 (no colon)
Component table headers: Chemical Name CAS Number Conc% Aquatic Acute Aquatic Chronic:	FIXED TEXT (TRANSLATED) Fixed column labels from the translation file. Conc% is the same label used in Section 8. <i>On a sheet: Chemical Name CAS Number Conc% Aquatic Acute Aquatic Chronic</i> You maintain: Nothing If missing: Not printed when the table is empty. On French and German PDFs the long acute/chronic headers run into the neighbouring cells. labels.chemical_name, cas_number, el_conc_pct, aquatic_acute, aquatic_chronic; PDFService.php:949-956 (fixed widths 68/25/17/32/33 mm, single line)
Component table: Chemical Name:	FORMULA / RAW MATERIALS Prop 65 list name > master CAS preferred name > the constituent name entered on the raw material. A trade-secret constituent shows its trade-secret description, or "Trade Secret". <i>On a sheet: Copper phthalocyanine</i> You maintain: CAS master preferred names and raw-material constituent names; trade-secret flag and description on withheld constituents (Raw materials screen). If missing: Blank cell when all name sources are empty. A trade-secret row with no description prints "Trade Secret" (translated). aquatic_components[].name = COALESCE(prop65_list.chemical_name, cas_master.preferred_name, rmc.chemical_name) (Formula.php:291); TS: tradeSecretName(rmc.trade_secret_description) > labels.trade_secret
Component table: CAS Number:	FORMULA / RAW MATERIALS The constituent's CAS number from the raw material. Trade-secret constituents are masked. <i>On a sheet: 147-14-8</i> You maintain: CAS numbers and trade-secret flags on raw-material constituents. If missing: Rows with no CAS are dropped from the table. Trade secret prints "TRADE SECRET". aquatic_components[].cas (rmc.cas_number); masked with labels.trade_secret_cas when cas=TRADE_SECRET or is_trade_secret; SDSGenerator.php:2732-2740
Component table: Conc%:	FORMULA / RAW MATERIALS The same prescribed concentration band that Section 3 uses for that constituent. The exact percentage is never printed. <i>On a sheet: 10 - 30%</i> You maintain: Constituent percentages (exact or min/max) on raw materials and formula line percentages. If missing: "<0.1%" for very small amounts; blank for a trade-secret row that cannot be matched to the composition. formatConcentration(compByCas[cas]) on concentration_min/max else concentration_pct → PRESCRIBED_RANGES; SDSGenerator.php:2735-2743, :3408-3453

Component table: Aquatic Acute / Aquatic Chronic:	HAZARD ENGINE Each component's aquatic acute and chronic category from current federal hazard data, or from its CAS determination only when no current federal data exists for that CAS. Category 1 also shows its M-factor. <i>On a sheet: Category 1 (M = 10) Category 1 (M = 1)</i> You maintain: Aquatic acute/chronic categories and M-factors on each constituent CAS (federal hazard data import, or the CAS determinations screen for CAS with no federal data). If missing: An em dash (—) when there is no class on that route. Category 1 with no M-factor on file prints "(M = 1)", which looks the same as supplier data. aquatic_components[].acute/chronic_category (hazard_classifications.category_canonical, hsr.is_current=1; else CPD selected_hazards) + m_factor_acute/chronic>0 else 1; formatAquaticCategory SDSGenera...
Component table: which rows print:	HAZARD ENGINE One row for each constituent at 0.01% or more that has any aquatic class, sorted by concentration from highest. Trade-secret manual-hazard rows always sort first. You maintain: Constituent aquatic data; current flags on hazard source records. If missing: Table omitted when there are no rows. HazardEngine::buildAquaticComponentSummary over aquaticBuffer (HazardEngine.php:596-660, :1287-1323); TS manual JSON buffered at 100%
Persistence and Degradability:	REGULATORY LIST Per-product override > a sentence naming each constituent flagged as PBT on the EPA SARA 313 list at 0.1% or more > "No data available." <i>On a sheet: Contains component(s) identified as ... PBT under SARA 313 (40 CFR 372.28): Lead (CAS 7439-92-1).</i> You maintain: SARA 313 list kept current with its PBT flag (Regulatory lists import); constituent CAS numbers. If missing: "No data available." text_overrides[12]['persistence'] > section12.pbt_components(:components = ' '; '-joined pbt_component_item) from SARA313Service::analyse is_pbt & conc>=0.1 > section12.persistence; SDSGenerator:2798-2...
PBT component item:	REGULATORY LIST For each PBT constituent: the SARA list name with its footnote markers stripped, or the composition name, followed by the CAS number. Trade-secret constituents are masked. <i>On a sheet: Mercury (CAS 7439-97-6)</i> You maintain: SARA 313 list chemical names and PBT flags. If missing: Not printed when there is no PBT constituent. section12.pbt_component_item ':name (CAS :cas)'; name = sara313_list.chemical_name > composition chemical_name, trailing +#+ stripped; SDSGenerator.php:2799-2814
Bioaccumulative Potential:	REGULATORY LIST Per-product override > "See Persistence and Degradability above." when the PBT sentence prints (or the full PBT sentence if Persistence is overridden) > "No data available." <i>On a sheet: See Persistence and Degradability above.</i> You maintain: Same SARA 313 PBT flags as Persistence. If missing: "No data available." No PBT: text_overrides[12]['bioaccumulation'] > section12.bioaccumulation. PBT: override > (persistence overridden ? pbtLine : section12.pbt_see_persistence); SDSGenerator:2819-2824
Mobility in Soil:	FIXED TEXT (TRANSLATED) Per-product override > fixed "No data available." No data source exists for this line. <i>On a sheet: No data available.</i> You maintain: Nothing (an override only). If missing: "No data available." On old snapshots the PDF leaves the line out, while the HTML preview prints an empty label. text_overrides[12]['mobility'] > section12.mobility; SDSGenerator.php:2832; PDFService.php:991; preview.php:425

SECTION 13: DISPOSAL CONSIDERATIONS

Section 13 prints one fixed disposal sentence, which can be replaced per product, and an RCRA hazardous-waste line built when the sheet is generated. That line comes from the product's hazard codes and Section 9 flash point, plus a match of each component's CAS number against the admin-maintained EPA RCRA waste-code list.

Disposal Methods:

PRODUCT OVERRIDE A fixed sentence from the language file, unless someone has typed replacement text for this product and language in the SDS editor. Per-product text > standard sentence. Resale raw-material sheets always print the standard sentence.
On a sheet: Dispose of in accordance with all applicable federal, state, and local regulations. Do not...
You maintain: Nothing for the standard text. Optional per-product wording in SDS editor > Section 13 > Disposal Methods (finished goods only)
If missing: The standard sentence: "Dispose of in accordance with all applicable federal, state, and local regulations. Do not dump into sewers, drains, or waterways."
`text_overrides(fg, lang, 13, 'methods', sds_version_id IS NULL) > translation section13.methods (lang > EN > key). SDSGenerator.php:2987; PDFService.php:63,1167; preview.php:585`

RCRA Hazardous Waste Classification (lead-in sentence):

TEXT CHOSEN BY HAZARD Fixed wording that prints only when at least one item applies to the product as sold: D001, D002, D003, or a component with a D code. In the PDF the long label sits on its own line, with the text below it.
On a sheet: If discarded as sold, this product may be regulated as hazardous waste under US EPA RCRA (40 CFR 261):
You maintain: Nothing directly. It depends on the hazard, flash point and RCRA-list inputs below
If missing: Replaced by one of the two 'No RCRA hazardous waste characteristic...' sentences
`section13.rcra_intro + ' '; '-joined items + '.' (SDSGenerator.php:2972-2974); label labels.rcra_classification (en.php:645); PDFService.php:64,1259`

Ignitable (D001) item:

HAZARD ENGINE Prints when the product has a flammable-liquid code (H224/H225/H226), a resolved flash point below 60 °C, an oxidizer code (H270-H272) or a pyrophoric/self-heating code (H250-H252). The reasons are joined with commas.
On a sheet: ignitable (D001) — flash point below 60 °C (140 °F), oxidizer
You maintain: Correct hazard inputs (CAS determinations, vendor hazard data, finished-good hazard override) and the flash point on each raw material (Raw materials screen)
If missing: No D001 item. Flammable gases, flammable aerosols and flammable solids (H220-H223, H228) never trigger it
`section13.rcra_d001 + rcra_reason_flash_point | rcra_reason_flash_point_gt | rcra_reason_oxidizer | rcra_reason_ignitable_solid; SDSGenerator.php:2894-2916`

D001 reason: flash point below 60 °C / not determined at or above 60 °C:

FORMULA / RAW MATERIALS Uses the same flash point as Section 9: Section 9 override text for the language > lowest flash point among all raw materials in the formula. 'Below 60 °C' also prints whenever H224-H226 is present, even if the flash point is 60 °C or higher.
On a sheet: flash point below 60 °C (140 °F)
You maintain: Flash point (°C) and the '>' flag on every raw material (Raw materials screen), trace ones included. Any Section 9 flash-point override needs '°C' or '°F' with the degree sign
If missing: No flash-point reason. An override with no number is ignored. An override like '75 F' (no ° sign) is read as 75 °C
`H224/5/6 or (fp<60 && !gt) > rcra_reason_flash_point; fp<60 && gt > rcra_reason_flash_point_gt. resolveFlashPointNumeric (:3037-3056) > formula_props.flash_point_c`

D001 reasons: oxidizer / pyrophoric or self-heating material:

HAZARD ENGINE Prints when the hazard result contains H270-H272 (oxidizer) or H250-H252 (pyrophoric or self-heating).
On a sheet: oxidizer
You maintain: Hazard classification inputs for oxidizing or self-heating constituents (CAS determinations)
If missing: Not printed
`section13.rcra_reason_oxidizer / rcra_reason_ignitable_solid; SDSGenerator.php:2908-2913`

Corrosive (D002) item:

HAZARD ENGINE Prints whenever the product is classified H314 (skin corrosion). pH, physical state and H290 are not considered.
On a sheet: corrosive (D002) — based on its skin corrosion classification (H314), pH not determined
You maintain: Skin-corrosion classification inputs (CAS determinations, vendor data, finished-good hazard override)
If missing: No D002 item
`H314 in engine H-codes > section13.rcra_d002; SDSGenerator.php:2918-2920`

Reactive (D003) item and reasons:

HAZARD ENGINE Prints when the product has a water-reactive code (H260/H261), an explosive code (H200-H205) or a self-reactive/organic peroxide code (H240-H242). The reasons are joined with commas.

On a sheet: reactive (D003) — reacts with water, explosive

You maintain: Hazard classification inputs (CAS determinations)

If missing: No D003 item

section13.rcra_d003 + rcra_reason_water_reactive | rcra_reason_explosive | rcra_reason_unstable;
SDSGenerator.php:2922-2934

Toxicity-characteristic component (D code) item:

REGULATORY LIST Each component's CAS number is matched exactly against the RCRA waste-code list. A D-code match prints at any concentration above zero, with the TCLP limit from the list. Trade secrets print as 'Trade Secret' or their description, but the code is still shown.

On a sheet: contains Lead (D008 if the toxicity characteristic regulatory level of 5 mg/L (TCLP) is exceeded)

You maintain: RCRA waste-code list (Admin > RCRA). Add compound CAS numbers for metals, because the seed covers element CAS only. CAS numbers and percentages on raw-material constituents

If missing: No item. A constituent with no percentage on file, or a metal present only as a compound CAS, is never reported

RCRAService::analyse/match on rcra_waste_codes (kind D); formatRcraCode section13.rcra_code_d / rcra_code_d_nolimit; name COALESCE(prop65_list, cas_master, rmc.chemical_name)

'No characteristic identified' sentence:

TEXT CHOSEN BY HAZARD Prints when no D001/D002/D003 item and no D-code component applies. If only listed (F/K/P/U) components matched, the shorter version prints without the 'no component is...' clause.

On a sheet: No RCRA hazardous waste characteristic (40 CFR 261 Subpart C) has been identified for this...

You maintain: A complete RCRA waste-code list and complete constituent CAS and percentage data, because this sentence is a positive claim based on finding no match

If missing: This sentence is itself the default when nothing matches

section13.rcra_none (nothing matched) / rcra_none_characteristic (only F/K/P/U matched);
SDSGenerator.php:2975-2977

Listed-waste codes of components, for reference:

REGULATORY LIST Prints when a component at 0.1 % or more (using its maximum percentage) matches an F, K, P or U code on the RCRA list. The wording says each listing's actual scope.

On a sheet: Listed-waste codes of components, for reference (they do not apply to this product as sold):

contains Toluene (U220...)

You maintain: F/K/P/U rows in the RCRA waste-code list (Admin > RCRA). Constituent CAS numbers and percentages

If missing: Sentence omitted

section13.rcra_listed_intro + rcra_component + rcra_code_f/k/p/u; RCRAService::MIN_CONCENTRATION_PCT 0.1 (pct_exact > range max); SDSGenerator.php:2948-2981, 3011-3018

Generator-responsibility sentence:

FIXED TEXT (TRANSLATED) Fixed wording from the language file. It is always the last sentence of the RCRA line and cannot be edited per product.

On a sheet: The waste generator is responsible for the final hazardous waste determination at the time...

You maintain: Nothing

If missing: The English text, or the raw key

section13.rcra_generator (en.php:378) appended unconditionally; SDSGenerator.php:2982

SECTION 14: TRANSPORT INFORMATION

Section 14 is now worked out for each product by the transport classifier. It uses the product's flash point and boiling point (from raw-material data or the Section 9 override), the hazard statements from the hazard engine, and the product's transport type, description and family. Per-product overrides, entered separately for each language, win line by line, and the fixed wording (entry names, notes, labels) comes from the translation files.

UN Number:

TRANSPORT RULES Worked out from the product's flash point, its hazard statements (flammable, corrosive, toxic, aquatic) and its product type (ink, paint, wash, or one of the related types). Per-product override > classifier result > "Not regulated" / "Not determined".

On a sheet: UN1210

You maintain: Flash point on every raw material that has one (Raw materials > Flash Point). Hazard data on each CAS (CAS determinations). Transport product type, or a descriptive description / family (Finished goods). Physical state (Finished goods).

If missing: "Not determined" when no raw has a flash point and the product has no hazard statements (publishing is blocked). "Not regulated" when no transport class is triggered.

text_overrides(14,un_number,lang) > TransportClassifier::classify()['un_number']; fp = resolveFlashPointNumeric (9/flash_point > formula_props); SDSGenerator.php:3077-3138

Proper Shipping Name:	TRANSPORT RULES The official DOT name for the chosen UN entry, taken from the translation file. ES/FR/DE keep the English name and add a local translation in brackets. Override > classifier entry name > "Not regulated" / "Not determined". <i>On a sheet: Printing ink, flammable</i> You maintain: Same inputs as UN Number. The names themselves are kept in the translation files. If missing: "Not regulated" / "Not determined". A name missing in one language falls back to the English text. <code>text_overrides(14,proper_shipping_name) > section14.psn_{psn_key} (en.php:404-424; es/fr/de:379-398) + technical names; SDSGenerator.php:3110,3139</code>
Proper Shipping Name: technical names in brackets (n.o.s. entries only):	FORMULA / RAW MATERIALS For n.o.s. entries the ingredients the hazard engine credits with the main transport hazard are added in brackets, at most two, highest wt% first. In practice a Class 3 entry usually names only one (the solvent with the most severe flammability category). Names are never translated. <i>On a sheet: Flammable liquids, n.o.s. (Toluene)</i> You maintain: Ingredient names (CAS determinations > preferred name; Raw materials > constituent name). Flammability, corrosion, toxicity and aquatic determinations on each CAS. Trade-secret flags. If missing: No brackets are printed. An English warning appears in the preview Warnings box, and publishing is not blocked. Never added to UN1210/UN1263 or when the PSN is overridden. <code>TransportClassifier::technicalNames()/casHCodeMap(hazard_classes), TC.php:345-371,475-491; names Formula.php:291 COALESCE(prop65,cas_master,rmc)</code>
Hazard Class:	TRANSPORT RULES The classifier picks the primary DOT class and any secondary classes using the 49 CFR 173.2a precedence rules. Override > classifier > "Not regulated" / "Not determined". <i>On a sheet: 3 (8)</i> You maintain: Flash point and boiling point on the raws. Skin-corrosion and acute-toxicity categories on each CAS. Physical state, which decides whether 8 or 6.1 comes first. If missing: "Not regulated" / "Not determined" <code>text_overrides(14,hazard_class) > TransportClassifier hazard_class = primary + ' (' + subsidiaries + ')'; TC.php:197-280,426-472; SDSGenerator.php:3114,3140</code>
Packing Group:	TRANSPORT RULES The most stringent group from the flammability test (boiling point, flash point), the corrosion test (first skin-corrosion category found) and the toxicity test. Override > classifier > "Not applicable" (not regulated) / "Not determined". <i>On a sheet: II</i> You maintain: Boiling point on every volatile raw (Raw materials). Flash point. Skin-corrosion sub-category (1A/1B/1C) and acute-toxicity category on each CAS. A Section 9 boiling-point override is NOT used here. If missing: With no boiling point, H224 gives I and otherwise the flash-point rules apply. "Not applicable" when not regulated; "Not determined" when undetermined. <code>text_overrides(14,packing_group) > mostStringent(pg3[formula_props.boiling_point_c,fp,H224/H225], pg8, pg61); TC.php:169-193,373-424</code>
Environmental Hazards:	TRANSPORT RULES "Marine pollutant: Yes" when the product carries H400, H410 or H411; otherwise "No". Override > engine aquatic codes > "Not determined". <i>On a sheet: Environmental Hazards: Marine pollutant: No</i> You maintain: Aquatic toxicity data and M-factors on each CAS (CAS determinations). If missing: "Marine pollutant: No" when no aquatic code is present. "Not determined" when the product is unclassified. The line is skipped on old snapshots. <code>text_overrides(14,environmental_hazards) > section14.marine_pollutant_yes no (en.php:425-426) on H400/H410/H411; SDSGenerator.php:3103-3116,3142</code>
Note: combustible-liquid sentence (conditional):	TRANSPORT RULES Printed only when the product is not regulated and its flash point is known to be from 60 °C up to (not including) 93 °C. It cannot be overridden. <i>On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only.</i> You maintain: Flash point on the raws (or the Section 9 flash-point override). If missing: Not printed <code>TransportClassifier notes[]='combustible' (TC.php:124,159-161) > section14.note_combustible (en.php:427); SDSGenerator.php:3119-3123</code>
Note: viscous Packing Group III sentence (conditional):	TRANSPORT RULES Printed only when the product is Class 3, Packing Group II, with no secondary class. It cannot be overridden. <i>On a sheet: If viscous, may be assigned to Packing Group III under 49 CFR 173.121(b)(1)...</i> You maintain: Same inputs as Packing Group. If missing: Not printed <code>TransportClassifier notes[]='viscous' (TC.php:299-301) > section14.note_viscous (en.php:429); SDSGenerator.php:3119-3123</code>

Note: carrier verification sentence:

FIXED TEXT (TRANSLATED) Fixed sentence from the translation file. It is always printed last on the Note line.
On a sheet: Transport classification should be verified with the carrier and against current 49 CFR...
You maintain: Nothing
If missing: Always printed
section14.note (en.php:401; es/fr/de:377), appended unconditionally at SDSGenerator.php:3123; PDFService.php:1003; preview.php:436-438

Outcome wording: Not regulated / Not applicable / Not determined:

FIXED TEXT (TRANSLATED) The fixed words printed on the UN, name, class, packing group and environmental lines when the classifier finds no class, or cannot decide. A per-line override still wins.
On a sheet: Not regulated
You maintain: Nothing (the outcome itself depends on flash point and hazard data)
If missing: n/a
labels.not_regulated / labels.not_applicable / labels.not_determined (en.php:681-683); SDSGenerator.php:3098-3107

SECTION 15: REGULATORY INFORMATION

Section 15 is assembled in SDSGenerator::section15() from the product's composition checked against the regulatory lists you maintain (TSCA inventory, SARA 313, HAP, SNUR, Prop 65), from the Section 2 hazard result, from a few per-product editor notes, and from fixed translated wording.

OSHA Status:

HAZARD ENGINE The sentence follows the same classification result that Section 2 prints: 'classified' when there is a signal word, pictogram, hazard class or H-statement, otherwise 'not classified'. Per-product editor note (per language) > automatic sentence.
On a sheet: This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).
You maintain: The hazard data behind Section 2 (constituent classifications on CAS Determinations, finished-good hazard override). Optionally a per-language note in the SDS editor.
If missing: No note and nothing classified: 'This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).' A frozen pre-update note ('...HazCom 2012...') prints word for word, whatever the classification.
text_overrides(fg,15,osha_status,lang) ?? isClassified(hazard) ? section15.osha_status : section15.osha_status_not_classified (SDSGenerator.php:3226-3232, :903-909)

TSCA Status:

REGULATORY LIST Every ingredient CAS is checked against the EPA TSCA inventory and the per-CAS TSCA review; the 'all listed' sentence prints only when every one is covered. Per-product editor note > per-CAS review (listed/exempt/not listed) > inventory match > not verified.
On a sheet: All components of this product are listed on or exempt from the TSCA inventory.
You maintain: TSCA Inventory loaded from EPA and kept current (Regulatory Data > TSCA Inventory). TSCA Review on CAS Determinations for exempt, confidential or crossover CAS. A CAS on every raw-material constituent, and at least one constituent on every raw material.
If missing: 'TSCA inventory status has not been verified for all components.' plus an operator warning (not printed). A raw material with no constituents, or a constituent without a CAS, is skipped silently, so 'all listed' can still print.
trim(text_overrides[15][tsca_status]) > TSCAService::analyse(composition): cas_master.tsca_status (<>'auto') > tsca_inventory row > not_verified; all_covered → section15.tsca_status

SARA 313 / TRI Reporting:

FIXED TEXT (TRANSLATED) Fixed heading from the translation file. Always printed on new sheets.
On a sheet: SARA 313 / TRI Reporting:
You maintain: Nothing
If missing: Older saved sheets without the SARA data omit the whole block.
labels.sara_313_title; printed when sections[15].sara_313.reportable is an array (PDFService.php:1017-1020; preview.php:448-451)

SARA 313 supplier-notification statement / 'none' sentence:

REGULATORY LIST Fixed wording. Which sentence prints depends on whether any ingredient CAS on the SARA 313 list is at or above its threshold.
On a sheet: This product does not contain any toxic chemicals subject to ... Section 313 ...
You maintain: SARA 313 list (Admin > SARA 313 List) and ingredient CAS numbers
If missing: No listed chemical at or above threshold: the 'does not contain any toxic chemicals ... de minimis concentration' sentence.
labels.sara_313_statement when reportable non-empty, else labels.sara_313_none (PDFService.php:1022-1041; preview.php:452-467)

SARA 313 chemical lines (name, CAS, band, threshold, PBT):	REGULATORY LIST Ingredients whose exact CAS is on the SARA 313 list at or above the list threshold. Name: SARA list name > ingredient name. Band: Section 3 range. Threshold: PBT threshold if PBT > de minimis %. <i>On a sheet:</i> • Toluene (CAS 108-88-3) — 10 - 30% (de minimis threshold: 1%) You maintain: SARA 313 list rows with CAS, name, de minimis % (0.1 for OSHA carcinogens), PBT flag and PBT threshold. Ingredient CAS and percentages on raw materials. If missing: Blank list name: the ingredient name prints. A CSV import with no de minimis column defaults to 1%. Older saved sheets show an empty band. SARA313Service::analyse (exact CAS on sara313_list; de minimis_pct / pbt_threshold_pct); bandRegulatoryEntries → formatConcentration (SDSGenerator.php:2533-2549, :3408)
SARA 313 range note:	FIXED TEXT (TRANSLATED) Fixed italic note. Printed only when SARA chemicals are listed. <i>On a sheet: Concentrations are stated as the prescribed concentration ranges of 29 CFR 1910.1200(i)(1)...</i> You maintain: Nothing If missing: Not printed when no SARA chemical is listed. labels.sara_313_range_note (PDFService.php:1036-1038; preview.php:465)
EPA HAPs (heading):	FIXED TEXT (TRANSLATED) Fixed heading. English says 'EPA HAPs'; Spanish, French and German use a long Clean Air Act 112(b) title. <i>On a sheet: EPA HAPs:</i> You maintain: Nothing If missing: Older saved sheets without HAP data omit the block. labels.hap_title (en.php:661; es/fr.php:634; de.php:631); PDFService.php:1046-1076; preview.php:472-494
HAP table: Triggering HAP Chemical Wt% in Formula:	REGULATORY LIST Ingredients whose exact CAS is on the HAP list at 0.01% or more, plus HAPs typed on raw materials. The amount is the Section 3 range band, not the exact %. <i>On a sheet: Toluene 10 - 30%</i> You maintain: HAP List (name filled in for every CAS). HAPs typed on the raw material when they cannot be matched by CAS (name and weight %). Ingredient percentages. If missing: Blank HAP-list name: empty name cell. Typed HAP with blank name or 0%: left out. No HAP at all: 'This product does not contain any EPA HAPs listed under Clean Air Act Section 112(b).' HAPService (hap_list exact CAS ≥0.01) + getManualHaps (raw_materials.haps_data); hap_name ?? chemical_name; bandRegulatoryEntries; sorted by exact % desc
Total HAP Content:	FORMULA / RAW MATERIALS Sum of the exact percentages of all HAP rows, printed to 2 decimals. <i>On a sheet: Total HAP Content: 12.35%</i> You maintain: Same as the HAP rows If missing: Only printed when there are HAPs. HAPService total_hap_pct (HAPService.php:147-153); labels.hap_total; PDFService.php:1069-1071; preview.php:486-490
EPA Significant New Use Rules (SNUR):	REGULATORY LIST Ingredients whose CAS is on the SNUR list, at any concentration. Name, CAS, rule citation and description all come from the SNUR list as typed (English). <i>On a sheet:</i> • Chemical X (CAS 123-45-6) — 40 CFR 721.1234 You maintain: SNUR List (Admin > SNUR List) with name, CAS, rule citation and description If missing: No match: no SNUR block at all. Blank name: the line starts '(CAS ...)'. Blank citation or description: that part is left out. SDSGenerator::analyseSnur (:4338-4389) on snur_list.cas_number; chemical_name, rule_citation, description; labels.snur_title; PDFService.php:1083-1099
California Proposition 65 (heading):	FIXED TEXT (TRANSLATED) Fixed heading, always printed. <i>On a sheet: California Proposition 65:</i> You maintain: Nothing If missing: English wording if the sheet language lacks it. labels.prop65_title (PDFService.php:1106-1145; preview.php:528,541)
Prop 65 warning symbol:	ADMIN SETTING Yellow warning triangle, shown only when a warning prints. Admin-uploaded PROP65 image > image shipped with the system. You maintain: Optional pictogram upload; otherwise the shipped image must exist If missing: No image file: the symbol is not printed; the warning text still prints. PictogramHelper getPngPath/getWebPath('PROP65'): uploads/pictograms/PROP65.(png jpg gif) > assets/pictograms/png/PROP65.png; alt labels.prop65_pictogram_alt

Prop 65 WARNING text:	REGULATORY LIST Cancer and reproductive chemicals found by matching ingredient CAS to the Prop 65 list, plus Prop 65 entries on raw materials, minus inhalation-only chemicals (e.g. carbon black) when the product is not all solid. The wording is the cancer, reproductive or combined template in the sheet language. <i>On a sheet: WARNING: This product can expose you to chemicals including Toluene, which is/are known...</i> You maintain: Prop 65 list (CAS, name, toxicity type in lower case). Prop 65 entries on raw materials (a listed CAS, or 'override' with name and types). The inhalation-only CAS setting. Physical state on every raw material. If missing: No cancer or reproductive chemical left: 'This product is not known to contain any chemicals listed under California Proposition 65.' A capitalised toxicity type ('Cancer'), or a pre-update name-only Prop 65 entry, is silently left out. <pre>Prop65Service::analyse (prop65_list ≥0.01%; raw_materials.prop65_data); removeInhalationOnlyFromResults (sds.inhalation_only_cas); rebuildProp65Warning (section15.prop65_warning_*)</pre>
Prop 65 '(trace)' marker:	ADMIN SETTING Added to a chemical name when every occurrence is below the admin trace threshold (CAS path) or ticked 'trace' on the raw material. <i>On a sheet: Benzene (trace)</i> You maintain: Admin Settings Prop 65 trace threshold; trace tick-boxes on raw-material Prop 65 entries If missing: Setting blank or 0: 0.1% is used. <pre>setting prop65.auto_trace_threshold_pct (default 0.1); prop65_data[.].is_trace; section15.prop65_trace_name (Prop65Service.php:71-86, 312-336)</pre>
Listed chemicals:	REGULATORY LIST One line per Prop 65 chemical with its listing types, printed under the warning. The types come from the Prop 65 list or the raw-material override. <i>On a sheet: • Toluene (CAS 108-88-3) — developmental toxicity, female reproductive toxicity</i> You maintain: Prop 65 list toxicity types spelled as cancer, developmental, female reproductive, male reproductive or reproductive If missing: No CAS: the line prints without '(CAS ...)'. A type not in that list: the line is dropped. No warning: no list. <pre>buildProp65ListedLines (SDSGenerator.php:2560-2617); labels.prop65_listed; section15.prop65_listed_line / _no_cas / prop65_type_*</pre>
State Regulations:	PRODUCT OVERRIDE Only from the note the operator types in the SDS editor for this product, separately for each language. Prints whenever present. <i>On a sheet: State Regulations: New Jersey Right-to-Know: Toluene (CAS 108-88-3)</i> You maintain: SDS editor > Section 15 > State Regulations, per product and per language If missing: Line omitted. Resale raw-material sheets can never carry one. <pre>trim(text_overrides(fg,15,state_regs,lang)) (SDSGenerator.php:3163); labels.state_regulations; PDFService.php:1151-1158; preview.php:543-551</pre>
Sections 12–15 footnote:	ADMIN SETTING Fixed translated note, switched on or off for all sheets by one admin checkbox (on by default). <i>On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...</i> You maintain: Admin Settings checkbox 'Print the Sections 12–15 footnote' If missing: Checkbox off: nothing prints. <pre>settings sds.show_ghs_section_note (missing = on, '0' = off) → document.ghs_section_note (SDSGenerator.php:3336-3346; PDFService.php:1189-1201; preview.php:607-609)</pre>

SECTION 16: OTHER INFORMATION

Section 16 is built almost entirely from publish metadata (version number and effective date written when a sheet is published), the per-language translation files (labels, draft text, abbreviation master table) and the Admin or manufacturer disclaimer text; nothing in it is typed per product any more.

Version:	PUBLISH RECORD The version number the system assigns when the sheet is published (next number for that product, alias, resale item or private-label item). Unpublished previews print the translated draft text. <i>On a sheet: Version: 3 (unpublished: Version: Draft (not yet published))</i> You maintain: Nothing to enter. Published SDS rows must never be hard-deleted (soft delete is fine; it still counts toward the next number). If missing: Preview/draft: 'Draft (not yet published)' (translated). Pre-update snapshot re-rendered without restamp: no Version line at all. <pre>sections[16].version: section16() = section16.draft; SDSGenerator::stampPublishedVersion() :758-777 sets (string)\$version + meta.sds_version; \$version = MAX(sds_versions.version per FG/RM/alias)+1 or...</pre>
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Effective Date:	PUBLISH RECORD The publish date written to the version record, printed as month/day/year in every language. <i>On a sheet: Effective Date: 10/09/2026</i> You maintain: Nothing If missing: Draft/preview: line not printed (value ""). Pre-update snapshot: line absent; the old 'Revision Date' line prints instead. sections[16].effective_date = format_date(\$effectiveDate,'m/d/Y') in stampPublishedVersion :764; publishers use date('Y-m-d'); scripts/publish-worker.php:76,:194 uses gmdate (UTC) for base/alias rows
Abbreviations:	FIXED TEXT (TRANSLATED) A master list of abbreviations per language, filtered to the terms that actually appear elsewhere on that sheet, printed in the list's file order. <i>On a sheet: Abbreviations: CAS = Chemical Abstracts Service registry number; GHS = ...; OSHA =</i> You maintain: Developers keep the four translation tables (en/es/fr/de section16.abbreviation_table) complete and alphabetical; any new abbreviation in generated text, disclaimers or overrides must be added to all four. If missing: No term matched: the line is not printed. Term missing from the table: it prints undefined (today IATA and IMDG on every sheet). AbbreviationService::build() (generate :250, generateFromBase :576, PL refilter :705-715); table = translations section16.abbreviation_table[lang] > EN table; PATTERNS :27-39; collectCorpus :107-213
DOCUMENT HEADER, FOOTER AND DISCLAIMER <i>Document chrome (the page-1 header, section banners, footer, disclaimer, PDF properties and file names) comes mainly from company or manufacturer settings, the per-language translation files and the version and date stamped at publish, with a small number of fixed layout constants.</i>	
Header logo (page 1) - standard, alias and resale SDS:	COMPANY / MANUFACTURER The company logo uploaded in Admin Settings. It prints on page 1 only. <i>On a sheet: Company logo top-left of page 1, about 51 mm wide</i> You maintain: Upload the company logo (PNG/JPG/GIF, 2 MB max) in Admin > Settings > Company Logo. The file must stay on the server. If missing: No logo prints. The title and rule still print. Re-renders of older snapshots show today's logo file, or none if the logo was removed or its file type changed. settings.company_logo_path ('/uploads/company-logo.<ext>', AdminController::processLogoUpload) > meta.company_logo_path > PDFService::resolveLogoPath > SDSTcpdf::Header
Header logo (page 1) - private label SDS:	COMPANY / MANUFACTURER The private label manufacturer's own logo. The company logo is never used instead. <i>On a sheet: Manufacturer's logo top-left of page 1</i> You maintain: Upload a logo on each private label manufacturer's form. If missing: No logo prints. The company logo is not substituted. manufacturers.logo_path ('/uploads/manufacturers/mfg-logo-<id>.<ext>') > createManufacturerVariant sets meta.company_logo_path (SDSGenerator.php:684-687)
SAFETY DATA SHEET (header title):	FIXED TEXT (TRANSLATED) Fixed title from the translation file for the sheet language. It prints bold and right-aligned on page 1. <i>On a sheet: SAFETY DATA SHEET / HOJA DE DATOS DE SEGURIDAD</i> You maintain: Nothing If missing: "SAFETY DATA SHEET" (the English default) translations document.title > meta.document.title > SDSDocumentStrings::resolve (English DEFAULTS) > SDSTcpdf::Header :88-95
Header rule; no header on pages 2+:	FIXED TEXT A navy rule sits under the page-1 header. Pages 2 onward have no header and the text starts near the top edge. You maintain: Nothing If missing: n/a SDSTcpdf::Header returns early when page > 1; PDFService MARGIN_TOP 30 / MARGIN_TOP_CONTINUATION 15 (:19-21, :296)
SECTION N: TITLE banners (all 16):	FIXED TEXT (TRANSLATED) The word 'SECTION' and each section title come from the translation file and print in capitals. Accented letters now capitalize correctly. <i>On a sheet: SECTION 1: IDENTIFICATION / SECCIÓN 1: IDENTIFICACIÓN</i> You maintain: Nothing If missing: "SECTION" plus the translation title meta.document.section_prefix + sections[N].title (translation sectionN.title); mb_strtoupper in PDFService::renderSection :314-324, preview.php :17/:43

Sections 12-15 footnote (after Section 15):	ADMIN SETTING One shared footnote in the sheet language, printed once after Section 15, if the admin toggle is on (default on). <i>On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...</i> You maintain: Admin > Settings: tick 'Print the Sections 12–15 footnote' to print it (on by default). If missing: Toggle off: nothing prints. <code>settings.sds.show_ghs_section_note ('0' = off, else on) gates translation document.ghs_section_note > sections[15].ghs_note; PDFService::renderGhsSectionNote :1189-1201</code>
DISCLAIMER (bar):	FIXED TEXT (TRANSLATED) A navy bar after Section 16, labelled from the translation file. It prints only when there is disclaimer text. <i>On a sheet: DISCLAIMER / DESCARGO DE RESPONSABILIDAD / HAFTUNGSAUSSCHLUSS</i> You maintain: Nothing If missing: Freshly generated sheets always print it. A re-render of a pre-#34 snapshot with no stored disclaimer prints no bar and no text. <code>meta.labels.disclaimer > translation labels.disclaimer (sheet lang, then EN) > key; PDFService::renderLegalDisclaimer :1297-1316; preview.php :615</code>
Disclaimer text - standard, alias and resale SDS:	ADMIN SETTING Admin per-language disclaimer > built-in translation text for that language. It is frozen into the snapshot at publish. <i>On a sheet: The information provided in this Safety Data Sheet is correct to the best of our knowledge...</i> You maintain: Optional. Admin > Settings > Legal / Disclaimer Statement, one box per language (EN was seeded by migration 052). If missing: The built-in translation paragraph for the sheet language (never blank on new generation). Re-renders of pre-#34 snapshots keep the old value, which may be blank or English. <code>settings.sds.legal_disclaimer.<lang> > translations section16.disclaimer; SDSGenerator::resolveLegalDisclaimer :3313-3320; migration 052 seeds .en only</code>
Disclaimer text - private label SDS:	COMPANY / MANUFACTURER Manufacturer's disclaimer for that language > admin per-language disclaimer > built-in translation text. <i>On a sheet: Manufacturer's own disclaimer paragraph in the sheet language</i> You maintain: Optional. A per-language disclaimer on the manufacturer form. If missing: The admin text for that language, else the built-in translation text. <code>manufacturers.disclaimer_json[lang] (blanks dropped) > settings.sds.legal_disclaimer.<lang> > section16.disclaimer; createManufacturerVariant :689-696</code>
Bold field labels throughout the sheet:	FIXED TEXT (TRANSLATED) Labels stored with the sheet > translation label in the sheet language > English translation > a capitalized key name. <i>On a sheet: Product Identifier: / Identificador del producto:</i> You maintain: Nothing If missing: The translation label (sheet language, then English) <code>meta.labels (SDSGenerator::getLabels :3460-3525) > labels.<key> via SDSDocumentStrings::translate > ucwords(key); PDFService::label :1240-1245</code>
Footer left: product code (every page):	FORMULA / RAW MATERIALS The code of the product the sheet is issued for, printed exactly as passed in. Finished-good alias sheets print the customer code including the pack extension. <i>On a sheet: UVNG009 / BK1080-5G</i> You maintain: The product code on the finished good, the customer code on each alias, the internal code on resale raw materials, and the custom code on private label items. If missing: The cell is empty. <code>meta.product_code: FG product_code; resale stripPack(internal_code); FG alias raw aliases.customer_code; PL custom_code > stripped alias code > FG code. SDSTcpdf::Footer :144-146</code>
Footer centre: Page X of Y:	FIXED TEXT (TRANSLATED) The words come from the translation file and the page numbers from the PDF engine. <i>On a sheet: Page 2 of 5 / Página 2 de 5 / Page 2 sur 5 / Seite 2 von 5</i> You maintain: Nothing If missing: "Page X of Y" in English <code>meta.document.page / page_of > SDSDocumentStrings DEFAULTS; SDSTcpdf::Footer :148-153</code>
Footer right: Rev. version and effective date (published):	PUBLISH RECORD The version number and publish date stamped automatically at publish, the same values as Section 16. The date always prints as m/d/Y. <i>On a sheet: Rev. 3 — 10/09/2026 / Rév. 3 — 10/09/2026</i> You maintain: Nothing. It is automatic at publish and depends on the server clock. If missing: Only the version prints: "Rev. 3". <code>PDFService::footerRevision :1210-1233: revision_prefix + meta.sds_version + (s16.effective_date > meta.effective_date > s16.revision_date); stampPublishedVersion :758-779</code>

Footer right: Draft (unpublished):	FIXED TEXT (TRANSLATED) Prints on previews and ad-hoc renders that have no version stamp. <i>On a sheet: Draft (not yet published) / Borrador (aún no publicado)</i> You maintain: Nothing If missing: n/a sections[16].version = translation section16.draft when meta.sds_version is 0/absent; footerRevision :1232
Footer right: legacy generation date (old snapshots):	PUBLISH RECORD Re-renders of snapshots made before #32 print the old generation date instead of a version. <i>On a sheet: Rev. 10/01/2026</i> You maintain: Nothing If missing: The Draft text footerRevision :1228-1230 uses sections[16].revision_date when meta.sds_version is absent
Footer rule and typography; page size and font:	FIXED TEXT Every sheet in every language uses a grey rule 15 mm from the bottom, 8 pt Helvetica in three equal cells, and US Letter portrait pages. You maintain: Nothing If missing: n/a SDSTcpdf::Footer :123-158; new SDSTcpdf('P','mm','LETTER') PDFService:258; core font helvetica; MARGIN_BOTTOM 20
PDF property: Title:	FIXED TEXT (TRANSLATED) A translated prefix plus the product code, identical to the footer code. <i>On a sheet: SDS - UVNG009 / HDS - UVNG009 / FDS - ... / SDB - ...</i> You maintain: Nothing beyond the product code If missing: English "SDS - <code>" (also on re-renders of non-English snapshots made before #37) meta.document.pdf_title with :code = meta.product_code; PDFService:267
PDF property: Author:	COMPANY / MANUFACTURER The supplier name printed in Section 1: the company name from Admin Settings, or the private label manufacturer's name. <i>On a sheet: Acme Inks, Inc.</i> You maintain: Company name in Admin > Settings, and the name on each manufacturer. If missing: "SDS System". If the company name was never saved, the config placeholder "Your Ink Company, Inc." is used. trim(sections[1].manufacturer_name): settings.company.name (config company.name only if the row was never stored) / manufacturers.name; PDFService:262-266
PDF property: Subject:	FIXED TEXT (TRANSLATED) Fixed text from the translation file. <i>On a sheet: Safety Data Sheet / Hoja de datos de seguridad</i> You maintain: Nothing If missing: English "Safety Data Sheet" meta.document.pdf_subject > DEFAULTS; PDFService:268
PDF properties: Creator, Producer, dates, XMP, language:	FIXED TEXT Fixed program names and the render time. An embedded XMP copy repeats Title, Author and Subject. No document language is declared. <i>On a sheet: Creator 'SDS System'; Producer 'TCPDF 6.11.3 (http://www.tcpdf.org)'</i> You maintain: Nothing If missing: n/a PDFService:261 SetCreator; TCPDF _putinfo :9601-9643, _putXMP :9836; no /Lang (setLanguageArray never called); no Keywords
Stored PDF file name (published):	PUBLISH RECORD The code without its pack extension, then the PL tag, the version and the language (default language has no language suffix). A clash adds _2, _3. <i>On a sheet: UVNG009_v3.pdf / UVNG009_v3_es.pdf / UVNG009_PL_Acme_Coatings_v2.pdf</i> You maintain: Product and alias codes, and the manufacturer name for private label. If missing: Empty code gives "unnamed_file". Blank manufacturer name gives the tag "PL_unnamed_file". PDFService::generate :101-124, versionedBaseName :159-168, reserveUniquePath :178-193; meta.filename_tag 'PL_'+sanitize(name[0..40]); sds.default_language
Stored PDF file name (unversioned render):	FIXED TEXT Code, an optional PL tag, the language and a timestamp, used when the sheet has no version. <i>On a sheet: UVNG009_SDS_en_20261009_143012.pdf</i> You maintain: Nothing If missing: n/a PDFService::generate :121-123 date('Ymd_His')
Download file names:	PUBLISH RECORD Built when a user downloads. The standard and private label names include the language. The public Lookup name does not. <i>On a sheet: SDS_UVNG009_v3_en.pdf / SDS_UVNG009_v3.pdf / PL_SDS_UVNG009_Acme_v2_en.pdf</i> You maintain: Nothing If missing: No code: "SDS_v3_en.pdf" / Lookup "SDS_SDS_v3.pdf" / preview "SDS_draft_en.pdf" SDSController::download :649-668; LookupController :84-95 (no lang); PrivateLabelController :913-919; SDSPreviewResponse::filename :56-68 (v{n} or draft, no PL tag)

Email attachment and report ZIP names:	PUBLISH RECORD Item code + '_SDS' (+ language when not English). These carry no version. The send queue uses the item name or code with no pack strip and no language suffix. <i>On a sheet: BK1080-5G_SDS.pdf / BK1080-5G_SDS_ES.pdf</i> You maintain: Nothing If missing: A language is skipped when no published PDF exists on disk. The report ZIP logs the item to _MISSING_ITEMS.csv. SDSAutoSendService :766-810; SDSSendQueueController::sendEmail :274-281 (EN only); ReportController ZIP :1024-1066
Alias PDFs rebuilt on the fly (send queue, report ZIP):	PUBLISH RECORD When an alias has no published PDF, the base snapshot is re-rendered with the alias code swapped in. The version, date, disclaimer and logo path come from the base snapshot. <i>On a sheet: Footer 'BK1080-5G' with the base 'Rev. 3 — 10/01/2026'</i> You maintain: Nothing If missing: n/a SDSSendQueueController :242-255 createAliasVariant(raw customer_code); ReportController::generateAliasPdf :1315-1362 (bypasses createAliasVariant; S1 identifier = code only)
Preview page: browser title, back links, status line, buttons:	FIXED TEXT English application text around the embedded real PDF. The PL back link names the manufacturer. <i>On a sheet: EN — Draft: this is the PDF Publish will produce...</i> You maintain: Nothing If missing: Links and buttons that do not apply are omitted. layouts/main.php:7; SDSController :97-102/:169-177; PrivateLabelController :298-307/:1005-1017; preview-pdf.php :16-61; SDSPreviewResponse :28-49
Preview page: Warnings box:	FORMULA / RAW MATERIALS Data-quality warnings from the generator. The finished-good preview adds the company emergency phone and transport checks. The PL stored preview shows the warnings frozen at publish. They never print on the PDF. <i>On a sheet: Warnings: TSCA status not verified for 1 component</i> You maintain: The underlying data: emergency phones, TSCA list, transport data, raw material data. If missing: The box is not shown. sds.warnings: FormulaCalcService + UV pack + TSCAService::warningText + S14 tech-names + PL mfg phone; FG preview adds SDSReadinessService (SDSController :79-89)

CALCULATED VALUES AND THEIR INPUTS

In this scope, every printed number, range and derived classification is calculated when the sheet is generated. The inputs are the formula lines and each raw material's own data (constituent percentages, flash and boiling points, specific gravity, VOC, solubility, physical state), passed through the hazard engine, the transport classifier and the admin-maintained regulatory lists (RCRA, SARA 313, HAP, carcinogens), with per-product text overrides winning wherever they exist.

Carbon black / titanium dioxide cancer hazard (Sections 2, 3, 11 and 15 Prop 65):	FORMULA / RAW MATERIALS The cancer hazard, carcinogen listing and Prop 65 warning for an inhalation-only substance are kept only when it is the sole ingredient or every other raw material in the formula is marked Solid or Powder. Otherwise they are suppressed. <i>On a sheet: H351 and the Prop 65 warning kept only when every other ingredient is a solid or powder</i> You maintain: Physical State on every raw material (Raw Materials form); the inhalation-only CAS list (Admin > Settings, default carbon black 1333-86-4 and TiO2 13463-67-7) If missing: A raw material with a blank Physical State counts as not a powder, so the hazard is suppressed SDSGenerator::applyCarbonBlackLogic / applySolidPowderLiquidFiltering :3546-3560, :3785-3810; raw_materials.physical_state per line; setting sds.inhalation_only_cas :3593
Section 3 Type (Substance / Mixture):	FORMULA / RAW MATERIALS Substance/Mixture setting on the finished good > on Auto, 'Substance' only for a single-ingredient formula whose raw material is marked Substance, otherwise 'Mixture'. Resale sheets use the raw material's own setting. <i>On a sheet: Mixture</i> You maintain: Substance/Mixture on the finished good, or on the raw material for single-ingredient and resale products If missing: 'Mixture' SubstanceMixtureResolver; finished_goods.substance_mixture > raw_materials.substance_mixture (single line); SDSGenerator :105-108, :370, :1111-1114; PDFService :730

Section 3 component rows (which ingredients are listed):	FIXED TEXT A constituent is listed when it is at or above 0.1%, is not flagged non-hazardous, and either is hazardous per the engine or has an exposure limit. Inhalation-only substances with no remaining hazard are dropped, and trade-secret rows are merged by their description. <i>On a sheet: Toluene 108-88-3 10 - 30% H225, H304, H315</i> You maintain: Non-hazardous flag, classifications and trade-secret description on each constituent (Raw Materials > Constituents; CAS determinations) If missing: Rows below 0.1% are left out, even when several trade-secret rows together reach 0.1% or more SDSGenerator::section3 :1026-1101; raw_material_constituents.is_non_hazardous; engine hazardous_cas / exposure_limits; sds.inhalation_only_cas; 0.1 cut-off per row
Section 3 Concentration column:	FORMULA / RAW MATERIALS Each constituent's percentage times its formula line percentage is summed per CAS and placed in the widest prescribed HazCom range that contains it. Ranged constituents use their min-max and exact ones a single point. <i>On a sheet: 10 - 30%</i> You maintain: Constituent % (exact, or min and max) on each raw material (Raw Materials > Constituents) and the line % on each formula If missing: A constituent with no % counts as 0 and is dropped. A value no range fits prints '<0.1%' (min below 0.1) or '80 - 100%' SDSGenerator::formatConcentration :3408-3453 on Formula::getExpandedComposition concentration_min/max else concentration_pct; raw_material_constituents.pct_exact/pct_min/pct_max
Section 5 Specific Hazards (flammable sentence and 'Flash point:' line); flammable media and firefighter advice:	HAZARD ENGINE Per-product override > a sentence chosen by the engine's H224-H227 code, with the Section 9 flash point appended. The extinguishing media and firefighter advice use the same H-code test. <i>On a sheet: Highly flammable liquid and vapor (Flammable Liquids, Category 2)... Flash point: 12 °C (53.6 °F).</i> You maintain: Flammable Liquid classifications on constituents (CAS determinations); flash points on raw materials If missing: No flammable sentence and no flash point line when the engine gives no H224-H227, even if the flash point is below 60 °C SDSGenerator::section5 :1368-1466; text_overrides(5,'specific_hazards'); section5.specific_hazards_flammable_cat1-4, section5.flash_point_line; resolveFlashPointDisplay
Section 6 Methods and Material for Containment and Cleaning Up:	FORMULA / RAW MATERIALS Per-product override > solid, paste or liquid wording chosen from the resolved physical state (finished good > main raw material > Liquid). <i>On a sheet: Absorb with inert material and place in a closed container for disposal.</i> You maintain: Physical State on the finished good, or on its highest-weight raw material If missing: Liquid wording SDSGenerator :1538-1549; text_overrides(6,'containment') > section6.containment_solid _paste _liquid via resolvePhysicalState
Section 8 Engineering Controls (dust sentence):	FORMULA / RAW MATERIALS A dust-control sentence is added when the resolved physical state is Solid or Powder. <i>On a sheet: Adds the dust-control sentence for solids and powders</i> You maintain: Physical State on the finished good, or on its highest-weight raw material If missing: No dust sentence (the state defaults to Liquid) SDSGenerator::engineeringControls :1894-1911; section8.engineering_dust; resolvePhysicalState
Section 8 exposure-limit table, Conc. % column:	FORMULA / RAW MATERIALS The same prescribed range as the CAS row in Section 3 > the limit's own concentration when the CAS is not in the composition. <i>On a sheet: 10 - 30%</i> You maintain: As Section 3 (constituent %) If missing: '<0.1%' for components between 0.01% and 0.1% SDSGenerator :1832-1848 formatConcentration(compositionByCas[cas]) else exposure_limits[].concentration_pct; PDFService :806; preview :289
Physical State:	FORMULA / RAW MATERIALS Finished good's Physical State > physical state of the raw material with the highest weight % in the formula > 'Liquid'. The same value also drives Sections 6, 8 and 14. <i>On a sheet: Liquid</i> You maintain: Physical State on the finished good (Finished Goods form) or on its highest-weight raw material (Raw Materials form) If missing: 'Liquid', silently. A blank on the main raw material does not fall through to the next one SDSGenerator::resolvePhysicalState :1954-1963; finished_goods.physical_state > FormulaCalcService formula_props.physical_state :498-545 > 'Liquid'

Color:	FORMULA / RAW MATERIALS The finished good's Color dropdown only. Resale raw-material sheets never show a color. <i>On a sheet: Black</i> You maintain: Color on the finished good (Finished Goods form) If missing: The line is omitted finished_goods.color (options from setting sds.color_options); SDSGenerator :2036-2038, :2111; resale :381 reads raw_materials.color, which does not exist
Appearance:	FORMULA / RAW MATERIALS Per-product override > color plus state when the finished good has a color > Appearance text of the highest-weight raw material > the physical state in lower case. <i>On a sheet: Black liquid</i> You maintain: Color on the finished good, or Appearance on its highest-weight raw material If missing: The lower-case physical state, e.g. 'liquid' SDSGenerator :2066-2091; text_overrides(9,'appearance') > section9.appearance_color_state > formula_props.appearance > section9.appearance_state
Odor:	FORMULA / RAW MATERIALS Per-product override > Odor of the raw material with the highest weight % > 'Not determined'. <i>On a sheet: Mild solvent</i> You maintain: Odor on the highest-weight raw material of each formula (Raw Materials form) If missing: 'Not determined' SDSGenerator :2093-2103; text_overrides(9,'odor') > formula_props.odor (raw_materials.odor of dominant RM) > labels.not_determined
Initial Boiling Point and Boiling Range:	FORMULA / RAW MATERIALS Per-product override > the lowest boiling point of any raw material in the formula (including sub-formulas), whatever its amount > 'Not determined'. <i>On a sheet: 78.4 °C (173.1 °F)</i> You maintain: Boiling Point (°C) on each volatile raw material (Raw Materials form) If missing: 'Not determined' SDSGenerator::resolveBoilingPointDisplay :1343-1366; text_overrides(9,'boiling_point') > MIN(raw_materials.boiling_point_c) FormulaCalcService :476-486
Flash Point:	FORMULA / RAW MATERIALS Per-product override > the lowest flash point of any raw material in the formula (including sub-formulas), with '>' when that raw's value is a 'greater than' > 'Not determined'. The same value feeds Sections 5, 13 and 14. <i>On a sheet: > 93 °C (199.4 °F)</i> You maintain: Flash Point (°C) and the '>' flag on every raw material (Raw Materials form) If missing: 'Not determined' SDSGenerator::resolveFlashPointDisplay :1310-1334; text_overrides(9,'flash_point') > MIN(raw_materials.flash_point_c) + flash_point_greater_than; FormulaCalc :466-474
Solubility:	FORMULA / RAW MATERIALS Per-product override > a band from the weight share of raw materials marked soluble (full weight) or partially soluble (half weight), among the raws that have any value: 90% or more soluble, 5% partial, 1% negligible, else not soluble. <i>On a sheet: Partially soluble in water</i> You maintain: Solubility on each raw material (Raw Materials form). Water, glycols and other water-soluble raws must read 'Soluble in water' If missing: 'Not determined' when no raw material has a value FormulaCalcService SOLUBILITY_WEIGHTS :28-38, :488-519, solubilityKeyForFraction :555-567; text_overrides(9,'solubility'); section9.solubility_*
Specific Gravity:	FORMULA / RAW MATERIALS A volume-weighted average of each raw material's specific gravity, rounded to 3 decimals. <i>On a sheet: 1.053</i> You maintain: Specific Gravity on every raw material (Raw Materials form). A lb/gal density is converted only when it is typed on the form If missing: Each raw missing a value counts as 1.000. The line never prints 'Not determined' VOC Calculator::getMixtureSG :236-267, applyDefaults :582-592; raw_materials.specific_gravity else 1.0; SDSGenerator :2117
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS VOC weight fraction (each raw's VOC % times its line %) times mixture specific gravity times 8.345. Exempt compounds and water are not subtracted. <i>On a sheet: 2.15</i> You maintain: VOC wt% (or the 'less than 1%' flag) and Specific Gravity on every raw material If missing: A missing VOC counts as 0 and a missing SG as 1.0, with no marker VOC Calculator::getVOC_LbPerGal :277-294, getTotalVOCWeightPercent :181-197; VOC_i = 0.99 if voc_less_than_one else voc_wt else 0; SDSGenerator :2118

VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw is flagged less than 1% VOC > otherwise the sum of each raw's VOC % times its line %. <i>On a sheet: 25.71 (or '<1')</i> You maintain: VOC wt% or the 'less than 1%' flag on every raw material If missing: A missing VOC counts as 0 SDSGenerator :2052-2056, :2120; FormulaCalcService all_voc_less_than_one :452-464; VOCCalculator :181-197
Solids (wt%):	FORMULA / RAW MATERIALS The sum of each raw's solids % times its line %. A raw with no solids % is estimated as 100 minus VOC, exempt VOC and water. <i>On a sheet: 74.3</i> You maintain: Solids wt% (or VOC, exempt VOC and water wt%) on each raw material; the Exempt VOC list (admin) If missing: Estimated. A raw with no VOC or solids data counts as 100% solids VOCcalculator :378-387, :643-694; raw_materials.solids_wt else 100-voc-exempt-water; exempt_voc_list table (FormulaCalcService :383-400)
Section 11 component heading (name, CAS, range):	FORMULA / RAW MATERIALS Printed for each component at or above 0.1% that has an exposure limit or a carcinogen listing, with the same range as Section 3. Trade-secret identities are not masked here. <i>On a sheet: Toluene (CAS 108-88-3) — 10 - 30%</i> You maintain: As Section 3; exposure limits and the carcinogen list If missing: No band on legacy snapshots SDSGenerator::section11 :2625-2662; formatConcentration(compositionByCas); requires engine exposure_limits or carcinogen_list hit; PDFService :883-889
Acute Toxicity (oral, dermal, inhalation lines):	HAZARD ENGINE Per-product override > for each route, the most severe Category 1-4 the hazard engine assigned, with its H-statement. <i>On a sheet: Acute toxicity (oral): Category 4 — Harmful if swallowed (H302).</i> You maintain: Acute toxicity categories per CAS (CAS determinations / CPD) If missing: '<Route>: Not classified based on available data.' SDSGenerator::buildAcuteToxicity :2353-2484, :2697; text_overrides(11,'acute_toxicity'); section11.acute_route_line; h_statements
ATEmix value (appended to each acute toxicity line):	HAZARD ENGINE Calculated with the GHS additivity formula from each acute-toxic component's LD50/LC50, or a default value for its category when none is entered. Printed only when the result matches the printed category. <i>On a sheet: ATEmix = 1255 mg/kg bw.</i> You maintain: Acute toxicity categories per CAS, and optionally the oral/dermal/inhalation LD50/LC50 values (CAS determinations) If missing: The category default ATE is used silently. No value is printed when the calculated category differs from the printed one HazardEngine::applyATECalculation :1939-2110; hazard_classifications.ate*_mg_kg / ate_inhalation_vapor_mg_l_4h else CATEGORY_DEFAULT_ATES; formatAte :2491
Section 11 Carcinogenicity:	FORMULA / RAW MATERIALS Per-product override > one line per component of at least 0.1% found on the carcinogen list, with its range > the 'none listed' sentence. <i>On a sheet: No components present at or above 0.1% are listed as carcinogens by IARC, NTP or OSHA.</i> You maintain: Carcinogen list (admin); constituent %; raw material physical state (carbon black / TiO2 suppression) If missing: The 'none listed at or above 0.1%' sentence text_overrides(11,'carcinogenicity') > CarcinogenService::buildSummaryText (carcinogen_list, LISTING_THRESHOLD_PCT 0.1) > section11.carcinogenicity
Section 12 component aquatic table (Conc. %, Aquatic Acute, Aquatic Chronic with M-factor):	HAZARD ENGINE Every component of at least 0.01% with an aquatic classification, its Section 3 range, and its M-factor for Category 1. Trade-secret identities are masked. <i>On a sheet: Zinc oxide 1314-13-2 1 - 5% Category 1 (M = 10) Category 1 (M = 1)</i> You maintain: Aquatic categories and M-factors per CAS (CAS determinations / CPD) If missing: '—' for no class. A missing M-factor prints '(M = 1)' HazardEngine aquatic_components :1287-1325, resolveMFactor :1229-1239; hazard_classifications.m_factor_acute/chronic else 1.0; SDSGenerator :2722-2752
Persistence and Degradability / Bioaccumulative Potential:	REGULATORY LIST Per-product override > a line naming every SARA 313 PBT chemical at or above 0.1% > 'No data available'. Bioaccumulation refers back to that line. <i>On a sheet: Contains a PBT chemical on the EPA TRI list: Lead</i> You maintain: SARA 313 list with the PBT flag (admin) If missing: The 'No data available' sentences SDSGenerator :2798-2824; text_overrides(12,'persistence' 'bioaccumulation') > section12.pbt_components (sara313_list.is_pbt) > section12.persistence

Section 13 RCRA ignitable (D001):	FORMULA / RAW MATERIALS Flash point from the Section 9 override (the number in it) > the formula flash point. D001 when below 60 °C, or H224-H226, plus oxidizer and pyrophoric reasons from H-codes. <i>On a sheet: ignitable (D001) — flash point below 60 °C (140 °F)</i> You maintain: Raw material flash points; flammable and oxidizer classifications If missing: No D001 from the flash point when none is known SDSGenerator::section13 :2893-2913, resolveFlashPointNumeric :3037-3058; text_overrides(9,'flash_point') per language > formula_props.flash_point_c
Section 13 RCRA corrosive (D002) / reactive (D003):	HAZARD ENGINE D002 when the engine gives H314. D003 from water-reactive, explosive or organic-peroxide H-codes. <i>On a sheet: corrosive (D002) — based on its skin corrosion classification (H314), pH not determined</i> You maintain: Constituent classifications (CAS determinations) If missing: Omitted SDSGenerator :2915-2932; H314 -> D002; H260/H261, H200-H205, H240-H242 -> D003
Section 13 RCRA toxicity-characteristic and listed-waste components:	REGULATORY LIST Composition CAS numbers matched to the admin RCRA code list. D codes match at any amount; F/K/P/U codes at 0.1% or more. <i>On a sheet: contains Lead (D008 if the TCLP regulatory level of 5 mg/L is exceeded)</i> You maintain: RCRA waste code list (Admin > RCRA) If missing: 'No RCRA hazardous waste characteristic...' sentence RCRAService :42-158 (rcra_waste_codes, migration 055); SDSGenerator :2934-2992, :2999-3021; section13.rcra_code_d / _d_nolimit
UN Number / UN Proper Shipping Name:	TRANSPORT RULES Per-field override > classifier from flash point (below 60 °C means Class 3), H-codes, physical state (solid entries) and product type (transport type, else keywords in the description). n.o.s. names add the top two hazardous constituents. <i>On a sheet: UN1210 / Printing ink, flammable; or 'Flammable liquids, n.o.s. (Toluene, Xylene)'</i> You maintain: Raw material flash points; constituent classifications; Transport Product Type or a clear description on the finished good; physical state If missing: 'Not regulated' when no class applies and any flash point or H-code exists. 'Not determined' (publishing blocked) when neither exists SDSGenerator::section14 :3073-3154; text_overrides(14,'un_number' 'proper_shipping_name') > TransportClassifier::classify :108-303; resolveProductType :310-336
Transport Hazard Class(es):	TRANSPORT RULES Per-product override > primary class plus subsidiaries ranked by the 49 CFR 173.2a precedence table. <i>On a sheet: 3 (6.1, 8)</i> You maintain: As UN Number If missing: 'Not regulated' / 'Not determined' SDSGenerator :3140; text_overrides(14,'hazard_class') > TransportClassifier :195-280
Packing Group:	TRANSPORT RULES Per-product override > most stringent of: Class 3 from boiling point (35 °C or below is I) and flash point (below 23 °C is II, else III); Class 8 and 6.1 from the classification sub-category; Class 9 III. <i>On a sheet: II</i> You maintain: Raw material boiling and flash points; constituent classifications If missing: 'Not applicable' (not regulated) / 'Not determined'. Class 8 and 6.1 default to II SDSGenerator :3141; TransportClassifier :170-193, corrosivePackingGroup :373, toxicPackingGroup :391; formula_props.boiling_point_c (S9 override not read)
Environmental Hazards (Marine pollutant):	TRANSPORT RULES Per-product override > 'Yes' when the engine gives H400, H410 or H411 on a regulated product, otherwise 'No'. <i>On a sheet: Marine pollutant: Yes</i> You maintain: Aquatic classifications per CAS If missing: 'Marine pollutant: No'; 'Not determined' when the classification is not determined SDSGenerator :3102-3116, :3142; text_overrides(14,'environmental_hazards') > section14.marine_pollutant_yes _no; labels.not_determined
Transport notes:	TRANSPORT RULES A combustible note when not regulated and the flash point is 60-93 °C, a viscous-liquid note for Class 3 PG II, then a fixed carrier-verification sentence. There is no override. <i>On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only. ...verify with your carrier.</i> You maintain: Raw material flash points If missing: Only the fixed carrier sentence SDSGenerator :3118-3123; TransportClassifier :124, :159-161, :299-301; section14.note_combustible, note_viscous, section14.note

SARA 313 reportable components:

REGULATORY LIST Composition CAS numbers on the SARA 313 list at or above their de minimis (or PBT) threshold, printed with the EPA list name, the Section 3 range and the threshold.
On a sheet: • Ethylene glycol (CAS 107-21-1) — 1 - 5% (de minimis threshold: 1%)
You maintain: SARA 313 list (admin); constituent %
If missing: The 'no SARA 313 chemicals' sentence
SARA313Service::analyse :37-88 (sara313_list de minimis_pct / pbt_threshold_pct, is_pbt); SDSGenerator :3176, bandRegulatoryEntries :2533; PDFService :1017-1040

SARA 313 range note:

FIXED TEXT (TRANSLATED) A fixed sentence printed under the list whenever at least one SARA 313 chemical is reportable.
On a sheet: Concentrations are stated as the prescribed ranges...; the upper end is the maximum present.
You maintain: Nothing
If missing: Not printed when nothing is reportable
labels.sara_313_range_note (en.php:671); PDFService :1035-1037

Hazardous Air Pollutants (rows and Total HAP Content):

REGULATORY LIST Composition CAS numbers on the HAP list (0.01% and up) plus HAPs entered by hand on raw materials, each printed with a range. The total is the exact sum.
On a sheet: Toluene | 10 - 30% ... Total HAP Content: 12.35%
You maintain: HAP list (admin); manual HAP entries on raw materials
If missing: The 'no HAPs' sentence
HAPService :79-148 (hap_list) + SDSGenerator::getManualHaps :4281-4330 (raw_materials.haps_data); rows banded; total number_format 2

PRODUCT FAMILIES AND UV PRODUCTS

In this scope, the text comes from the product family resolved for each item: its UV/LED flag, its per-language default text and its name. It is combined with the acrylate detection in the UV Acrylate Rule Pack, the admin on/off switch, per-product settings and overrides, and fixed translation sentences.

Recommended Use (Section 1, finished good / intermediate / alias / private-label sheet):

PRODUCT OVERRIDE Per-product, per-language text override > the product's own Recommended Use field > the family's default text in the sheet language > the family's English text > canned translation sentence. The label 'Recommended Use' comes from the translation file.
On a sheet: Recommended Use: UV-curable offset ink for packaging.
You maintain: Family default text in every language (Settings > Product Families). Keep the product's Recommended Use field blank, because it has no language and prints as typed on EN/ES/FR/DE. Use the SDS editor override for per-language product wording. The product must resolve to a family.
If missing: "Printing ink for commercial and industrial applications." (ES/FR/DE translated). If the family has English text but none in the sheet language, the English family text prints on the foreign sheet.
SDSGenerator::resolveUseText: text_overrides(1, 'recommended_use', fg, lang) > finished_goods.recommended_use > product_families.recommended_use_json[lang] > ['en'] > section1.recommended_use

Recommended Use (Section 1, resale raw-material sheet):

PRODUCT FAMILY Family default text in the sheet language > family English text > canned resale sentence. Resale sheets have no per-product override and no product column.
On a sheet: Recommended Use: Raw material for industrial formulation.
You maintain: The raw material's family (rule or manual pick on the raw-material form) and that family's text. A raw placed in a product family prints that family's product text.
If missing: "Raw material for industrial formulation." (ES 'Materia prima para formulación industrial.', FR/DE translated)
resolveUseText on synthetic resale row (recommended_use=null, id=null → text_overrides fg 0): product_families.recommended_use_json[lang] > ['en'] > section1.recommended_use_resale

Restrictions on Use (Section 1, finished good / intermediate / alias / private-label sheet):

PRODUCT OVERRIDE Per-product, per-language override > the product's own Restrictions field > family restriction text in the sheet language > family English text > canned translation sentence.
On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.
You maintain: Family restriction text in every language (Settings > Product Families). Keep the product's Restrictions field blank: it has no language and masks the family text.
If missing: "For professional/industrial use only. Not for household consumer use." (translated per language)
resolveUseText(field='restrictions'): text_overrides(1, 'restrictions') > finished_goods.restrictions_on_use > product_families.restrictions_json[lang] > ['en'] > section1.restrictions

Restrictions on Use (Section 1, resale raw-material sheet):

PRODUCT FAMILY Family restriction text in the sheet language > family English text > canned resale restriction sentence.
On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.
You maintain: Raw material's family and that family's restriction text.
If missing: "For professional/industrial use only. Not for household consumer use." (resale key, same English wording)
product_families.restrictions_json[lang] > ['en'] > section1.restrictions_resale (SDSGenerator.php:855-862, resale row :369-391)

PPE Recommendations: Respiratory / Hand / Eye / Skin (Section 2, on UV acrylate products):	TEXT CHOSEN BY HAZARD The same hazard-tier sentence Section 8 starts with (or the Section 8 override), printed only for hazard-driven tiers. The UV supplement sentences that Section 8 adds are NOT included, so on UV sheets the Section 2 text is shorter than Section 8's. <i>On a sheet: Hand Protection: Chemical-resistant gloves (nitrile recommended) meeting 29 CFR 1910.138...</i> You maintain: Nothing in this scope (H-code driven) If missing: Unclassified product: no PPE lines in Section 2 (Section 8 still prints the baseline plus UV sentences) <code>SDSGenerator::section2 → resolvePPE (text_overrides(8,field) > section8.ppe.<field>.<tier>); UV append only in section8() :1856-1866</code>
Type (Section 3: Substance / Mixture):	FORMULA / RAW MATERIALS Product's Substance/Mixture setting > Auto: 'Substance' only if the formula has exactly one line and it is a raw material marked Substance > 'Mixture'. Resale sheets use the raw material's setting. The product family has no influence. <i>On a sheet: Type: Mixture</i> You maintain: Mark single-substance raw materials as Substance (raw-material form). Set the finished-good setting only when the automatic rule is wrong. If missing: "Mixture" <code>SubstanceMixtureResolver; finished_goods.substance_mixture > single raw_materials line with substance_mixture='substance' > mixture; labels.type / labels.substance / labels.mixture</code>
UV Acrylate Information (Section 4, printed last):	PRODUCT FAMILY Prints only when the UV rule pack is switched on, the product's family is flagged UV/LED, and an acrylate is found among the constituents (14 listed CAS numbers or names containing acrylate/acrylic). The sensitizer wording is used when the mixture carries H317, the unclassified wording otherwise; component names come from the composition. <i>On a sheet: UV Acrylate Information: UV/EB curable product containing acrylate monomers/oligomers (...)</i> You maintain: UV/LED flag on the right families. Raws resolved into those families with enough weight % to win, or a family pick on the product. Pack enabled in Admin Settings. Correct chemical names / CAS for acrylate constituents (raw-material constituents, CAS master). Trade-secret descriptions on secret acrylates. If missing: Paragraph and label omitted <code>UVAcrylateRulePack::isApplicable (settings.uv_acrylate_rule_pack, product_families.is_uv) + detectAcrylates; section4.uv_acrylate_note / _unclassified; names COALESCE(prop65_list, cas_master.preferre...</code>
UV Acrylate Information (Section 5):	PRODUCT FAMILY Same three-way gate as Section 4. Fixed translated sentence when it fires. <i>On a sheet: UV Acrylate Information: UV-curable formulations may generate acrid smoke if involved in a fire...</i> You maintain: As Section 4 UV paragraph If missing: Omitted <code>UVAcrylateRulePack::getSafeHandlingLanguage → section5.uv_acrylate_note; rendered via PDFService FIELD_LABEL_MAP/renderGenericSection, preview genericLabelMap</code>
UV Acrylate Information (Section 6):	PRODUCT FAMILY Same gate as Section 4. Fixed translated sentence. <i>On a sheet: UV Acrylate Information: Avoid release to drains. Uncured acrylate monomers must not enter...</i> You maintain: As Section 4 UV paragraph If missing: Omitted <code>section6.uv_acrylate_note (UVAcrylateRulePack.php:195; en.php:112)</code>
UV Acrylate Information (Section 7):	PRODUCT FAMILY Same gate as Section 4. Fixed translated sentence. The Handling and Storage fields themselves do not depend on the family. <i>On a sheet: UV Acrylate Information: Store away from UV light sources, direct sunlight and heat...</i> You maintain: As Section 4 UV paragraph If missing: Omitted <code>section7.uv_acrylate_note (UVAcrylateRulePack.php:196; en.php:156)</code>
Respiratory Protection (Section 8, UV supplement sentence):	PRODUCT FAMILY The hazard-tier sentence, followed by the UV respirator sentence when the Section 4 gate fires. A per-product Section 8 override replaces the whole field, and nothing is appended to it. <i>On a sheet: ... Use local exhaust ventilation when handling uncured product; if mist or vapor is generated...</i> You maintain: As Section 4 UV paragraph If missing: Only the hazard-tier sentence prints <code>SDSGenerator::section8(\$uvPack = uvAcrylates !== []) appends section8.uv_respiratory unless text_overrides(8,'respiratory') set (:1856-1866)</code>
Hand Protection (Section 8, UV supplement sentence):	PRODUCT FAMILY Hazard-tier sentence + UV glove sentence when the gate fires; override replaces the field. <i>On a sheet: ... For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm...)</i> You maintain: As Section 4 UV paragraph If missing: Only the hazard-tier sentence prints <code>section8.uv_hand_protection appended in section8() (:1860-1866)</code>

Eye Protection (Section 8, UV supplement sentence):	PRODUCT FAMILY Hazard-tier sentence + UV eye sentence when the gate fires; override replaces the field. <i>On a sheet: ... Chemical splash goggles or a face shield are required when handling uncured acrylate product.</i> You maintain: As Section 4 UV paragraph If missing: Only the hazard-tier sentence prints section8.uv_eye_protection appended in section8() (:1860-1866)
Skin Protection (Section 8, UV supplement sentence):	PRODUCT FAMILY Hazard-tier sentence + UV skin sentence when the gate fires; override replaces the field. <i>On a sheet: ... Avoid prolonged or repeated skin contact with uncured product. A barrier cream is recommended...</i> You maintain: As Section 4 UV paragraph If missing: Only the hazard-tier sentence prints section8.uv_skin_protection appended in section8() (:1860-1866)
Conditions to Avoid (Section 10, UV light item):	PRODUCT FAMILY Added whenever the product's family is flagged UV/LED. It does not depend on the rule-pack switch or on finding an acrylate. A per-product override of Conditions to Avoid replaces the whole list. <i>On a sheet: Conditions to Avoid: Excessive heat, ..., exposure to UV light and direct sunlight (...)</i> You maintain: UV/LED flag on UV families (Settings > Product Families); product resolved to that family If missing: UV item omitted; rest of list (heat / ignition / oxidizers ...) prints section10(\$isUv = UVAcrylateRulePack::familyIsUv) adds 'uv' → section10.cond_uv in COND_ORDER; skipped if text_overrides(10, 'conditions_avoid')
UV Acrylate Information (Section 11, after the component table):	PRODUCT FAMILY Same gate as Section 4. Sensitizer wording when H317 is present, unclassified wording otherwise. <i>On a sheet: UV Acrylate Information: Contains acrylate monomers that are known skin sensitizers...</i> You maintain: As Section 4 UV paragraph If missing: Omitted section11.uv_acrylate_note / _unclassified; PDFService.php:922-927, preview.php:390-392; label labels.uv_acrylate_note
UV Acrylate Information (Sections 4-7 and 11 on re-rendered pre-batch-C snapshots):	FIXED TEXT Alias auto-send, the report alias PDF and the private-label view re-render a stored base snapshot. Snapshots published before batch C may hold an English-only UV note (old family-name test, no H317 variant), which the renderers now print. <i>On a sheet: UV Acrylate Information: (English text from the old snapshot)</i> You maintain: Republish base products that have old snapshots If missing: Nothing prints (most legacy snapshots never carried the key) snapshot sections[4-7,11].uv_acrylate_note; SDSAutoSendService.php:205-211, ReportController.php:1318-1340, PrivateLabelController.php:958-962
UN Number / Proper Shipping Name (Section 14, ink / paint / n.o.s. choice):	TRANSPORT RULES The product's Transport Product Type setting > keyword search in the description plus the family name (related > paint > wash/cleaner > ink). This is the only place the family name, which is never printed, affects the sheet. Resale sheets have no setting and no override. <i>On a sheet: UN Number: UN1210 / Proper Shipping Name: Printing ink, flammable</i> You maintain: Transport Product Type on finished goods when the guess is wrong. Family names and descriptions that do not contain keywords by accident (plain substring match). Supplier product name on resale raw materials. If missing: No family and no flag: description keywords only, default ink → "UN1210" "Printing ink, flammable" TransportClassifier::resolveProductType(finished_goods.transport_product_type, description, fg.family=product_families.name); str_contains on UPPER text; section14.psn_*
Abbreviations (Section 16, UV-related entries):	FIXED TEXT (TRANSLATED) An entry from the master abbreviation table prints when its term appears in printed text, including the UV notes, UV PPE sentences, the Section 10 UV item and family Section 1 text. The family name is no longer printed, so it no longer adds entries. <i>On a sheet: UV: Ultraviolet (energy-curable); EB: Electron Beam (energy-curable); SCBA: ...</i> You maintain: Nothing If missing: Entry omitted AbbreviationService::build over printed corpus; section16.abbreviation_table (en.php:503 'UV' => 'Ultraviolet (energy-curable)', EB, LED, SCBA, NIOSH, CFR)

DATA YOU MUST KEEP POPULATED

Grouped by the screen where the information is entered. Blank entries do not stop most sheets from publishing; they change what prints, as shown in the last column.

Where	Keep populated	Feeds	If left blank
Raw materials	CAS number and percentage (exact, or min and max) for every constituent. Use the same style for a CAS across raw materials. Tick non-hazardous ingredients.	Nearly everything: Sections 2, 3, 4–8, 10, 11, 12, 13 and 15, plus Section 9 Solids and UV acrylate detection. The non-hazardous flag hides an ingredient from Section 3.	The constituent is silently dropped and hazards are under-reported. Generic text prints, e.g. Section 10 'Stable'. No check catches it. Mixed styles understate SARA 313 and HAP bands.
Raw materials	Flash point (°C) and its 'greater than' flag on every raw material that has one, especially solvents. Boiling point on volatile raw materials.	Section 9 Flash Point and Initial Boiling Point; Section 5 Specific Hazards; Section 13 D001; Section 14 UN number, shipping name, class and packing group.	Section 9 prints "Not determined". With no flash point and no hazard statements, Section 14 prints "Not determined" and publishing is blocked. With any hazard statement, it wrongly prints "Not regulated".
Raw materials	Manual hazard classification on hazardous raw materials that have no CAS, and H-codes on trade-secret or withheld constituents.	Section 2 signal word, pictograms and H/P codes; Section 3 trade-secret row; hazard text in Sections 4–7 and 10; Section 11 acute lines; Section 12 summation.	No hazards are counted and no trade-secret row prints. Publishing is blocked ("TRADE_SECRET (x%)") unless another raw material covers that group, in which case the hazard is silently missed.
Raw materials	Save (review) each raw material, and set its product family (by rule or by manual pick).	Whether bulk publish can include it and the readiness page; the family vote for finished goods (Section 1 use text, UV notes, Section 14 keywords).	Every finished good, alias and private-label item that uses it is skipped by bulk publish, with no reason given. A raw material with no family counts toward no family.
Raw materials	Constituent chemical name, trade-secret flag and trade-secret description.	Section 3 Chemical Name; the Section 12 table; Section 13 RCRA names; Section 14 technical names; the Section 4 UV note; acrylate detection by name.	An unflagged secret prints its real CAS and name. A blank description prints "Trade Secret" (in English on every language). A blank name leaves an empty cell and no Section 14 technical name.
Raw materials	Physical state, odor and appearance (entered in English), and Substance/Mixture.	Carbon black and TiO2 rules (Sections 2, 3, 8, 11 and the Section 15 Prop 65 warning); Section 9 State, Odor and Appearance; Section 6 Containment; Section 3 Type.	The raw material counts as a liquid: carbon black and TiO2 lose H351, exposure limits and Prop 65. If it is the largest raw material, 'Liquid' and 'Not determined' (odor) can print.
Raw materials	For resale raw materials: supplier product name and internal code.	Section 1 Product Identifier, footer, PDF title and file names on resale SDSs; Section 14 keyword rule for UN number and shipping name.	Prints "CODE — " and leaves the footer cell empty. Section 14 defaults to Printing ink (UN1210), and resale sheets cannot take Section 14 overrides.
Raw materials	Prop 65 entries (listed CAS, or a name and types, plus the trace tick) and HAP entries (name and weight %).	Section 15 Prop 65 warning, listed chemicals and '(trace)' marker; Section 15 HAP table and total.	Only chemicals matched by constituent CAS print; others are missed. Old single-name Prop 65 entries and blank or 0% HAP entries are ignored.
Raw materials	Specific gravity, VOC % (or the 'less than 1% VOC' flag), solids, exempt VOC and water %, and solubility.	Section 9 Specific Gravity, VOC (lb/gal and wt%), Solids (wt%) and Solubility.	Specific gravity is silently treated as 1.0 and VOC as 0%. Solids are estimated and overstated (100% if nothing is entered). Solubility prints 'Not determined' if no raw material has a value.
CAS determinations	A competent-person determination for every ingredient CAS without current federal data: classes, H/P codes, pictograms, signal word, categories and LD50/LC50 values.	All of Section 2; Sections 3–8 and 10; Section 11 (including ATEmix); Section 12; Section 13 D-codes; Section 14 class; Section 15 OSHA Status.	At or above the threshold, publishing is blocked (manual and private-label only). Below it, generic text prints, e.g. 'Not a hazardous substance or mixture.' Missing LD50/LC50 values silently use defaults.
CAS determinations	Exposure limits on active determinations, and the TSCA Review status (listed, exempt or not listed) for any CAS not on the public inventory.	Section 8 exposure table; Section 11 component limit lines; Section 3 listing; Section 15 TSCA Status and the TSCA warning.	Only federal limits print. 'TSCA inventory status has not been verified for all components.' prints, and a warning shows on manual publish and in previews.

Where	Keep populated	Feeds	If left blank
CAS determinations	Preferred chemical name, plus the nitrogen, sulfur and halogen flags and molecular formula (CAS Descriptions tab). Re-run the flag fill after adding new CAS records.	Chemical names in Sections 3, 8 and 11–15 and the Section 4 UV note; Section 14 technical names; Section 10 Hazardous Decomposition Products.	The constituent name is used instead. Flags left at 0 print only "Carbon monoxide, carbon dioxide, and other toxic gases may be released upon thermal decomposition."
Finished goods	A current formula with correct percentages, including on every finished good used as a component, and the Active flag.	Every section: Section 3 bands and cut-off, Section 2 and all hazard text, and Section 12 summation. Also whether bulk and private-label publishing can include it.	Publishing is blocked: 'No current formula found'. A component with no formula silently adds nothing. Inactive items are skipped by bulk publish but can still be published manually.
Finished goods	Hazard Classification Override: add or replace mode, classes with their H-codes, P-codes, pictograms, signal word and rationale (needs the override permission).	All of Section 2 and the hazard text in Sections 3–15 (not on resale sheets); Section 7 aerosol sentences.	The calculated classification is used. When it is filled, a class without its H-code prints with no phrase, an override with no mode is ignored, and replace mode clears the calculated codes.
Finished goods	Product code and description (including keywords like COATING, VARNISH, WASH or REDUCER), alias customer code and description, and Transport Product Type.	Section 1 Product Identifier, footer code, PDF title and file names; Section 14 UN number and Proper Shipping Name.	A blank description prints "BK1080 — ". A blank code gives the file name 'unnamed_file'. With no keyword or type set, Section 14 defaults to Printing ink (UN1210).
Finished goods	Physical state, color and Substance/Mixture.	Section 9 Physical State, Color and Appearance; Section 6 Containment; Section 8 dust sentence; Section 14 solid or liquid entry; Section 3 Type.	The largest raw material's state is used, else 'Liquid'. The Color line is left out. Type prints "Mixture" unless the formula is one raw material marked Substance.
Finished goods	Product family pick (Auto unless deliberately fixed). Leave the product's own Recommended Use and Restrictions blank unless the product really differs.	Section 1 use text; UV notes in Sections 4–8 and 11; Section 10 'UV light'; Section 14 keywords.	A blank use field is correct, so the family text prints. A filled one prints in every language and hides the family text. A manual pick ignores UV raw materials added later.
Product families	Membership rules, active flag, family name, and Recommended Use and Restrictions text in each language. Run Recompute after rule changes, switching a family on or off, or a CMS import.	The family of every product and raw material: Section 1 Recommended Use and Restrictions, UV logic, and Section 14 shipping-name keywords (the family name does not print).	The standard sentence prints, e.g. "Printing ink for commercial and industrial applications." A blank language prints the English family text. The old family keeps printing until Recompute.
Product families	The UV/LED flag on UV families.	UV notes in Sections 4, 5, 6, 7 and 11; UV PPE sentences on the Section 8 protection lines; Section 10 'UV light' item.	No UV text anywhere on the sheet, even for products full of acrylates.
Manufacturers	Emergency phone for each private-label manufacturer.	Section 1 Emergency line on that manufacturer's private-label SDSs.	Publishing is blocked for that manufacturer's private-label items. The line is left out and the preview shows a warning.
Manufacturers	Name, address, phone, email, website, logo and a disclaimer in each language, plus each private-label item's custom code and description or shared alias.	Private-label Section 1 Company, supplier lines and Product Identifier; page 1 logo; Disclaimer block; footer, PDF author and file names.	Blank lines are left out and no logo prints (the company's is never used). The disclaimer falls back to Admin settings. If an alias now points to another product, publishing is blocked.
Admin settings	Emergency Phone on Admin > Settings.	Section 1 Emergency line on every standard, resale and alias SDS.	Publishing is blocked on every path except private-label. A bulk job fails entirely, private-label items included. The settings save refuses a blank value.
Admin settings	The 'block publishing on missing hazard data' switch and the missing-data threshold %.	The publishing check for any CAS with no federal data and no determination (nothing prints from these).	If never set, blocking is on at 1.0%. A threshold saved blank becomes 0%, so every CAS with no data blocks publishing.
Admin settings	Company name, address, phone, email, website, logo, and the Legal/Disclaimer statement in each language.	Section 1 Company, Address and contact lines; page 1 logo; Disclaimer after Section 16; PDF author.	If never saved, 'Your Ink Company, Inc.' prints. If saved blank, the line is left out, Author shows 'SDS System' and no logo prints. A blank disclaimer prints the built-in translated text.

Where	Keep populated	Feeds	If left blank
Admin settings	Hazard rule settings: the inhalation-only CAS list, the UV acrylate rule pack switch, the Prop 65 trace threshold % and the GHS Sections 12–15 footnote.	Carbon black and TiO2 rules (Sections 2, 3, 4, 8, 11 and 15); UV notes in Sections 4–8 and 11; Section 15 '(trace)'; footnote after Section 15.	Defaults apply: carbon black and TiO2, rule pack on, 0.1% trace threshold, and the footnote prints. Any rule-pack value other than 'enabled' turns the UV notes off.
Admin settings	Default Product Use (keep it blank), physical state and color lists, pictogram artwork, time zone, and the bulk publish blackout windows and auto-run switch.	Section 1 use text on new finished goods; Section 9 State and Color; GHS, PPE and Prop 65 symbols; Effective Date; when bulk publish runs.	A filled Default Use stops family text on new finished goods. Custom states and colors print untranslated in ES/FR/DE. Shipped images and America/Chicago are the defaults.
Regulatory lists	Federal hazard data import and refresh for each CAS: classifications, acute toxicity estimates and aquatic M-factors.	Section 2 and all hazard text; Section 3 listing; Section 11 acute and chronic lines and ATEMix; Section 12 M-factors; the missing-data publishing check.	The CAS counts as missing data, so publishing is blocked at or above the threshold unless a determination exists. Aquatic prints "(M = 1)". An import does not trigger bulk republish.
Regulatory lists	Federal exposure limits (NIOSH, ACGIH, OSHA), the carcinogen list (one row per CAS and agency; it has no admin screen) and the EPA TSCA inventory.	Section 8 exposure table; carcinogen lines in Sections 2, 3, 4 and 11; Section 15 TSCA Status.	The Section 8 table is left out and Section 11 prints 'No components ... listed'. 'TSCA inventory status has not been verified for all components.' prints, with a warning.
Regulatory lists	RCRA codes, SARA 313 (de minimis and PBT flag), Prop 65 (lower-case toxicity types), HAP, SNUR and exempt VOC lists.	Section 13 waste codes; Section 15 SARA 313, Prop 65, HAP and SNUR; Section 12 Persistence and Bioaccumulation; Section 9 Solids.	'None' lines print, e.g. 'not known to contain any chemicals listed under California Proposition 65'. Section 12 prints "No data available." Lead, chromium and cadmium compounds need their own RCRA rows.
SDS editor	Section 14 overrides (UN number, shipping name, class, packing group, environmental hazards) and the Section 9 Flash Point override, entered in every language.	Section 14 lines. The flash point override also drives Sections 5, 9, 13 (D001) and 14. A UN number or class override lifts the 'Not determined' block.	Derived values print, possibly "Not determined". An English-only override leaves ES/FR/DE printing "Not determined". Type the degree sign, or '75 F' is read as 75 °C.
SDS editor	Review old frozen overrides and click 'Reset to automatic'. Republish after any change to hazards, overrides, families or settings, including base products for aliases.	Every overridden line, especially in Sections 4, 5, 7, 8, 10, 12 and 15. Published PDFs and alias sends print the stored snapshot.	Old wording keeps printing, e.g. 'Stable' on a reactive product, 'HazCom 2012' on unclassified ones, or a blanket TSCA claim. Sheets that are not republished keep their publish-time text.
SDS editor	Other text overrides in each language: Section 1 use, Section 2 Other Hazards, and Sections 4–13, plus the Section 15 OSHA, TSCA and State Regulations notes.	Replaces the whole matching line on standard, alias and private-label sheets. Section 8 protection overrides also feed Section 2, and Section 10 incompatibles feed Section 7.	Automatic text prints, e.g. Section 2 'None known.', and the State Regulations line is left out. English overrides do not carry over to ES/FR/DE. Overrides suppress the UV sentences.

PUBLISH GATES

Conditions checked when an SDS is published. A block stops that publish; a warning lets it through and tells the operator.

Condition	Effect	Applies to	How to clear it
Company emergency phone is blank	Blocks publishing	Manual finished-good, resale and alias publish; SDS Update 'Republish Selected' (whole selection refused); a bulk or scheduled job fails entirely. Not private-label republish.	Enter and save Emergency Phone in Admin > Settings. The built-in placeholder does not count, and the save refuses a blank.
An ingredient CAS at or above the missing-data threshold (default 1%) has no federal hazard data and no active competent-person determination	Blocks publishing	Manual finished-good and resale publish, private-label republish, and the private-label follow-on after manual publish. Not SDS Update republish or bulk/scheduled runs.	Import federal hazard data or add a CAS Determinations entry; or reduce the ingredient below threshold; or change the threshold or blocking option in Admin > Settings.
Section 14 transport would print "Not determined": no raw material has a flash point and there is no hazard classification data	Blocks publishing	Manual finished-good and resale publish, SDS Update republish, private-label republish (first language only). Bulk and scheduled runs check every language.	Add a raw-material flash point (the only option for resale), or in the SDS editor a numeric flash point, UN Number or Hazard Class override.

Condition	Effect	Applies to	How to clear it
Private-label manufacturer's emergency phone is blank	Blocks publishing	Private-label 'Publish now', Republish, Republish stale, follow-on after manual or SDS Update publish (base sheet still publishes), and bulk private-label items.	Enter the Emergency Phone on that manufacturer's record under Manufacturers.
Private-label item retired, finished good inactive, no published base SDS in a language, shared alias moved to another product, or manufacturer missing	Blocks publishing	Private-label republish and follow-on publishes; bulk/scheduled private-label items. Frozen items are skipped automatically, but manual Republish still publishes them.	Restore the item or activate the finished good; publish the base SDS in every language; re-point to a valid alias or enter a custom code.
TSCA inventory status not verified for every ingredient	Warning only	Shown after manual finished-good and resale publish, in every preview and on the readiness page. Silent on SDS Update, bulk and private-label publish.	Import the CAS into the TSCA inventory, set a TSCA Review on CAS Determinations, or add a Section 15 TSCA Status override in the SDS editor.
Shipping name ends "n.o.s." but no technical (chemical) names print, as US transport rules require	Warning only	Preview Warnings box only. Not shown when publishing and never printed on the sheet.	Add a Section 14 Proper Shipping Name override in the SDS editor, or ingredient names plus hazard data identifying the main hazard ingredient.
Other preview warnings: blank phone, transport not determined, calculation issues, UV acrylate formulator notes	Warning only	Finished-good preview shows the most; resale and private-label previews show fewer. Never printed on the sheet.	Fix the underlying data (phone, TSCA, transport, raw material data), then preview again.
Not checked at all: product family not found, UV product with no acrylate found, missing specific gravity, VOC or solids	Warning only	In practice no warning appears: every publish path publishes normally, using defaults (specific gravity 1.0, VOC 0, solids back-calculated).	Nothing to clear. Fix the data, then republish queued products or run the family Recompute on Product Families.
Routine controls: user permission, scheduled-run blackout window, auto bulk publish switch, another run in progress, bulk eligibility (unreviewed raw materials, unpublished intermediates, inactive products)	Blocks publishing	Permission covers all manual actions. The rest affect only bulk and scheduled runs; ineligible products are left out, not failed.	Grant edit permission; wait or change schedule settings; review and save raw materials, publish intermediates, activate finished goods.

WHAT CAN BE CHANGED PER PRODUCT

Text typed in the SDS editor for one product and one language replaces the automatic value for that product and language only.

Field	What a per-product edit does	Limits
Section 1 Product Identifier, company details, Emergency Phone, logo	No override in the SDS editor. These come from the product code and description, Admin > Settings, or the Manufacturers record.	No SDS text edit can change these lines. If the emergency phone is blank, the line is left out and publishing is blocked.
Section 1 Recommended Use / Restrictions on Use	Replaces the automatic text for this product and language, on standard and private-label sheets. It outranks the product's own text, the family text and the standard sentence.	
Section 2 Signal Word, Pictograms, Hazard Classes, Hazard and Precautionary Statements	No SDS editor override. The only route is the Hazard Classification Override (add or replace) on the Finished Goods screen, which needs a special permission.	Not available on resale sheets. In replace mode the calculated codes are cleared. If the override names a class without its H-code, the class prints with no phrase.
Section 2 Personal Protective Equipment lines	No separate override. The Section 8 protection overrides feed these lines.	Prints only when the protection level is driven by hazards. Never prints with the UV acrylate sentence, even if a Section 8 override is stored.
Section 2 Other Hazards	Replaces "None known." for this product and language.	A stored blank override hides the line.

Field	What a per-product edit does	Limits
Section 3 Type and Composition / Information on Ingredients table	No override. The table comes from the formula, the ingredient data and the hazard result. Type comes from the Substance/Mixture field.	To change it, fix the formula, constituents, trade-secret flags, or the inhalation-only CAS list in Admin > Settings.
Section 4 first aid paragraphs (Inhalation, Skin, Eyes, Ingestion, Symptoms, Notes to Physician)	Each one replaces the whole automatic paragraph for this product and language, including its alias and private-label sheets.	The UV Acrylate Note still prints. An English override does not carry over to ES/FR/DE.
Section 5 Suitable / Unsuitable Media, Specific Hazards, Firefighter advice	Each one replaces the whole automatic paragraph for this product and language.	A Section 9 flash point override changes the Specific Hazards flash point sentence but does not change Suitable Extinguishing Media.
Section 6 Personal Precautions, Environmental Precautions, Containment and Cleaning Up	Each one replaces the whole automatic paragraph for this product and language.	The UV Acrylate Note still prints.
Section 7 Handling and Storage	Replaces the whole automatic paragraph for this product and language. A Storage override also replaces the cross-reference to Section 10's incompatible materials.	Without a Storage override, a Section 10 Incompatible Materials override changes the Storage line too.
Section 8 Exposure Limits table	No override. The table comes from federal exposure data and competent-person determinations.	Change it through data imports or CAS Determinations.
Section 8 Engineering Controls, Respiratory, Hand, Eye, Skin Protection	Replaces the whole line for this product and language. The protection overrides also drive the Section 2 PPE lines when those lines are driven by hazards.	A filled override suppresses the UV PPE sentences.
Section 9 Physical State, Color, Specific Gravity, VOC, Solids	No override. These come from the finished good's fields and calculations from raw material data.	An SG that rounds to 0 prints "Not determined"; missing data silently defaults to SG 1.0 and 0% VOC.
Section 9 Appearance, Odor, Boiling Point, Solubility	Replaces the automatic value for this product and language. The boiling point override prints exactly as typed.	A boiling point override does not change Section 14.
Section 9 Flash Point	Replaces the value for this language. A numeric override also feeds the Section 5 flash point line, Section 13 D001 and Section 14 class and packing group.	Text with no number (e.g. 'None') is ignored by those other sections. Always type the degree sign: '75 F' is read as 75 °C.
Section 10 Reactivity, Stability, Conditions to Avoid, Incompatible Materials, Decomposition	Replaces the automatic text for this product and language. Incompatible Materials also feeds the Section 7 storage list.	A Conditions to Avoid override suppresses the 'UV light' item.
Section 11 Acute Toxicity, Chronic Effects, Carcinogenicity	Each one replaces the whole automatic paragraph for this product and language.	The component toxicology table has no override. The UV Acrylate Note still prints.
Section 12 Ecotoxicity, Persistence, Bioaccumulation, Mobility in Soil	Each one replaces the whole automatic paragraph for this product and language.	The component aquatic toxicity table still prints and has no override.
Section 13 Disposal Methods / Waste Classification (RCRA)	A Disposal Methods override replaces the standard 'Dispose of in accordance with all applicable federal, state, and local regulations' sentence.	RCRA Waste Classification has no override. Only a Section 9 flash point override can affect D001.
Section 14 UN Number, Proper Shipping Name, Hazard Class, Packing Group, Environmental Hazards	Replaces the derived line for this language. A UN Number or Hazard Class override also lifts the "Not determined" publish block.	A Shipping Name override alone does not lift the block. Resale sheets cannot be overridden. The transport note has no override. An English-only override leaves ES/FR/DE showing "Not determined".
Section 15 OSHA Status and TSCA Status	Replaces the automatic sentence for this product and language. A TSCA Status override also silences the TSCA warning.	Old stored 'HazCom 2012' or 'All components are listed...' notes keep printing until reset to automatic.
Section 15 State Regulations; SARA 313, HAPs, SNUR, Prop 65	State Regulations prints only from an override, after the Prop 65 block. Without one the line is left out.	SARA 313, HAP, SNUR and Prop 65 have no override. They come from the regulatory lists and manual entries on raw materials.

Field	What a per-product edit does	Limits
Section 16 Version, Effective Date, Abbreviations, Disclaimer; GHS footnote	No override. Version and date are stamped at publish. Abbreviations are built from the printed text. The disclaimer comes from Admin > Settings or the manufacturer.	The GHS footnote can only be switched off in Admin > Settings.
All overrides (general)	Overrides are saved per product and per language. Republish to apply them, because published PDFs print the snapshot stored at publish time.	Old frozen overrides keep printing over the new logic, and a stored blank override hides the line. Use 'Reset to automatic' to clear them.

PRIVATE LABEL SHEETS

- Product identifier: a custom code and description from the private-label registry. If none is set, the shared alias code (cut at the first hyphen) and its description are used, then the base product's. The identifier is frozen at publish and also prints in the footer.
- Supplier block (company, address, phone, email, website): from the manufacturer record only. No company value is ever inherited.
- Emergency phone: the manufacturer's own number only. If it is blank, the preview shows a warning and publishing is refused.
- Logo: the manufacturer's logo, or no logo. The company logo is never used.
- Disclaimer: the manufacturer's text for that language. If it is blank, the standard admin or translated disclaimer prints, which may name the base company.
- Version number: counted per private-label item, so its first publish prints version 1 whatever the base product's version. Every configured language is published under the same number.
- Abbreviations: filtered again against the branded sheet, so acronyms in manufacturer details can add definitions.
- PDF Author and file names: the manufacturer name (stored as {code}_PL_{Manufacturer}_v{n}[_lang].pdf; downloads use the PL_SDS_ prefix).
- Everything else (Sections 2-16, use text, per-product overrides) is inherited from the base product by design. There is no manufacturer-level content override.

LANGUAGES

- Translated in ES/FR/DE: titles, labels and banners (accented capitals now correct), footer words (FR 'Page x sur y', 'Rév.'), GHS statements, all hazard-derived paragraphs in Sections 2-15, Section 9 standard words, the Prop 65 warning, RCRA, PBT and SARA wording, abbreviation tables, PDF Title and Subject, and the default disclaimer.
- English by design on every sheet: regulatory citations (29 CFR, 49 CFR, ANSI, NFPA, NIOSH), agency acronyms, 'CAS', 'EPA Method 24', and 49 CFR proper shipping names (with a translated gloss added on ES/FR/DE).
- Data stored in one language, which prints as entered (English) on every sheet: chemical names, trade-secret descriptions, list names (SARA, HAP, SNUR, Prop 65), exposure-limit notes, and raw-material odor and appearance text.
- Product-level use text typed in the product's own columns prints on every language sheet. Use per-language family defaults or per-language editor entries to localise it.
- Per-product overrides are stored per language. An English override does not carry over to ES/FR/DE, and State Regulations has no fallback to other languages.
- Not localised: month/day/year dates, '.' decimals in ranges and Section 9 values, and US Letter page size.
- A missing translation falls back to English, then to the raw key. Sheets published before the translation update still show some English document strings until they are republished.

TEXT THAT STAYS FIXED

Content that is the same on every sheet, and why.

Item	Why it is fixed
Section titles, field labels, table headings, 'SAFETY DATA SHEET', page counter and footer words	Required SDS layout (29 CFR 1910.1200 App. D). These are format constants, translated per language.
GHS hazard (H) and precautionary (P) statement wording, pictogram names and signal words	Official GHS/HazCom phrases. Only which ones print depends on the data.
0.1% disclosure cut-off and the prescribed concentration ranges (0.1-1% ... 80-100%)	Policy fixed by decision #8 and the 1910.1200(i) prescribed ranges. Deliberately not settings.
Trade-secret / withheld-percentage statement citing 29 CFR 1910.1200(i)	Regulatory statement. It prints whenever an ingredient row is listed, because every row shows a range.
Standard sentences in Sections 4-13 (first aid, fire, spill, handling, PPE, stability, toxicology, ecology, disposal)	Fixed wording chosen by hazard code, physical state or product family. The text is the same for every product that triggers it.
Base defaults: 'None known.' (Other Hazards), 'Mobility in Soil: No data available.' and the generic disposal sentence	The system holds no data for these. Each can be replaced per product.
Carcinogen-list mapping and the H-code-to-PPE mapping	Regulatory interpretations kept in code. They could become data later.
49 CFR proper shipping names, the carrier-verification note and the combustible/viscous notes	Regulatory entry names and wording. ES/FR/DE sheets add a translated gloss.
Prop 65 warning templates and the Sections 12-15 footnote	Safe-harbor wording and a 1910.1200(g)(2) citation. The footnote has only an on/off admin switch.
UV acrylate sentences and the acrylate detection list	One fixed rule pack for all UV families. Detection is a hard-coded list of CAS numbers and name patterns.
US Letter page size, Helvetica, month/day/year dates and '.' decimals on every language	Format constants. There is no A4 or local number or date option for ES/FR/DE today.
Missing sub-sections: Section 4 'General advice', Section 7(c) specific end uses, Section 14(f)/(g), and unsupported Section 9 properties (pH, vapour pressure, etc.)	Not generated. There is no data field for them.

CHANGES SINCE THE FIRST AUDIT

Before this update roughly a third of the sheet came from product data. These are the main fields that moved from fixed text to data, or are new.

Field	Change	Note
S1 Recommended Use / Restrictions on Use	Now from data	New per-language product-family default between the product's own text and the generic sentence (#3).
S1 Emergency phone	Changed	Comes from Admin Settings only, with no config fallback. Private label uses the manufacturer's number. Blank blocks publishing (#2).
S1 Email / Website / Country	Changed	Now printed on the PDF. The address no longer has stray commas (#1, #38).
S1 Product Family line	Removed	No longer printed. The family is still used for use text, UV and transport.
S2 PPE recommendations	Now from data	Translated tier sentences chosen from H-codes, shared with Section 8. Products with only physical hazards no longer get PPE icons (#15).
S2 Other Hazards	Changed	Always printed: the override if one exists, otherwise 'None known.' (#4).
S3 Type (Mixture / Substance)	Now from data	Was always 'Mixture'. Now from the Substance/Mixture setting, with an automatic rule for single-ingredient formulas (#6).
S3/S8/S11/S12/S15 concentrations	Changed	A prescribed range replaces the exact % everywhere. The HAP total stays exact (#8).
S4 First-aid routes and Notes to Physician	Now from data	Built from H-codes, with aspiration, corrosive and delayed-inhalation wording (#9).

Field	Change	Note
S4 Most Important Symptoms (acute / delayed)	New	Restates health hazard statements; editable (#10).
S5 Extinguishing media and firefighting advice	Changed	New flammable and water-reactive tiers. The flash point uses the same value as Section 9.
S6 Spill precautions and containment	Changed	Aquatic wording follows the classification tier. Containment follows the physical state, falling back to the dominant raw material's state.
S7 Handling / Storage	Now from data	Built up from every matching hazard instead of one canned paragraph. Storage names the Section 10 incompatible materials (#13).
S8 Engineering controls and PPE	Now from data	Base sentence plus dust, flammable and corrosive additions. Translated PPE tiers include a 'no special PPE' tier (#14, #15).
S9 Boiling Point, Odor, Appearance, Physical State, Solubility	Now from data	Calculated from raw-material data (new boiling point field, migration 054). VOC and solids lines are no longer printed (#16-#18).
S10 Reactivity, stability, conditions, incompatibles, decomposition	Now from data	Built up from hazard classes and the nitrogen, sulfur and halogen flags on ingredient CAS numbers (#19).
S11 Acute toxicity, chronic effects, carcinogenicity	Now from data	Lines per route with category and ATEmix, chronic text from H-codes, and banded carcinogen findings with the 0.1% threshold (#20-#22).
S12 Ecotoxicity, PBT, component aquatic table, Mobility in Soil	Now from data	Repeats the product's aquatic H-statements, adds a table with M-factors and names SARA 313 PBT chemicals (#23, #24).
S13 RCRA hazardous-waste line	New	D001-D003 from H-codes and flash point, plus component codes from the admin-maintained RCRA list (#26).
S14 UN number / shipping name / class / packing group	Now from data	Classified for the mixture by TransportClassifier, with a marine pollutant line and notes. Overrides now take priority. 'Not determined' blocks publishing (#27).
S15 OSHA and TSCA status	Now from data	Follows the classification (HazCom 2024) and a per-CAS TSCA check, with a 'not verified' warning (#28, #29).
S15 SARA 313 block and Prop 65 listed chemicals	New	SARA 313 now actually prints. Prop 65 chemicals are listed under a translated warning (#30, #42).
S16 Version / Effective Date and footer revision	New	Stamped at publish. Replace the generation date. Previews print 'Draft (not yet published)' (#32).
Abbreviations and Disclaimer	Now from data	A per-language abbreviation table filtered to the terms on each sheet. Disclaimer set per language in Admin Settings and per manufacturer (#33, #34).
Per-section 'not required by OSHA HazCom' notes	Removed	Replaced by one switchable footnote after Section 15 (#25).